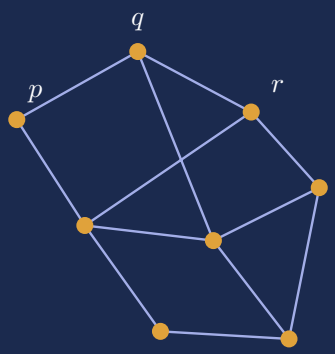
$\neg \quad \wedge \quad \vee \quad \rightarrow \quad \leftrightarrow \quad \forall \quad \exists$



DISCRETE MATHEMATICS AND GRAPH THEORY

Unit 3
Propositional Logic

Syntax and semantics · Connectives and truth tables · Validity and satisfiability · Laws of logic · Logical implication · Rules of inference · Quantifiers

LECTURE NOTES · SOLVED PROBLEMS · EXERCISES WITH SOLUTIONS

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Syllabus hours	6 hrs (Unit 3.0)
Prepared for	B.Tech CSE students, GEC Munger
Exam pattern	University theory paper: 2, 5/7 and 10/15-mark questions

References: K. H. Rosen, *Discrete Mathematics and its Applications*; R. P. Grimaldi, *Discrete and Combinatorial Mathematics*; J. P. Tremblay & R. Manohar, *Discrete Mathematical Structures with Applications to Computer Science*; C. L. Liu & D. P. Mohapatra, *Elements of Discrete Mathematics*; S. Lipschutz & M. Lipson, *Discrete Mathematics* (Schaum's Outlines).

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How to use these notes

Every topic follows the same pattern: **definition** (teal) → **laws and theorems** (indigo) → **worked examples** → **pitfalls** (red) and **exam tips** (green). Sections 2–7 also contain exercise sets (2A, 2B, 3A, 4A–4C, 5A, 6A, 7A–7C), each followed at once by complete solutions in teal boxes. Section 6.6 has ten extra solved examples on rules of inference; Section 8 contains 12 fully solved university-pattern problems; Section 9 is a graded exercise set (Parts A–D) whose complete solutions are in Section 10. A one-page revision sheet closes the notes.

Notation: T / 1 = true, F / 0 = false; $\neg p$ (also $\sim p$) = not p; T_0 = any tautology; F_0 = any contradiction; $\equiv$ (or $\Leftrightarrow$) = logical equivalence; $\Rightarrow$ = logical implication; $\therefore$ = therefore.

Syllabus item (Unit 3.0)	Section	Suggested hours
Syntax, Semantics	1	0.5
Basic Connectives and Truth Tables	2	1.0
Validity and Satisfiability	3	0.5
Logical Equivalence: The Laws of Logic	4	1.0
Logical Implication	5	0.5
Rules of Inference	6	1.5
The use of Quantifiers	7	1.0

Proof techniques (direct proof, proof by contraposition, proof by contradiction, and proof of necessity and sufficiency) are covered in the separate Unit 4 notes.

1 Introduction and Core Definitions

Mathematical logic is the study of correct reasoning. It has two parts. In **propositional logic** we treat each statement as a single unit that is either true or false. In **predicate logic** we look inside a statement at objects and their properties, and use quantifiers such as "for all" and "for some". Sections 1–6 of this unit cover propositional logic; Section 7 covers quantifiers.

1.1 Proposition

Definition: Proposition

A **proposition** (statement) is a declarative sentence that is either true or false, but not both. Its truth value is written T (or 1) and F (or 0).

Sentence	Proposition?	Reason
New Delhi is the capital of India.	Yes	Declarative; value T
$1 > 2$.	Yes	Declarative; value F
$x + 3 = 6$	No	Value depends on x (open statement)
Close the door.	No	Command
What time is it?	No	Question
How beautiful the Ganga is!	No	Exclamation
This statement is false.	No	Paradox: no consistent truth value

Classical logic rests on two laws:

- **Law of excluded middle:** every proposition is either true or false.
- **Law of non-contradiction:** no proposition is both true and false.

1.2 Atomic and compound propositions

- An **atomic** (simple, primitive) proposition cannot be broken into smaller propositions, e.g. p: "It rains today".
- A **compound** proposition is built from atomic ones with connectives, e.g. "It rains today *or* the match is played" $= p \vee q$.
- **Truth-functionality:** the truth value of a compound proposition is completely determined by the truth values of its parts and the connectives used.

1.3 Syntax: well-formed formulas

Syntax is the set of rules that tells us which strings of symbols are correctly written formulas; it does not deal with meaning. The symbols used are propositional variables $p, q, r, \ldots$; constants T, F; connectives $\neg, \wedge, \vee, \rightarrow, \leftrightarrow$; and brackets.

Definition: Well-formed formula (wff) — recursive

(i) Every propositional variable and the constants T, F are wffs.

(ii) If A is a wff, then $\neg A$ is a wff.

(iii) If A and B are wffs, then $(A \wedge B)$, $(A \vee B)$, $(A \rightarrow B)$ and $(A \leftrightarrow B)$ are wffs.

(iv) Nothing else is a wff.

For example, $(p \to (q \vee \neg r))$ is a wff, while $p \vee \wedge q$ and $(p \to q$ are not.

Order of precedence (highest first): $\neg,\ \wedge,\ \vee,\ \to,\ \leftrightarrow$. Thus $\neg p \vee q \wedge r \to s$ means $((\neg p) \vee (q \wedge r)) \to s$. In examinations, always use full brackets to avoid confusion.

1.4 Semantics: interpretations

Semantics deals with meaning. An **interpretation** (truth assignment) gives each variable the value T or F; the truth tables of the connectives then give the value of the whole formula.

$$\text{Rows in a truth table} = 2^n, \qquad \text{Distinct truth functions of } n \text{ variables} = 2^{2^n}$$

With $n = 2$ there are 4 rows and 16 possible truth functions; $\wedge, \vee, \to, \leftrightarrow, \oplus, \uparrow, \downarrow$ are seven of them. With $n = 3$ there are 8 rows; with $n = 4$, 16 rows.

1.5 More on propositions

Logic is the study of the methods of reasoning. When it is written in a precise symbolic language it is called **symbolic logic** or **mathematical logic**.

Definition: Proposition and truth value

A **proposition** is a statement (declaration) which, in a given context, can be said to be either true or false, but not both. The truth or falsity of a proposition is its **truth value**: 1 if it is true and 0 if it is false (or T and F).

- "Bengaluru is in Karnataka" and "Three is a prime number" are true propositions; "Seven is divisible by 3" and "Every rectangle is a square" are false propositions.
- "Take a triangle ABC" is not a proposition: it is not a statement at all.
- "$xy = yx$" is a statement, but we cannot say whether it is true or false until we know what x and y are; so it is not a proposition. If the context tells us that x, y are real numbers, it becomes a (true) proposition.

Words such as *not, and, or, if...then, if and only if* are **logical connectives**. The propositions from which a compound proposition is built are its **components** or **primitives**; propositions containing no connective are **simple** (atomic) propositions.

2 Basic Connectives and Truth Tables

2.1 The five basic connectives

Connective	Symbol	Read as	Rule to memorise
Negation	$\neg p$	not p	True exactly when p is false
Conjunction	$p \wedge q$	p and q	True only when both are true
Disjunction	$p \vee q$	p or q	False only when both are false
Implication	$p \to q$	if p then q	False only when p is T and q is F
Biconditional	$p \leftrightarrow q$	p iff q	True when p, q have the same value

2.2 Master truth table

p	q	$\neg p$	$p \wedge q$	$p \vee q$	$p \to q$	$p \leftrightarrow q$	$p \oplus q$	$p \uparrow q$	$p \downarrow q$
T	T	F	T	T	T	T	F	F	F
T	F	F	F	T	F	F	T	T	F
F	T	T	F	T	T	F	T	T	F
F	F	T	F	F	T	T	F	T	T

Row-ordering convention. For n variables, the first column alternates in blocks of 2^{n-1}, the next in blocks of 2^{n-2}, and the last column alternates every row. Always count your rows: 2^n.

2.3 Understanding the conditional $p \to q$

p is the **antecedent** (hypothesis) and q the **consequent** (conclusion). Read it as a promise: "If you score 90, I will give you a prize." The promise is broken only when you score 90 and receive no prize.

- If p is false, $p \to q$ is true whatever q is (**vacuous truth**).
- If q is true, $p \to q$ is true whatever p is.
- $p \to q \equiv \neg p \vee q$ — the single most used identity in this unit.

English form	Symbol
If p then q; p implies q; q follows from p	$p \to q$
p only if q	$p \to q$
p is sufficient for q; q is necessary for p	$p \to q$
q if p; q whenever p	$p \to q$
q unless r	$\neg r \to q \equiv r \vee q$
p is necessary and sufficient for q	$p \leftrightarrow q$

Common pitfall

"p **only if** q" is $p \to q$, *not* $q \to p$. "You can clear GATE only if you appear for GATE" means clear $\to$ appear. "Unless" simply becomes $\vee$: "You will not crack GATE unless you study" $= \neg c \vee s \equiv c \to s$.

2.4 Converse, inverse and contrapositive

Name	Form	Equivalent to $p \to q$?
Converse	$q \to p$	No
Inverse	$\neg p \to \neg q$	No
Contrapositive	$\neg q \to \neg p$	**Yes**

p	q	$p \to q$	$q \to p$	$\neg p \to \neg q$	$\neg q \to \neg p$
T	T	T	T	T	T
T	F	F	T	T	F
F	T	T	F	F	T
F	F	T	T	T	T

Result

A conditional is equivalent to its contrapositive; the converse is equivalent to the inverse (each is the contrapositive of the other).

2.5 Biconditional

$$p \leftrightarrow q \;\equiv\; (p \to q) \wedge (q \to p) \;\equiv\; (p \wedge q) \vee (\neg p \wedge \neg q), \qquad \neg(p \leftrightarrow q) \equiv p \oplus q.$$

2.6 NAND and NOR — functionally complete connectives

$p \uparrow q = \neg(p \wedge q)$ and $p \downarrow q = \neg(p \vee q)$. Each alone can express every connective, which is why NAND or NOR gates alone can build any digital circuit.

Expression	NAND only	NOR only
$\neg p$	$p \uparrow p$	$p \downarrow p$
$p \wedge q$	$(p \uparrow q) \uparrow (p \uparrow q)$	$(p \downarrow p) \downarrow (q \downarrow q)$
$p \vee q$	$(p \uparrow p) \uparrow (q \uparrow q)$	$(p \downarrow q) \downarrow (p \downarrow q)$
$p \to q$	$p \uparrow (q \uparrow q)$	$((p \downarrow p) \downarrow q) \downarrow ((p \downarrow p) \downarrow q)$

NAND and NOR are commutative but **not associative**: $p \uparrow (q \uparrow r) \equiv \neg p \vee (q \wedge r)$ whereas $(p \uparrow q) \uparrow r \equiv (p \wedge q) \vee \neg r$.

2.7 Worked examples

Example 2.1 – Truth value from a given assignment

If p, q are true and r, s are false, find the truth value of
(i) $p \vee (q \wedge r)$ (ii) $(p \wedge (q \wedge r)) \vee \neg((p \vee q) \wedge (r \vee s))$ (iii) $(p \to r) \leftrightarrow (\neg q \to s)$.

Solution. (i) $q \wedge r = \mathrm{T} \wedge \mathrm{F} = \mathrm{F}$, so $p \vee \mathrm{F} = \mathrm{T}$.
(ii) $p \wedge (q \wedge r) = \mathrm{F}$; $(p \vee q) \wedge (r \vee s) = \mathrm{T} \wedge \mathrm{F} = \mathrm{F}$, so its negation is T. Hence $\mathrm{F} \vee \mathrm{T} = \mathrm{T}$.
(iii) $p \to r = \mathrm{T} \to \mathrm{F} = \mathrm{F}$; $\neg q \to s = \mathrm{F} \to \mathrm{F} = \mathrm{T}$. Hence $\mathrm{F} \leftrightarrow \mathrm{T} = \mathrm{F}$.

Example 2.2 – Working backwards

If $p \to q$ is false, find the truth value of $\neg(p \wedge q) \to q$.

Solution. $p \to q$ is false only when $p = \mathrm{T}$, $q = \mathrm{F}$. Then $p \wedge q = \mathrm{F}$, $\neg(p \wedge q) = \mathrm{T}$, and $\mathrm{T} \to \mathrm{F} = \mathrm{F}$. The value is **F**.

Example 2.3 – Constructing a three-variable table

Construct the truth table of $(p \wedge q) \to (p \vee r)$.

Solution. Three variables give $2^3 = 8$ rows.

p	q	r	$p \wedge q$	$p \vee r$	$(p \wedge q) \to (p \vee r)$
T	T	T	T	T	**T**
T	T	F	T	T	**T**
T	F	T	F	T	**T**
T	F	F	F	T	**T**
F	T	T	F	T	**T**
F	T	F	F	F	**T**
F	F	T	F	T	**T**
F	F	F	F	F	**T**

Every entry of the last column is T, so the formula is a tautology (Section 3).

Example 2.4 – Translation

Let p: Ravi studies hard; q: Ravi passes. Symbolise: (a) Ravi passes only if he studies hard. (b) Ravi will not pass unless he studies hard. (c) Studying hard is sufficient for Ravi to pass. (d) Ravi passes if and only if he studies hard, but he does not study hard.

Solution. (a) $q \to p$ (b) $\neg q \vee p \equiv q \to p$ (c) $p \to q$ (d) $(q \leftrightarrow p) \wedge \neg p$ ("but" is always $\wedge$).

Example 2.5 – Converse, inverse, contrapositive

State the converse, inverse and contrapositive of: "If a triangle is not isosceles, then it is not equilateral."

Solution. Let p: the triangle is isosceles; q: it is equilateral. The statement is $\neg p \to \neg q$.

- Converse $\neg q \to \neg p$: If a triangle is not equilateral, then it is not isosceles.
- Inverse $p \to q$: If a triangle is isosceles, then it is equilateral.
- Contrapositive $q \to p$: If a triangle is equilateral, then it is isosceles. (True — and equivalent to the original.)

Example 2.6 – NAND identity

Show that $p \to q \equiv p \uparrow (q \uparrow q)$.

Solution. $q \uparrow q = \neg(q \wedge q) = \neg q$. Then $p \uparrow \neg q = \neg(p \wedge \neg q) = \neg p \vee q \equiv p \to q$ (De Morgan, double negation, conditional law).

2.8 Exclusive disjunction

In $p \vee q$ the word "or" is used in the **inclusive** sense: $p \vee q$ is true when p or q *or both* are true. Sometimes "or" is meant in the **exclusive** sense — one or the other, but not both.

Definition: Exclusive disjunction

The compound proposition $p \veebar q$ (read "either p or q but not both") is the **exclusive disjunction** of p and q. It is true exactly when one of p, q is true and the other is false. (It is also written $p \oplus q$.)

p	q	$p \vee q$	$p \veebar q$
T	T	T	F
T	F	T	T
F	T	T	T
F	F	F	F

The two columns differ only in the first row, where both p and q are true. Note that $p \veebar q \equiv \neg(p \leftrightarrow q)$.

2.9 The connectives on concrete propositions

Example 2.7 – Conjunction and disjunction

Let p: $\sqrt{2}$ is an irrational number, q: 9 is a prime number, r: All triangles are equilateral, s: $2+5=7$. Find the truth values of $p \wedge q$, $q \wedge r$, $r \wedge s$, $s \wedge p$ and of $p \vee q$, $q \vee r$, $r \vee s$, $s \vee p$.

Solution. p and s are true; q and r are false.
$p \wedge q$: "$\sqrt{2}$ is an irrational number and 9 is a prime number" — **false** (q is false). $q \wedge r$ — **false**. $r \wedge s$ — **false**. $s \wedge p$: "$2+5=7$ and $\sqrt{2}$ is an irrational number" — **true**.
$p \vee q$ — **true**. $q \vee r$: "9 is a prime number or all triangles are equilateral" — **false** (both parts false). $r \vee s$ — **true**. $s \vee p$ — **true**.

Example 2.8 – Exclusive disjunction

Let p: $\sqrt{2}$ is an irrational number, q: $2+3=5$, r: 9 is a prime number, s: All triangles are isosceles. Find the truth values of $p \veebar q$, $q \veebar r$, $r \veebar s$, $s \veebar p$.

Solution. p, q are true and r, s are false. $p \veebar q$: "Either $\sqrt{2}$ is irrational or $2+3=5$, but not both" — **false** (both true). $q \veebar r$ — **true**. $r \veebar s$ — **false** (both false). $s \veebar p$ — **true**.

Example 2.9 – Conditional and biconditional

Let p: 2 is a prime number, q: 3 is a prime number, r: 6 is a perfect square, s: 9 is a multiple of 6. Find the truth values of $p \to q$, $p \to r$, $r \to p$, $r \to s$ and of $p \leftrightarrow q$, $r \leftrightarrow s$, $p \leftrightarrow r$, $p \leftrightarrow s$, $q \leftrightarrow r$, $q \leftrightarrow s$.

Solution. p, q are true and r, s are false.
$p \to q$ ("If 2 is a prime number, then 3 is a prime number") is **true**; $p \to r$ ("If 2 is a prime number then 6 is a perfect square") is **false**; $r \to p$ and $r \to s$ are **true** (false antecedent).
$p \leftrightarrow q$ and $r \leftrightarrow s$ are **true** (equal values); $p \leftrightarrow r$, $p \leftrightarrow s$, $q \leftrightarrow r$, $q \leftrightarrow s$ are **false**.

2.10 More worked examples on connectives

Example 2.10 – Compound propositions in words

Let p: A circle is a conic, q: $\sqrt{5}$ is a real number, r: Exponential series is convergent. Express in words: (i) $p \wedge \neg q$ (ii) $\neg p \vee q$ (iii) $p \veebar \neg q$ (iv) $q \to \neg p$ (v) $p \to (q \veebar r)$ (vi) $\neg p \leftrightarrow q$.

Solution. (i) A circle is a conic and $\sqrt{5}$ is not a real number.
(ii) A circle is not a conic or $\sqrt{5}$ is a real number.
(iii) Either a circle is a conic or $\sqrt{5}$ is not a real number (but not both).
(iv) If $\sqrt{5}$ is a real number, then a circle is not a conic.
(v) If a circle is a conic, then either $\sqrt{5}$ is a real number or the exponential series is convergent (but not both).
(vi) If a circle is not a conic then $\sqrt{5}$ is a real number, and if $\sqrt{5}$ is a real number then a circle is not a conic.

Example 2.11 – Four two-variable tables

Construct the truth tables of (i) $p \wedge \neg q$ (ii) $\neg p \vee q$ (iii) $p \to \neg q$ (iv) $\neg p \veebar \neg q$.

Solution. Considering all four combinations of truth values of p and q:

p	q	$\neg p$	$\neg q$	$p \wedge \neg q$	$\neg p \vee q$	$p \to \neg q$	$\neg p \veebar \neg q$
T	T	F	F	F	T	F	F
T	F	F	T	T	F	T	T
F	T	T	F	F	T	T	T
F	F	T	T	F	T	T	F

Example 2.12 – Working back from a false conditional

Let p and q be primitive statements for which $p \to q$ is false. Find the truth values of (i) $p \wedge q$ (ii) $\neg p \vee q$ (iii) $q \to p$ (iv) $\neg q \to \neg p$.

Solution. $p \to q$ is false only when p is true and q is false; so $\neg p$ is false and $\neg q$ is true. Hence (i) $p \wedge q$ is **F**; (ii) $\neg p \vee q = \text{F} \vee \text{F}$ is **F**; (iii) $q \to p = \text{F} \to \text{T}$ is **T**; (iv) $\neg q \to \neg p = \text{T} \to \text{F}$ is **F**.

Example 2.13 – Six values from a given assignment

Let p, q, r have truth values 0, 0, 1 respectively. Find the truth values of (1) $(p \vee q) \vee r$ (2) $(p \wedge q) \wedge r$ (3) $(p \wedge q) \to r$ (4) $p \to (q \wedge r)$ (5) $p \wedge (r \to q)$ (6) $p \to (q \to \neg r)$.

Solution. (1) $p \vee q$ is 0 and r is 1, so the value is **1**. (2) $p \wedge q$ is 0, so **0**. (3) $0 \to 1$ is **1**. (4) p is 0, so the conditional is **1**. (5) $r \to q = 1 \to 0 = 0$, so $p \wedge 0$ is **0**. (6) p is 0, so the conditional is **1**.

Example 2.14 – Finding possible values

Find the possible truth values of p, q, r when (i) $p \to (q \vee r)$ is false; (ii) $p \wedge (q \to r)$ is true.

Solution. (i) $p \to (q \vee r)$ is false only when p is true and $q \vee r$ is false, i.e. q and r are both false. So p, q, r are $1, 0, 0$.
(ii) $p \wedge (q \to r)$ is true only when p is true and $q \to r$ is true. $q \to r$ is true when (a) r is true (q either way) or (b) r is false and q is false. So (p, q, r) is one of $(1, 1, 1)$, $(1, 0, 1)$, $(1, 0, 0)$.

Example 2.15 – Grouping matters

Construct the truth tables of (i) $(p \vee q) \wedge r$ (ii) $p \vee (q \wedge r)$.

Solution. Each formula has three primitives, so there are $2^3 = 8$ rows.

p	q	r	$p \vee q$	$(p \vee q) \wedge r$	$q \wedge r$	$p \vee (q \wedge r)$
T	T	T	T	T	T	T
T	T	F	T	F	F	T
T	F	T	T	T	F	T
T	F	F	T	F	F	T
F	T	T	T	T	T	T
F	T	F	T	F	F	F
F	F	T	F	F	F	F
F	F	F	F	F	F	F

The truth values of $(p \vee q) \wedge r$ and $p \vee (q \wedge r)$ are *not* identical (compare rows 2 and 4, where p is true and r is false): brackets cannot be moved freely.

Example 2.16 – Two more three-variable tables

Construct the truth tables of (i) $(p \wedge q) \to \neg r$ (ii) $q \wedge (\neg r \to p)$.

Solution.

p	q	r	$\neg r$	$p \wedge q$	$(p \wedge q) \to \neg r$	$\neg r \to p$	$q \wedge (\neg r \to p)$
T	T	T	F	T	F	T	T
T	T	F	T	T	T	T	T
T	F	T	F	F	T	T	F
T	F	F	T	F	T	T	F
F	T	T	F	F	T	T	T
F	T	F	T	F	T	F	F
F	F	T	F	F	T	T	F
F	F	F	T	F	T	F	F

Example 2.17 – Working backwards with four primitives

If q has the truth value 1, find all truth value assignments of p, r, s for which $[q \to \{(\neg p \vee r) \wedge \neg s\}] \wedge \{\neg s \to (\neg r \wedge q)\}$ is true.

Solution. The proposition is $u \wedge v$ with $u \equiv q \to \{(\neg p \vee r) \wedge \neg s\}$ and $v \equiv \neg s \to (\neg r \wedge q)$; both must be 1.
Since $q = 1$ and $u = 1$, $(\neg p \vee r) \wedge \neg s$ must be 1. So $\neg p \vee r = 1$ and $\neg s = 1$, i.e. $s = 0$.
Since $\neg s = 1$ and $v = 1$, $\neg r \wedge q$ must be 1, so $\neg r = 1$, i.e. $r = 0$.
Since $\neg p \vee r = 1$ and $r = 0$, $\neg p = 1$, i.e. $p = 0$. Thus p, r, s must all have the value 0 (the only assignment).

Example 2.18 – Rows needed and a value

Indicate how many rows are needed in the truth table of $(p \vee \neg q) \leftrightarrow \{(\neg r \wedge s) \to t\}$, and find its truth value if p and r are true and q, s, t are false.

Solution. There are five primitives p, q, r, s, t, so $2^5 = \mathbf{32}$ rows are needed.
With the given values, $\neg q = 1$ and $\neg r = 0$. $p \vee \neg q = 1$. $\neg r \wedge s = 0$, so $(\neg r \wedge s) \to t = 1$. Hence the biconditional is $1 \leftrightarrow 1$, value **1**.

2.11 Exercise Set 2A — connectives and truth tables

2A.1 State which of the following sentences are propositions: (1) A triangle contains three sides. (2) $x + 2$ is a positive integer. (3) 5 divides x. (4) Is $\sqrt{2}$ a rational number? (5) 21 is an even number. (6) India had a woman prime minister.

2A.2 Indicate the negation of each proposition: (1) $2 + 3 = 5$ (2) 5 divides 27 (3) Computer Science is a hard subject. (4) Bangalore has pleasant weather.

2A.3 Give the conjunction and disjunction of p and q and their truth values: (1) p: 4 is a perfect square, q: 27 is a prime number. (2) p: 5 is divisible by 2, q: 7 is a multiple of 5.

2A.4 Indicate the truth value of: (1) If $3 + 4 = 7$, then $5 + 3 = 8$. (2) If $3 + 4 = 5$, then $4 + 6 = 10$. (3) If $4 + 2 = 7$, then $3 + 2 = 4$. (4) If $4 + 5 = 9$, then $3 + 1 = 6$.

2A.5 For a certain triangle ABC let p: ABC is isosceles, q: ABC is equilateral, r: ABC is equiangular. Write in words: (1) $p \wedge \neg q$ (2) $\neg p \vee q$ (3) $p \to q$ (4) $q \to p$ (5) $\neg r \to \neg q$ (6) $p \leftrightarrow \neg q$.

2A.6 Determine the truth value required: (1) $p \wedge q$ is false and p is true; find q. (2) $p \vee q$ is false and q is false; find p. (3) $p \to q$ is true and q is false; find p. (4) $p \leftrightarrow q$ is true and p is false; find q.

2A.7 Given that p is true and q is false, find the truth values of (1) $\neg p \wedge q$ (2) $\neg(p \to \neg q)$ (3) $(p \wedge q) \to (p \vee q)$ (4) $\neg(p \wedge q) \vee \{\neg(q \leftrightarrow p)\}$ (5) $(p \to q) \vee \{\neg(p \leftrightarrow \neg q)\}$ (6) $(p \to \neg q) \vee (q \to \neg p)$.

2A.8 Construct the truth tables of (1) $p \vee \neg q$ (2) $p \wedge (q \wedge p)$ (3) $p \wedge (q \vee p)$ (4) $(p \vee q) \wedge \neg p$ (5) $\neg(p \vee \neg q)$ (6) $q \leftrightarrow (\neg p \vee \neg q)$ (7) $p \to (q \to r)$ (8) $(p \to q) \to r$ (9) $[(p \wedge q) \vee \neg r] \leftrightarrow p$.

2.12 Solutions to Exercise Set 2A

2A.1 – Which sentences are propositions?

(1) A true proposition. (2) Not a proposition — its truth depends on x. (3) Not a proposition (depends on x). (4) Not a proposition — it is a question. (5) A false proposition. (6) A true proposition.

2A.2 – Negations

(1) $2 + 3 \neq 5$. (2) 5 does not divide 27. (3) Computer Science is not a hard subject. (4) Bangalore does not have pleasant weather.

2A.3 – Conjunction and disjunction

(1) p is true, q is false. $p \wedge q$: "4 is a perfect square and 27 is a prime number" — value **0**. $p \vee q$: "4 is a perfect square or 27 is a prime number" — value **1**.
(2) p and q are both false. $p \wedge q$: "5 is divisible by 2 and 7 is a multiple of 5" — **0**. $p \vee q$: "5 is divisible by 2 or 7 is a multiple of 5" — **0**.

2A.4 – Truth values of conditionals

(1) T $\to$ T: **1**. (2) The antecedent $3 + 4 = 5$ is false: **1**. (3) Antecedent false: **1**. (4) $4 + 5 = 9$ is true but $3 + 1 = 6$ is false: T $\to$ F, **0**.

2A.5 – Triangle statements in words

(1) ABC is isosceles and is not equilateral. (2) ABC is not isosceles or ABC is equilateral. (3) If ABC is isosceles, then it is equilateral. (4) If ABC is equilateral, then it is isosceles. (5) If ABC is not equiangular, then it is not equilateral. (6) If ABC is isosceles then it is not equilateral, and if ABC is not equilateral then it is isosceles.

2A.6 – Recovering a truth value

(1) $p \wedge q$ false with p true forces $q =$ **0**. (2) $p \vee q$ false forces both false: $p =$ **0**. (3) $p \to q$ true with q false forces $p =$ **0** (otherwise T $\to$ F). (4) $p \leftrightarrow q$ true means equal values: $q =$ **0**.

2A.7 – Values with $p = 1, q = 0$

(1) $\neg p \wedge q = 0 \wedge 0 =$ **0**. (2) $p \to \neg q = 1 \to 1 = 1$, so its negation is **0**. (3) $(p \wedge q) \to (p \vee q) = 0 \to 1 =$ **1**. (4) $\neg(p \wedge q) = 1$, so the disjunction is **1**. (5) $p \to q = 0$; $p \leftrightarrow \neg q = 1 \leftrightarrow 1 = 1$, its negation 0; so $0 \vee 0 =$ **0**. (6) $p \to \neg q = 1 \to 1 = 1$, so the disjunction is **1**.

2A.8 – Nine truth tables

p	q	$p \vee \neg q$	$p \wedge (q \wedge p)$	$p \wedge (q \vee p)$
T	T	T	T	T
T	F	T	F	T
F	T	F	F	F
F	F	T	F	F

p	q	$(p \vee q) \wedge \neg p$	$\neg(p \vee \neg q)$	$q \leftrightarrow (\neg p \vee \neg q)$
T	T	F	F	F
T	F	F	F	F
F	T	T	T	T
F	F	F	F	F

p	q	r	$p \to (q \to r)$	$(p \to q) \to r$	$[(p \wedge q) \vee \neg r] \leftrightarrow p$
T	T	T	T	T	T
T	T	F	F	F	T
T	F	T	T	T	F
T	F	F	T	T	T
F	T	T	T	T	T
F	T	F	T	F	F
F	F	T	T	T	T
F	F	F	T	F	F

2.13 NAND and NOR in detail

For any p, q the De Morgan laws give $\neg(p \wedge q) \equiv \neg p \vee \neg q$ and $\neg(p \vee q) \equiv \neg p \wedge \neg q$. The proposition $\neg(p \wedge q)$, read "not both p and q", is written $p \uparrow q$; the symbol $\uparrow$ is the **NAND** connective (*Not* + *AND*). The proposition $\neg(p \vee q)$, read "neither p nor q", is written $p \downarrow q$; $\downarrow$ is the **NOR** connective (*Not* + *OR*). Thus

$$(p \uparrow q) \equiv \neg(p \wedge q) \equiv \neg p \vee \neg q, \qquad (p \downarrow q) \equiv \neg(p \vee q) \equiv \neg p \wedge \neg q,$$

and $p \uparrow q$, $p \downarrow q$ are duals of each other.

p	q	$p \wedge q$	$p \uparrow q$	$p \vee q$	$p \downarrow q$
T	T	T	F	T	F
T	F	F	T	T	F
F	T	F	T	T	F
F	F	F	T	F	T

Example 2.19 – Negating NAND and NOR

Prove that (i) $\neg(p \downarrow q) \equiv (\neg p \uparrow \neg q)$ (ii) $\neg(p \uparrow q) \equiv (\neg p \downarrow \neg q)$.

Solution. (i) $\neg(p \downarrow q) \equiv \neg\{\neg(p \vee q)\} \equiv \neg(\neg p \wedge \neg q) \equiv (\neg p) \uparrow (\neg q)$.
(ii) $\neg(p \uparrow q) \equiv \neg\{\neg(p \wedge q)\} \equiv \neg(\neg p \vee \neg q) \equiv (\neg p) \downarrow (\neg q)$.

Example 2.20 – NAND and NOR are not associative

Prove that (i) $p \uparrow (q \uparrow r) \equiv \neg p \vee (q \wedge r)$ (ii) $(p \uparrow q) \uparrow r \equiv (p \wedge q) \vee \neg r$ (iii) $p \downarrow (q \downarrow r) \equiv \neg p \wedge (q \vee r)$ (iv) $(p \downarrow q) \downarrow r \equiv (p \vee q) \wedge \neg r$.

Solution. (i) $p \uparrow (q \uparrow r) \equiv \neg[p \wedge (q \uparrow r)] \equiv \neg\{p \wedge [\neg(q \wedge r)]\} \equiv \neg p \vee (q \wedge r)$.
(ii) $(p \uparrow q) \uparrow r \equiv \neg[(p \uparrow q) \wedge r] \equiv \neg[\neg(p \wedge q) \wedge r] \equiv (p \wedge q) \vee \neg r$.
(iii) $p \downarrow (q \downarrow r) \equiv \neg[p \vee (q \downarrow r)] \equiv \neg[p \vee \neg(q \vee r)] \equiv \neg p \wedge (q \vee r)$.
(iv) $(p \downarrow q) \downarrow r \equiv \neg[(p \downarrow q) \vee r] \equiv \neg[\neg(p \vee q) \vee r] \equiv (p \vee q) \wedge \neg r$.
Remark. Since (i) $\not\equiv$ (ii) and (iii) $\not\equiv$ (iv), $p \uparrow (q \uparrow r) \not\equiv (p \uparrow q) \uparrow r$ and $p \downarrow (q \downarrow r) \not\equiv (p \downarrow q) \downarrow r$: the connectives $\uparrow$ and $\downarrow$ are **not associative**.

Example 2.21 – Only NAND, only NOR

Express in terms of only NAND and only NOR: (i) $\neg p$ (ii) $p \wedge q$ (iii) $p \vee q$ (iv) $p \to q$.

Solution. (i) Since $p \wedge p \equiv p$ and $p \vee p \equiv p$: $\neg p \equiv \neg(p \wedge p) \equiv p \uparrow p$ and $\neg p \equiv \neg(p \vee p) \equiv p \downarrow p$.
(ii) $p \wedge q \equiv \neg\neg(p \wedge q) \equiv \neg(p \uparrow q) \equiv (p \uparrow q) \uparrow (p \uparrow q)$. Also $p \wedge q \equiv \neg(\neg p) \wedge \neg(\neg q) \equiv (\neg p) \downarrow (\neg q) \equiv (p \downarrow p) \downarrow (q \downarrow q)$.
(iii) $p \vee q \equiv \neg\neg p \vee \neg\neg q \equiv (\neg p) \uparrow (\neg q) \equiv (p \uparrow p) \uparrow (q \uparrow q)$. Also $p \vee q \equiv \neg\neg(p \vee q) \equiv \neg(p \downarrow q) \equiv (p \downarrow q) \downarrow (p \downarrow q)$.
(iv) $p \to q \equiv \neg(p \wedge \neg q) \equiv p \uparrow (\neg q) \equiv p \uparrow (q \uparrow q)$. Also $p \to q \equiv \neg p \vee q \equiv (\neg p \downarrow q) \downarrow (\neg p \downarrow q) \equiv \{(p \downarrow p) \downarrow q\} \downarrow \{(p \downarrow p) \downarrow q\}$, using (iii).

2.14 Exercise Set 2B — NAND and NOR

2B.1 Prove that $(p \uparrow q) \equiv (q \uparrow p)$ and $(p \downarrow q) \equiv (q \downarrow p)$.

2B.2 Prove that $[p \to (\neg p \to q)] \equiv [p \uparrow (p \downarrow q)]$.

2B.3 Prove: (i) $(p \uparrow q) \equiv (\neg p \wedge q) \vee (p \wedge \neg q) \vee (\neg p \wedge \neg q)$ (ii) $(p \downarrow q) \equiv (\neg p \vee q) \wedge (p \vee \neg q) \wedge (\neg p \vee \neg q)$.

2B.4 Prove: (i) $(p \leftrightarrow q) \equiv (r \uparrow s) \uparrow (r \uparrow s)$, where $r = p \uparrow (q \uparrow q)$ and $s = q \uparrow (p \uparrow p)$; (ii) $(p \leftrightarrow q) \equiv (r \downarrow r) \downarrow (s \downarrow s)$, where $r = \{(p \downarrow p) \downarrow q\} \downarrow \{(p \downarrow p) \downarrow q\}$ and $s = \{(q \downarrow q) \downarrow p\} \downarrow \{(q \downarrow q) \downarrow p\}$.

2.15 Solutions to Exercise Set 2B

2B.1 – Commutativity

$p \uparrow q \equiv \neg(p \wedge q) \equiv \neg(q \wedge p) \equiv q \uparrow p$ and $p \downarrow q \equiv \neg(p \vee q) \equiv \neg(q \vee p) \equiv q \downarrow p$, by the commutative laws for $\wedge$ and $\vee$.

2B.2 – A NAND–NOR identity

Left side. $p \to (\neg p \to q) \equiv \neg p \vee (\neg\neg p \vee q) \equiv \neg p \vee p \vee q \equiv T_0$ (inverse and domination laws).
Right side. $p \uparrow (p \downarrow q) \equiv \neg[p \wedge \neg(p \vee q)] \equiv \neg[p \wedge \neg p \wedge \neg q] \equiv \neg F_0 \equiv T_0$.
Conclusion. Both sides are tautologies, hence equivalent. ■

2B.3 – NAND and NOR in normal form

(i) $(\neg p \wedge q) \vee (p \wedge \neg q) \vee (\neg p \wedge \neg q)$: the three bracketed terms between them are true in every row except $p = q =$ T, so the expression is false only there — exactly like $\neg(p \wedge q) = p \uparrow q$. By laws: $(\neg p \wedge q) \vee (\neg p \wedge \neg q) \equiv \neg p \wedge (q \vee \neg q) \equiv \neg p$, and $\neg p \vee (p \wedge \neg q) \equiv (\neg p \vee p) \wedge (\neg p \vee \neg q) \equiv \neg p \vee \neg q \equiv p \uparrow q$.
(ii) This is the dual of (i) (each $\wedge \leftrightarrow \vee$), and the dual of $p \uparrow q \equiv \neg p \vee \neg q$ is $\neg p \wedge \neg q \equiv p \downarrow q$. By the principle of duality, (ii) holds. Directly: $(\neg p \vee q) \wedge (\neg p \vee \neg q) \equiv \neg p \vee (q \wedge \neg q) \equiv \neg p$, and

$\neg p \wedge (p \vee \neg q) \equiv \neg p \wedge \neg q \equiv p \downarrow q$. ■

2B.4 – The biconditional with one connective

(i) $r = p \uparrow (q \uparrow q) = p \uparrow \neg q \equiv \neg(p \wedge \neg q) \equiv p \to q$; similarly $s = q \uparrow (p \uparrow p) \equiv q \to p$. Then $(r \uparrow s) \uparrow (r \uparrow s) \equiv r \wedge s$ (Example "Only NAND, only NOR", (ii)) $\equiv (p \to q) \wedge (q \to p) \equiv p \leftrightarrow q$.
(ii) $(p \downarrow p) \downarrow q \equiv \neg(\neg p \vee q)$, so $r \equiv \neg\neg(\neg p \vee q) \equiv p \to q$; similarly $s \equiv q \to p$. Then $(r \downarrow r) \downarrow (s \downarrow s) \equiv \neg r \downarrow \neg s \equiv \neg(\neg r \vee \neg s) \equiv r \wedge s \equiv p \leftrightarrow q$. ■

3 Validity and Satisfiability

Every formula is exactly one of three types: a tautology, a contradiction or a contingency. Once we know the type, we know whether the formula is valid and whether it is satisfiable.

Definition: Tautology, contradiction, contingency

- **Tautology (valid formula):** true in every row of its truth table, e.g. $p \vee \neg p$.
- **Contradiction (unsatisfiable formula):** false in every row, e.g. $p \wedge \neg p$.
- **Contingency:** true in at least one row and false in at least one row, e.g. $p \to q$.
- **Satisfiable:** true in at least one row (i.e. not a contradiction).
- **Falsifiable (invalid):** false in at least one row (i.e. not a tautology).

Property	Tautology	Contradiction	Contingency
Final column	All T	All F	Mixed
Valid?	Yes	No	No
Satisfiable?	Yes	No	Yes
Falsifiable?	No	Yes	Yes
Negation is a	Contradiction	Tautology	Contingency
Example	$p \to (q \to p)$	$(p \vee q) \wedge \neg p \wedge \neg q$	$p \wedge q$

Key relationships

1. Every tautology is satisfiable; not every satisfiable formula is a tautology.
2. Every contingency is satisfiable; not every satisfiable formula is a contingency.
3. A is a tautology $\iff$ $\neg A$ is a contradiction.
4. A is satisfiable $\iff$ $\neg A$ is not a tautology.

3.1 Short-cut test (assume false)

To test whether $A \to B$ is a tautology, try to make it **false**: force $A =$ T and $B =$ F. If this leads to a clash, the formula is a tautology; if it succeeds, you have a falsifying row.

Example 3.1 – Classify three formulas

Classify: (a) $[(p \to q) \wedge (q \to r)] \to (p \to r)$ (b) $(p \vee q) \wedge \neg p \wedge \neg q$ (c) $(p \to q) \to (q \to p)$.

Solution. (a) Assume it is F: then $p \to r =$ F, so $p =$ T, $r =$ F; and $p \to q =$ T forces $q =$ T; but then $q \to r =$ T $\to$ F = F, so $(p \to q) \wedge (q \to r)$ is F, not T. The assumption fails, so (a) is a **tautology**.

(b) $\neg p \wedge \neg q \equiv \neg(p \vee q)$, so the formula is $(p \vee q) \wedge \neg(p \vee q) \equiv F_0$: a **contradiction**.
(c) Truth table:

p	q	$p \to q$	$q \to p$	$(p \to q) \to (q \to p)$
T	T	T	T	**T**
T	F	F	T	**T**
F	T	T	F	**F**
F	F	T	T	**T**

Mixed final column: a **contingency** (satisfiable but not valid).

Example 3.2 – Satisfiable but not valid

Is $\{p \to (q \vee r)\} \to \{(p \wedge q) \to r\}$ valid?

Solution. Make the consequent F: $p = q =$ T, $r =$ F. The antecedent is T $\to$ (T $\vee$ F) = T, so the whole formula is T $\to$ F = F. It is **not valid**. With $r =$ T it is true, so it is **satisfiable** — a contingency.

3.2 More examples on tautologies and contradictions

Recall: a compound proposition that is *true* for all possible truth values of its components is a **tautology** (also called a *logical truth*); one that is *false* for all of them is a **contradiction** (also called an *absurdity*); and one that can be either true or false is a **contingency**.

Example 3.3 – The simplest tautology and contradiction

Prove that, for any proposition p, $p \vee \neg p$ is a tautology and $p \wedge \neg p$ is a contradiction.

p	$\neg p$	$p \vee \neg p$	$p \wedge \neg p$
T	F	T	F
F	T	T	F

Solution. $p \vee \neg p$ is always true, so it is a tautology; $p \wedge \neg p$ is always false, so it is a contradiction.

Example 3.4 – Using exclusive or

Show that for any p, q: (i) $(p \veebar q) \vee (p \leftrightarrow q)$ is a tautology; (ii) $(p \veebar q) \wedge (p \leftrightarrow q)$ is a contradiction; (iii) $(p \veebar q) \wedge (p \to q)$ is a contingency.

p	q	$p \veebar q$	$p \leftrightarrow q$	(i)	(ii)	$p \to q$	(iii)
T	T	F	T	T	F	T	F
T	F	T	F	T	F	F	F
F	T	T	F	T	F	T	T
F	F	F	T	T	F	T	F

Solution. $p \veebar q$ and $p \leftrightarrow q$ always have opposite truth values. So their disjunction (i) is always T — a tautology — and their conjunction (ii) is always F — a contradiction. Column (iii) takes both values (T when p is false and q is true), so it is neither: a contingency.

Example 3.5 – One of each

Show that $p \to (p \vee q)$ is a tautology and $p \wedge (\neg p \wedge q)$ is a contradiction.

p	q	$p \vee q$	$p \to (p \vee q)$	$\neg p \wedge q$	$p \wedge (\neg p \wedge q)$
T	T	T	T	F	F
T	F	T	T	F	F
F	T	T	T	T	F
F	F	F	T	F	F

Solution. The fourth column is all T and the last column is all F.

Example 3.6 – Values independent of the components

Show that the truth values of (i) $\{p \wedge (p \to q)\} \to q$ (ii) $(p \to q) \leftrightarrow (\neg p \vee q)$ are independent of the truth values of their components.

Solution. (i)

p	q	$p \to q$	$r = p \wedge (p \to q)$	$r \to q$
T	T	T	T	**T**
T	F	F	F	**T**
F	T	T	F	**T**
F	F	T	F	**T**

(ii)

p	q	$u = p \to q$	$v = \neg p \vee q$	$u \leftrightarrow v$
T	T	T	T	**T**
T	F	F	F	**T**
F	T	T	T	**T**
F	F	T	T	**T**

The last column of each table is all T: whatever the truth values of p and q, the value is 1. Both propositions are **tautologies**.

Example 3.7 – Modus tollens as a tautology

Prove that $[\neg q \wedge (p \to q)] \to \neg p$ is a tautology.

p	q	$p \to q$	$r = \neg q \wedge (p \to q)$	$r \to \neg p$
T	T	T	F	**T**
T	F	F	F	**T**
F	T	T	F	**T**
F	F	T	T	**T**

Solution. The proposition $r \to \neg p$, where $r = \neg q \wedge (p \to q)$, is always true: a tautology.

Example 3.8 – The syllogism tautology

Prove that $[(p \to q) \wedge (q \to r)] \to (p \to r)$ is a tautology.

p	q	r	$p \to q$	$q \to r$	$(p \to q) \wedge (q \to r)$	$p \to r$	Given
T	T	T	T	T	T	T	**T**
T	T	F	T	F	F	F	**T**
T	F	T	F	T	F	T	**T**
T	F	F	F	T	F	F	**T**
F	T	T	T	T	T	T	**T**
F	T	F	T	F	F	T	**T**
F	F	T	T	T	T	T	**T**
F	F	F	T	T	T	T	**T**

Example 3.9 – Distributing a conditional

Prove that $\{p \to (q \to r)\} \to \{(p \to q) \to (p \to r)\}$ is a tautology.

p	q	r	$p \to (q \to r)$	$p \to q$	$p \to r$	$(p \to q) \to (p \to r)$	Given
T	T	T	T	T	T	T	**T**
T	T	F	F	T	F	F	**T**
T	F	T	T	F	T	T	**T**
T	F	F	T	F	F	T	**T**
F	T	T	T	T	T	T	**T**
F	T	F	T	T	T	T	**T**
F	F	T	T	T	T	T	**T**
F	F	F	T	T	T	T	**T**

Example 3.10 – Proof by cases as a tautology

Prove that $[(p \vee q) \wedge \{(p \to r) \wedge (q \to r)\}] \to r$ is a tautology.

p	q	r	$p \vee q$	$(p \to r) \wedge (q \to r)$	$(p \vee q) \wedge \{\cdots\}$	Given
T	T	T	T	T	T	**T**
T	T	F	T	F	F	**T**
T	F	T	T	T	T	**T**
T	F	F	T	F	F	**T**
F	T	T	T	T	T	**T**
F	T	F	T	F	F	**T**
F	F	T	F	T	F	**T**
F	F	F	F	T	F	**T**

3.3 Exercise Set 3A — tautologies and contradictions

3A.1 Determine whether the following are tautologies: (1) $p \vee [\neg(p \wedge q)]$ (2) $(p \vee q) \vee \neg p$ (3) $p \to (p \wedge q)$ (4) $p \to (p \vee q)$ (5) $\{\neg(p \to q)\} \to \neg q$.

3A.2 Prove that the following are tautologies:
(1) $(p \wedge q) \to p$ (2) $\neg(p \vee \neg q) \to \neg p$ (3) $\neg p \to (p \to q)$ (4) $[p \wedge (p \to q)] \to q$
(5) $[(p \to q) \vee (p \to r)] \leftrightarrow [p \to (q \vee r)]$ (6) $[(p \to q) \wedge (p \to \neg q)] \leftrightarrow \neg p$
(7) $[(p \wedge \neg q) \to r] \to [p \to (q \vee r)]$ (8) $\neg(p \vee q) \vee [(\neg p) \wedge q] \vee p$
(9) $[(p \to r) \wedge (q \to r)] \to [(p \vee q) \to r]$ (10) $\{[(p \vee q) \to r] \wedge (\neg p)\} \to (q \to r)$.

3A.3 Find the possible truth values of p, q, r, s, t for which the following are contradictions: (1) $[(p \wedge q) \wedge r] \to (s \vee t)$ (2) $[p \wedge (q \wedge r)] \to (s \veebar t)$.

3.4 Solutions to Exercise Set 3A

3A.1 – Which are tautologies?

p	q	(1)	(2)	(3)	(4)	(5)
T	T	T	T	T	T	T
T	F	T	T	F	T	T
F	T	T	T	T	T	T
F	F	T	T	T	T	T

Conclusion. (1), (2), (4), (5) are **tautologies**; (3) is **not** (it is false when p is true and q is false).

3A.2 – Ten tautologies

Each is shown by the "assume false" short-cut or by the laws of logic.
(1) To be false, $p \wedge q$ must be T and p F — impossible.
(2) To be false, $\neg(p \vee \neg q)$ must be T (so $p =$ F, $q =$ T) and $\neg p$ F (so $p =$ T) — impossible.
(3) To be false, $\neg p$ is T ($p =$ F) and $p \to q$ is F (needs $p =$ T) — impossible.
(4) This is modus ponens; see the example "Values independent of the components" (i).
(5) $(p \to q) \vee (p \to r) \equiv \neg p \vee q \vee \neg p \vee r \equiv \neg p \vee (q \vee r) \equiv p \to (q \vee r)$; a biconditional between equivalent formulas is a tautology.
(6) $(p \to q) \wedge (p \to \neg q) \equiv \neg p \vee (q \wedge \neg q) \equiv \neg p \vee F_0 \equiv \neg p$.
(7) $(p \wedge \neg q) \to r \equiv \neg p \vee q \vee r \equiv p \to (q \vee r)$; so the conditional is $A \to A$, a tautology.
(8) $\neg(p \vee q) \vee (\neg p \wedge q) \equiv (\neg p \wedge \neg q) \vee (\neg p \wedge q) \equiv \neg p \wedge (\neg q \vee q) \equiv \neg p$; then $\neg p \vee p \equiv T_0$.
(9) $(p \to r) \wedge (q \to r) \equiv (\neg p \vee r) \wedge (\neg q \vee r) \equiv (\neg p \wedge \neg q) \vee r \equiv (p \vee q) \to r$; again $A \to A$.
(10) To be false, $q \to r$ is F ($q =$ T, $r =$ F) and both $(p \vee q) \to r$ and $\neg p$ are T. But $q =$ T makes $p \vee q =$ T, so $(p \vee q) \to r =$ T $\to$ F $=$ F — a clash. Hence it is a tautology. ■

3A.3 – When is a conditional a contradiction?

The question asks for the truth values that make each proposition *false*. A conditional is false only when its antecedent is true and its consequent is false.
(1) $(p \wedge q) \wedge r$ true needs p, q, r all true; $s \vee t$ false needs s, t both false. **Answer:** p, q, r true; s and t false.
(2) $p \wedge (q \wedge r)$ true needs p, q, r true; $s \veebar t$ false needs s, t equal. **Answer:** p, q, r true; s and t both true or both false.

4 Logical Equivalence: The Laws of Logic

Definition: Logical equivalence

A and B are **logically equivalent**, $A \equiv B$, when they have identical truth values in every row; equivalently $A \leftrightarrow B$ is a tautology.

Note: $\leftrightarrow$ is a connective used to build a formula; $\equiv$ is not a connective — it says that two formulas always have the same truth value.

4.1 The laws of logic

No.	Law	Form (i)	Dual form (ii)
1	Double negation	$\neg\neg p \equiv p$	—
2	Idempotent	$p \vee p \equiv p$	$p \wedge p \equiv p$
3	Identity	$p \vee F_0 \equiv p$	$p \wedge T_0 \equiv p$
4	Inverse (complement)	$p \vee \neg p \equiv T_0$	$p \wedge \neg p \equiv F_0$
5	Domination	$p \vee T_0 \equiv T_0$	$p \wedge F_0 \equiv F_0$
6	Commutative	$p \vee q \equiv q \vee p$	$p \wedge q \equiv q \wedge p$
7	Associative	$p \vee (q \vee r) \equiv (p \vee q) \vee r$	$p \wedge (q \wedge r) \equiv (p \wedge q) \wedge r$
8	Distributive	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
9	De Morgan	$\neg(p \vee q) \equiv \neg p \wedge \neg q$	$\neg(p \wedge q) \equiv \neg p \vee \neg q$
10	Absorption	$p \vee (p \wedge q) \equiv p$	$p \wedge (p \vee q) \equiv p$

4.2 Equivalences involving $\rightarrow$ and $\leftrightarrow$

No.	Equivalence	Name
11	$p \rightarrow q \equiv \neg p \vee q$	Conditional law
12	$p \rightarrow q \equiv \neg q \rightarrow \neg p$	Contrapositive
13	$\neg(p \rightarrow q) \equiv p \wedge \neg q$	Negation of conditional
14	$(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$	—
15	$(p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$	Proof by cases
16	$(p \rightarrow q) \vee (p \rightarrow r) \equiv p \rightarrow (q \vee r)$	—
17	$(p \rightarrow r) \vee (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$	—
18	$p \rightarrow (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$	Exportation
19	$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$	Biconditional
20	$\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q \equiv p \oplus q$	Negated biconditional

Common pitfall

Rows 15 and 17 are routinely confused. $(p \wedge q) \rightarrow r$ is **not** equivalent to $(p \rightarrow r) \wedge (q \rightarrow r)$; it equals $(p \rightarrow r) \vee (q \rightarrow r)$. Test: $p = \text{T}, q = \text{F}, r = \text{F}$.

4.3 Substitution rules

1. **Transitive rule:** if $u \equiv v$ and $v \equiv w$, then $u \equiv w$.
2. **Substitution rule I:** replacing every occurrence of a variable in a tautology by any formula gives a tautology.
3. **Substitution rule II:** replacing a sub-formula by an equivalent one gives an equivalent formula.

4.4 Duality

Definition: Dual

For a formula s containing only $\neg, \wedge, \vee, T_0, F_0$, its **dual** s^d is obtained by interchanging $\wedge \leftrightarrow \vee$ and $T_0 \leftrightarrow F_0$. Variables and $\neg$ are unchanged. Eliminate $\rightarrow$ and $\leftrightarrow$ *before* taking the dual.

Principle of duality

If $u \equiv v$, then $u^d \equiv v^d$. Also $(s^d)^d \equiv s$. This is why every law in the table of Section 4.1 comes in a dual pair.

Example: dual of $(p \to q) \to r$. First $(p \to q) \to r \equiv \neg(\neg p \vee q) \vee r \equiv (p \wedge \neg q) \vee r$; its dual is $(p \vee \neg q) \wedge r$.

4.5 Method for proving an equivalence without truth tables

1. Start from the more complicated side.
2. Eliminate $\to$ and $\leftrightarrow$ (laws 11, 19).
3. Push $\neg$ inward (De Morgan, double negation).
4. Collapse with distributive, inverse, identity, domination and absorption laws.
5. **Name the law at every step** — examiners award step marks for it.

Example 4.1 – Basic simplification

Prove $(p \vee q) \wedge \neg(\neg p \wedge q) \equiv p$.

Solution.

$$\begin{aligned}
(p \vee q) \wedge \neg(\neg p \wedge q) &\equiv (p \vee q) \wedge (\neg\neg p \vee \neg q) && \textit{De Morgan}\\
&\equiv (p \vee q) \wedge (p \vee \neg q) && \textit{Double negation}\\
&\equiv p \vee (q \wedge \neg q) && \textit{Distributive}\\
&\equiv p \vee F_0 \equiv p && \textit{Inverse, Identity}
\end{aligned}$$

Example 4.2 – Equivalence and its dual

Prove $\neg(p \vee q) \vee [(\neg p \wedge q) \vee \neg q] \equiv \neg(q \wedge p)$ and verify the principle of duality for it.

Solution.

$$\begin{aligned}
(\neg p \wedge q) \vee \neg q &\equiv (\neg p \vee \neg q) \wedge (q \vee \neg q) && \textit{Distributive}\\
&\equiv (\neg p \vee \neg q) \wedge T_0 \equiv \neg p \vee \neg q && \textit{Inverse, Identity}\\
\text{LHS} &\equiv (\neg p \wedge \neg q) \vee (\neg p \vee \neg q) && \textit{De Morgan}\\
&\equiv \neg p \vee \neg q && \textit{Absorption (since } \neg p \wedge \neg q \textit{ is absorbed)}\\
&\equiv \neg(p \wedge q) \equiv \neg(q \wedge p) && \textit{De Morgan, Commutative}
\end{aligned}$$

Dual: $\neg(p \wedge q) \wedge [(\neg p \vee q) \wedge \neg q] \equiv \neg(q \vee p)$. Check:

$$\begin{aligned}
(\neg p \vee q) \wedge \neg q &\equiv (\neg p \wedge \neg q) \vee (q \wedge \neg q) \equiv \neg p \wedge \neg q && \textit{Distributive, Inverse, Identity}\\
\text{LHS}^d &\equiv (\neg p \vee \neg q) \wedge (\neg p \wedge \neg q) \equiv \neg p \wedge \neg q && \textit{De Morgan, Absorption}\\
&\equiv \neg(p \vee q) \equiv \neg(q \vee p) && \textit{De Morgan, Commutative}
\end{aligned}$$

Both hold, verifying the principle of duality.

Example 4.3 – Using absorption early

Prove $(p \to q) \wedge [\neg q \wedge (r \vee \neg q)] \equiv \neg(q \vee p)$.

Solution.

$$\begin{aligned}
\neg q \wedge (r \vee \neg q) &\equiv \neg q && \textit{Commutative, Absorption}\\
(p \to q) \wedge \neg q &\equiv (\neg p \vee q) \wedge \neg q && \textit{Conditional law}\\
&\equiv (\neg p \wedge \neg q) \vee (q \wedge \neg q) && \textit{Distributive}\\
&\equiv (\neg p \wedge \neg q) \vee F_0 \equiv \neg p \wedge \neg q && \textit{Inverse, Identity}\\
&\equiv \neg(p \vee q) \equiv \neg(q \vee p) && \textit{De Morgan, Commutative}
\end{aligned}$$

Example 4.4 – An identity with $\to$

Prove $(p \to q) \wedge (p \to r) \equiv p \to (q \wedge r)$ without truth tables.

Solution.

$$\begin{aligned}
(p \to q) \wedge (p \to r) &\equiv (\neg p \vee q) \wedge (\neg p \vee r) && \textit{Conditional law}\\
&\equiv \neg p \vee (q \wedge r) && \textit{Distributive}\\
&\equiv p \to (q \wedge r) && \textit{Conditional law}
\end{aligned}$$

Example 4.5 – Splitting into halves

Without truth tables, prove that $(p \to q) \wedge (\neg p \to q) \equiv q$.

Solution.

$$\begin{aligned}
(\neg p \vee q) \wedge (\neg\neg p \vee q) &\equiv (\neg p \vee q) \wedge (p \vee q) && \textit{Conditional law, Double negation}\\
&\equiv (\neg p \wedge p) \vee q && \textit{Distributive}\\
&\equiv F_0 \vee q \equiv q && \textit{Inverse, Identity}
\end{aligned}$$

This is the idea of proof by cases: if q follows both when p is true and when p is false, then q is true.

4.6 More examples on logical equivalence

Two propositions u and v are **logically equivalent** whenever they have the same truth value for every combination of truth values of their components — equivalently, whenever $u \leftrightarrow v$ is a tautology. We write $u \Leftrightarrow v$ (or $u \equiv v$), and $u \not\Leftrightarrow v$ when they are not equivalent. Logically equivalent propositions can be used in place of each other.

Example 4.6 – Equivalent simple propositions

Let x be a specified positive integer, p: x is an odd integer, q: x is not divisible by 2. Are p and q logically equivalent?

Solution. An integer is odd exactly when it is not divisible by 2, so p and q always have the same truth value: $p \Leftrightarrow q$.

Example 4.7 – The conditional law by truth table

Prove that $(p \to q) \Leftrightarrow (\neg p) \vee q$.

p	q	$p \to q$	$\neg p$	$\neg p \vee q$
T	T	T	F	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

Solution. The columns of $p \to q$ and $\neg p \vee q$ agree in every row.

Example 4.8 – A tautology from an equivalence

Prove that $\{(p \to q) \to r\} \leftrightarrow \{(\neg p \vee q) \to r\}$ is a tautology.

Solution. Since $(p \to q) \Leftrightarrow (\neg p \vee q)$, replacing $p \to q$ by $\neg p \vee q$ in $(p \to q) \to r$ gives an equivalent proposition (substitution rule). So $\{(p \to q) \to r\} \Leftrightarrow \{(\neg p \vee q) \to r\}$, and the biconditional between them is a tautology.

Example 4.9 – Recognising a known tautology

Prove that $[(r \to s) \wedge \{(r \to s) \to (t \to u)\}] \to (\neg t \vee u)$ is a tautology.

Solution. Let p denote $r \to s$ and q denote $t \to u$. Since $(t \to u) \Leftrightarrow (\neg t \vee u)$, we have $q \Leftrightarrow \neg t \vee u$. So the given proposition is equivalent to $\{p \wedge (p \to q)\} \to q$, which is a tautology (Section 3). Therefore the given proposition is a tautology.

Example 4.10 – Exclusive or without $\vee$

Prove that $p \veebar q \Leftrightarrow (p \vee q) \wedge (\neg p \vee \neg q)$.

p	q	$p \vee q$	$p \veebar q$	$\neg p \vee \neg q$	$(p \vee q) \wedge (\neg p \vee \neg q)$
T	T	T	F	F	F
T	F	T	T	T	T
F	T	T	T	T	T
F	F	F	F	T	F

Solution. The columns of $p \veebar q$ and $(p \vee q) \wedge (\neg p \vee \neg q)$ agree in every row.

Example 4.11 – Distributing a conditional over $\wedge$

Prove that $[p \to (q \wedge r)] \Leftrightarrow [(p \to q) \wedge (p \to r)]$.

p	q	r	$q \wedge r$	$p \to (q \wedge r)$	$p \to q$	$p \to r$	$(p \to q) \wedge (p \to r)$
T	T	T	T	T	T	T	T
T	T	F	F	F	T	F	F
T	F	T	F	F	F	T	F
T	F	F	F	F	F	F	F
F	T	T	T	T	T	T	T
F	T	F	F	T	T	T	T
F	F	T	F	T	T	T	T
F	F	F	F	T	T	T	T

Solution. The second and last columns agree in all eight rows.

Example 4.12 – Proof by cases by truth table

Prove that $[(p \vee q) \to r] \Leftrightarrow [(p \to r) \wedge (q \to r)]$.

p	q	r	$p \vee q$	$(p \vee q) \to r$	$p \to r$	$q \to r$	$(p \to r) \wedge (q \to r)$
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	F
T	F	T	T	T	T	T	T
T	F	F	T	F	F	T	F
F	T	T	T	T	T	T	T
F	T	F	T	F	T	F	F
F	F	T	F	T	T	T	T
F	F	F	F	T	T	T	T

Example 4.13 – Contrapositive of a compound conditional

Examine whether $[(p \vee q) \to r] \leftrightarrow [\neg r \to \neg(p \vee q)]$ is a tautology.

p	q	r	$(p \vee q) \to r$	$\neg(p \vee q)$	$\neg r \to \neg(p \vee q)$	Given
T	T	T	T	F	T	**T**
T	T	F	F	F	F	**T**
T	F	T	T	F	T	**T**
T	F	F	F	F	F	**T**
F	T	T	T	F	T	**T**
F	T	F	F	F	F	**T**
F	F	T	T	T	T	**T**
F	F	F	T	T	T	**T**

Solution. $(p \vee q) \to r$ and $\neg r \to \neg(p \vee q)$ have the same truth values in all rows (a conditional and its contrapositive). Hence the biconditional is a **tautology**.

Example 4.14 – Not equivalent

Show that $p \wedge (\neg q \vee r)$ and $p \vee (q \wedge \neg r)$ are not logically equivalent.

p	q	r	$\neg q \vee r$	$q \wedge \neg r$	$p \wedge (\neg q \vee r)$	$p \vee (q \wedge \neg r)$
T	T	T	T	F	T	T
T	T	F	F	T	F	T
T	F	T	T	F	T	T
T	F	F	T	F	T	T
F	T	T	T	F	F	F
F	T	F	F	T	F	T
F	F	T	T	F	F	F
F	F	F	T	F	F	F

Solution. The last two columns differ (for example when p is true, q is true and r is false, the first is F and the second is T). So the propositions are **not** logically equivalent.

4.7 Exercise Set 4A — logical equivalence

4A.1 For any two propositions p, q prove:
(1) $(p \to \neg q) \Leftrightarrow (q \to \neg p)$ (2) $\neg(p \to \neg q) \Leftrightarrow (p \wedge q)$ (3) $(p \vee q) \Leftrightarrow \neg(\neg p \wedge \neg q)$
(4) $(p \vee q) \wedge \neg(\neg p \wedge q) \Leftrightarrow p$ (5) $(p \leftrightarrow \neg q) \Leftrightarrow (\neg p \leftrightarrow q)$ (6) $[p \to (q \to p)] \Leftrightarrow [\neg p \to (p \to q)]$
(7) $\neg(p \leftrightarrow q) \Leftrightarrow (p \vee q) \wedge [\neg(p \wedge q)]$.

4A.2 For any propositions p, q, r prove:
(1) $[(p \to q) \wedge (p \to r)] \Leftrightarrow p \to (q \wedge r)$ (2) $[(p \to q) \wedge (r \to q)] \Leftrightarrow [(p \vee r) \to q]$
(3) $[(p \to q) \wedge (p \to \neg q)] \Leftrightarrow \neg p$ (4) $[p \to (q \to r)] \Leftrightarrow [(p \wedge \neg r) \to \neg q]$
(5) $[p \wedge (\neg r \vee q \vee \neg q)] \vee [(r \vee t \vee \neg r) \wedge \neg q] \Leftrightarrow p \vee \neg q$.

4A.3 Prove the following logical equivalences:
(1) $[(p \leftrightarrow q) \wedge (q \leftrightarrow r) \wedge (r \leftrightarrow p)] \Leftrightarrow [(p \to q) \wedge (q \to r) \wedge (r \to p)]$
(2) $p \to (q \to r) \Leftrightarrow p \to (\neg q \vee r) \Leftrightarrow (p \wedge q) \to r \Leftrightarrow (p \to r) \vee (q \to r)$
(3) $(p \veebar q) \Leftrightarrow [(p \wedge \neg q) \vee (\neg p \wedge q)] \Leftrightarrow \neg(p \leftrightarrow q)$
(4) $[p \to (q \vee r)] \Leftrightarrow [(p \to q) \vee (p \to r)] \Leftrightarrow [\neg r \to (p \to q)] \Leftrightarrow [(p \wedge \neg q) \to r]$.

4A.4 Prove that each of the following is a tautology:
(1) $[p \vee (q \wedge r)] \vee \neg[p \vee (q \wedge r)]$ (2) $[p \wedge (p \to q)] \to q$ (3) $p \to [q \to (p \wedge q)]$
(4) $[(p \vee q) \wedge \neg(\neg p \wedge (\neg q \vee \neg r))] \vee (\neg p \wedge \neg q) \vee (\neg p \wedge \neg r)$.

4.8 Solutions to Exercise Set 4A

4A.1 – Seven two-variable equivalences

(1) $p \to \neg q \equiv \neg p \vee \neg q \equiv \neg q \vee \neg p \equiv q \to \neg p$ (conditional, commutative).
(2) $\neg(p \to \neg q) \equiv p \wedge \neg\neg q \equiv p \wedge q$ (negation of a conditional, double negation).
(3) $\neg(\neg p \wedge \neg q) \equiv \neg\neg p \vee \neg\neg q \equiv p \vee q$ (De Morgan, double negation).
(4) $(p \vee q) \wedge \neg(\neg p \wedge q) \equiv (p \vee q) \wedge (p \vee \neg q) \equiv p \vee (q \wedge \neg q) \equiv p \vee F_0 \equiv p$ (De Morgan, distributive, inverse, identity).
(5) $p \leftrightarrow \neg q$ is true exactly when p and $\neg q$ agree, i.e. when p and q differ; $\neg p \leftrightarrow q$ is true exactly when $\neg p$ and q agree, i.e. again when p and q differ. So both are equivalent to $\neg(p \leftrightarrow q)$.
(6) $p \to (q \to p) \equiv \neg p \vee \neg q \vee p \equiv T_0$ and $\neg p \to (p \to q) \equiv p \vee \neg p \vee q \equiv T_0$; both are tautologies, hence equivalent.
(7) $\neg(p \leftrightarrow q) \equiv p \veebar q \equiv (p \vee q) \wedge (\neg p \vee \neg q)$ (example "Exclusive or without $\veebar$") $\equiv (p \vee q) \wedge \neg(p \wedge q)$ (De Morgan). ■

4A.2 – Three-variable equivalences

(1) $(\neg p \vee q) \wedge (\neg p \vee r) \equiv \neg p \vee (q \wedge r) \equiv p \to (q \wedge r)$ (distributive).
(2) $(\neg p \vee q) \wedge (\neg r \vee q) \equiv (\neg p \wedge \neg r) \vee q \equiv \neg(p \vee r) \vee q \equiv (p \vee r) \to q$.
(3) $(\neg p \vee q) \wedge (\neg p \vee \neg q) \equiv \neg p \vee (q \wedge \neg q) \equiv \neg p \vee F_0 \equiv \neg p$.
(4) The left side is $\neg p \vee \neg q \vee r$; the right side is $\neg(p \wedge \neg r) \vee \neg q \equiv \neg p \vee r \vee \neg q$. They are equal.
(5) $\neg r \vee q \vee \neg q \equiv T_0$, so $p \wedge T_0 \equiv p$; $r \vee t \vee \neg r \equiv T_0$, so $T_0 \wedge \neg q \equiv \neg q$. The whole is $p \vee \neg q$. ■

4A.3 – Chains of equivalences

(1) The left side is the conjunction of all six conditionals between p, q, r. On the right, $(q \to r) \wedge (r \to p) \Rightarrow q \to p$, $(r \to p) \wedge (p \to q) \Rightarrow r \to q$ and $(p \to q) \wedge (q \to r) \Rightarrow p \to r$ (syllogism), so the right side implies the other three conditionals and hence the left side; the converse is clear. (A truth-table proof is Part C1(a) of the exercises at the end.)
(2) $p \to (q \to r) \equiv p \to (\neg q \vee r)$ (conditional law inside) $\equiv \neg p \vee \neg q \vee r \equiv \neg(p \wedge q) \vee r \equiv (p \wedge q) \to r$. Also $\neg p \vee \neg q \vee r \equiv (\neg p \vee r) \vee (\neg q \vee r) \equiv (p \to r) \vee (q \to r)$ (idempotent $r \vee r \equiv r$, commutative,

associative).
(3) $p \veebar q$ is true exactly when one of p, q is true and the other false, i.e. when $(p \wedge \neg q) \vee (\neg p \wedge q)$ is true. Also $\neg(p \leftrightarrow q) \equiv \neg[(p \wedge q) \vee (\neg p \wedge \neg q)] \equiv (\neg p \vee \neg q) \wedge (p \vee q) \equiv (\neg p \wedge q) \vee (\neg q \wedge p)$ (De Morgan, then distributive, dropping the contradictions $\neg p \wedge p$ and $\neg q \wedge q$).
(4) $p \to (q \vee r) \equiv \neg p \vee q \vee r \equiv (\neg p \vee q) \vee (\neg p \vee r) \equiv (p \to q) \vee (p \to r)$. Also $\neg p \vee q \vee r \equiv r \vee (p \to q) \equiv \neg r \to (p \to q)$, and $\neg p \vee q \vee r \equiv \neg(p \wedge \neg q) \vee r \equiv (p \wedge \neg q) \to r$. ■

4A.4 – Four tautologies

(1) It has the form $u \vee \neg u$ with $u = p \vee (q \wedge r)$: a tautology by the inverse law.
(2) $p \wedge (p \to q) \equiv p \wedge (\neg p \vee q) \equiv (p \wedge \neg p) \vee (p \wedge q) \equiv p \wedge q$; and $(p \wedge q) \to q \equiv \neg p \vee \neg q \vee q \equiv T_0$.
(3) $p \to [q \to (p \wedge q)] \equiv \neg p \vee \neg q \vee (p \wedge q) \equiv \neg(p \wedge q) \vee (p \wedge q) \equiv T_0$.
(4) This is the example "Proving a tautology by splitting" below (and Problem 2 of the solved university problems): the first part simplifies to $u = p \vee (q \wedge r)$ and the rest to $\neg u$. ■

4.9 The laws of logic: proofs and consequences

In the laws of logic, T_0 denotes a tautology and F_0 a contradiction. The laws of double negation, idempotence and commutativity are easy to see; the others can be verified with truth tables. For example:

- **Proof of De Morgan's law** $\neg(p \vee q) \Leftrightarrow \neg p \wedge \neg q$**:**

p	q	$\neg(p \vee q)$	$\neg p$	$\neg q$	$\neg p \wedge \neg q$
T	T	F	F	F	F
T	F	F	F	T	F
F	T	F	T	F	F
F	F	T	T	T	T

- **Proof of the distributive law** $p \vee (q \wedge r) \Leftrightarrow (p \vee q) \wedge (p \vee r)$**:**

p	q	r	$q \wedge r$	$p \vee (q \wedge r)$	$p \vee q$	$p \vee r$	$(p \vee q) \wedge (p \vee r)$
T	T	T	T	T	T	T	T
T	T	F	F	T	T	T	T
T	F	T	F	T	T	T	T
T	F	F	F	T	T	T	T
F	T	T	T	T	T	T	T
F	T	F	F	F	T	F	F
F	F	T	F	F	F	T	F
F	F	F	F	F	F	F	F

Law for the negation of a conditional

For any p, q: $\neg(p \to q) \Leftrightarrow p \wedge \neg q$.

p	q	$p \to q$	$\neg(p \to q)$	$\neg q$	$p \wedge \neg q$
T	T	T	F	F	F
T	F	F	T	T	T
F	T	T	F	F	F
F	F	T	F	T	F

So, for any p and q: (1) $\neg(p \vee q) \equiv \neg p \wedge \neg q$; (2) $\neg(p \wedge q) \equiv \neg p \vee \neg q$; (3) $\neg(p \to q) \equiv p \wedge \neg q$; (4) $(p \to q) \equiv \neg\neg(p \to q) \equiv \neg(p \wedge \neg q) \equiv \neg p \vee q$.

Proposition	Negation
$\neg p$	p
$p \wedge q$	$\neg p \vee \neg q$
$p \vee q$	$\neg p \wedge \neg q$
$p \to q$	$p \wedge \neg q$

Table: negation of compound propositions

Example 4.15 – Negating a conditional in words

Let x be a specified number. Write down the negation of "If x is an integer, then x is a rational number."

Solution. The statement is $p \to q$ with p: x is an integer, q: x is a rational number. Since $\neg(p \to q) \equiv p \wedge \neg q$, the negation is "$x$ is an integer and x is not a rational number."

Example 4.16 – Negating a conditional with a compound consequent

Let x be a specified number. Write down the negation of "If x is not a real number, then it is not a rational number and not an irrational number."

Solution. Let p: x is a real number, q: x is a rational number, r: x is an irrational number. The proposition is $\neg p \to (\neg q \wedge \neg r)$, so its negation is

$$\neg[\neg p \to (\neg q \wedge \neg r)] \equiv \neg p \wedge \neg(\neg q \wedge \neg r) \equiv \neg p \wedge (q \vee r).$$

In words: "x is not a real number and it is a rational number or an irrational number."

Example 4.17 – Simplifying with the laws

Simplify: (i) $(p \vee q) \wedge [\neg\{(\neg p) \wedge q\}]$ (ii) $(p \vee q) \wedge [\neg\{(\neg p) \vee q\}]$ (iii) $\neg[\neg\{(p \vee q) \wedge r\} \vee \neg q]$.

Solution. (i)

$$\begin{aligned} (p \vee q) \wedge [\neg\{(\neg p) \wedge q\}] &\equiv (p \vee q) \wedge (p \vee \neg q) && \textit{De Morgan, double negation} \\ &\equiv p \vee (q \wedge \neg q) \equiv p \vee F_0 \equiv p && \textit{Distributive, inverse, identity} \end{aligned}$$

(ii)

$$\begin{aligned} (p \vee q) \wedge \neg\{(\neg p) \vee q\} &\equiv (p \vee q) \wedge (p \wedge \neg q) && \textit{De Morgan} \\ &\equiv \{(p \vee q) \wedge p\} \wedge \neg q \equiv p \wedge \neg q && \textit{Associative, commutative, absorption} \end{aligned}$$

(iii)

$$\begin{aligned} \neg[\neg\{(p \vee q) \wedge r\} \vee \neg q] &\equiv \{(p \vee q) \wedge r\} \wedge q && \textit{De Morgan, double negation} \\ &\equiv \{(p \vee q) \wedge q\} \wedge r \equiv q \wedge r && \textit{Associative, commutative, absorption} \end{aligned}$$

Example 4.18 – Three equivalences without truth tables

Prove: (i) $p \vee [p \wedge (p \vee q)] \Leftrightarrow p$ (ii) $[p \vee q \vee (\neg p \wedge \neg q \wedge r)] \Leftrightarrow (p \vee q \vee r)$ (iii) $[(\neg p \vee \neg q) \to (p \wedge q \wedge r)] \Leftrightarrow p \wedge q$.

Solution. (i) $p \vee [p \wedge (p \vee q)] \Leftrightarrow p \vee p \Leftrightarrow p$ (absorption, idempotent).

(ii) $\neg p \wedge \neg q \wedge r \Leftrightarrow (\neg p \wedge \neg q) \wedge r \Leftrightarrow \neg(p \vee q) \wedge r$ (associative, De Morgan). Therefore

$$p \vee q \vee (\neg p \wedge \neg q \wedge r) \Leftrightarrow (p \vee q) \vee [\neg(p \vee q) \wedge r] \Leftrightarrow [(p \vee q) \vee \neg(p \vee q)] \wedge [(p \vee q) \vee r]$$
$$\Leftrightarrow T_0 \wedge (p \vee q \vee r) \Leftrightarrow p \vee q \vee r.$$

(iii) $(\neg p \vee \neg q) \rightarrow (p \wedge q \wedge r) \Leftrightarrow \neg(\neg p \vee \neg q) \vee (p \wedge q \wedge r) \Leftrightarrow (p \wedge q) \vee [(p \wedge q) \wedge r] \Leftrightarrow p \wedge q$ (conditional law, De Morgan, associative, absorption). (This is also B3 of the exercises at the end.)

Example 4.19 – Two more equivalences

Prove: (i) $[(p \vee q) \wedge (p \vee \neg q)] \vee q \Leftrightarrow p \vee q$ (ii) $(p \rightarrow q) \wedge [\neg q \wedge (r \vee \neg q)] \Leftrightarrow \neg(q \vee p)$.

Solution. (i) $(p \vee q) \wedge (p \vee \neg q) \Leftrightarrow p \vee (q \wedge \neg q) \Leftrightarrow p \vee F_0 \Leftrightarrow p$ (distributive, inverse, identity). Therefore $[(p \vee q) \wedge (p \vee \neg q)] \vee q \Leftrightarrow p \vee q$.
(ii) This is the example "Using absorption early" earlier in this section. Another route: $\neg q \wedge (r \vee \neg q) \Leftrightarrow \neg q$ (commutative, absorption), and then

$$(p \rightarrow q) \wedge \neg q \Leftrightarrow \neg[(p \rightarrow q) \rightarrow q] \Leftrightarrow \neg[\neg(p \rightarrow q) \vee q] \Leftrightarrow \neg[(p \wedge \neg q) \vee q]$$
$$\Leftrightarrow \neg[(q \vee p) \wedge (q \vee \neg q)] \Leftrightarrow \neg[(q \vee p) \wedge T_0] \Leftrightarrow \neg(q \vee p).$$

(Here we used $\neg(u \rightarrow v) \Leftrightarrow u \wedge \neg v$ and $u \rightarrow v \Leftrightarrow \neg u \vee v$.)

Example 4.20 – Exportation and a three-term simplification

Prove: (i) $p \rightarrow (q \rightarrow r) \Leftrightarrow (p \wedge q) \rightarrow r$ (ii) $[\neg p \wedge (\neg q \wedge r)] \vee (q \wedge r) \vee (p \wedge r) \Leftrightarrow r$.

Solution. (i) $p \rightarrow (q \rightarrow r) \Leftrightarrow \neg p \vee (\neg q \vee r) \Leftrightarrow (\neg p \vee \neg q) \vee r \Leftrightarrow \neg(p \wedge q) \vee r \Leftrightarrow (p \wedge q) \rightarrow r$.
(ii) $\neg p \wedge (\neg q \wedge r) \Leftrightarrow (\neg p \wedge \neg q) \wedge r \Leftrightarrow r \wedge \neg(p \vee q)$, and $(q \wedge r) \vee (p \wedge r) \Leftrightarrow r \wedge (q \vee p) \Leftrightarrow r \wedge (p \vee q)$. Therefore the left side is

$$\{r \wedge \neg(p \vee q)\} \vee \{r \wedge (p \vee q)\} \Leftrightarrow r \wedge [\neg(p \vee q) \vee (p \vee q)] \Leftrightarrow r \wedge T_0 \Leftrightarrow r.$$

(Also Problem 9 of the solved university problems.)

Example 4.21 – Negation of a conditional, simplified

Prove that $\neg[\{(p \vee q) \wedge r\} \rightarrow \neg q] \Leftrightarrow q \wedge r$.

Solution.

$$\begin{aligned} \neg[\{(p \vee q) \wedge r\} \rightarrow \neg q] &\Leftrightarrow \neg[\neg\{(p \vee q) \wedge r\} \vee \neg q] && \textit{Conditional law} \\ &\Leftrightarrow \{(p \vee q) \wedge r\} \wedge q \Leftrightarrow (p \vee q) \wedge (q \wedge r) && \textit{De Morgan; assoc., comm.} \\ &\Leftrightarrow [(p \vee q) \wedge q] \wedge r \Leftrightarrow q \wedge r && \textit{Associative, absorption} \end{aligned}$$

Example 4.22 – Proving a tautology by splitting

Prove that $[(P \vee Q) \wedge \neg\{\neg P \wedge (\neg Q \vee \neg R)\}] \vee (\neg P \wedge \neg Q) \vee (\neg P \wedge \neg R)$ is a tautology.

Solution. Let w denote the proposition; then $w \equiv u \vee v$ with $u \equiv (P \vee Q) \wedge \neg\{\neg P \wedge (\neg Q \vee \neg R)\}$ and $v \equiv (\neg P \wedge \neg Q) \vee (\neg P \wedge \neg R)$. By the laws of logic,

$$u \Leftrightarrow (P \vee Q) \wedge \neg\{\neg P \wedge \neg(Q \wedge R)\} \Leftrightarrow (P \vee Q) \wedge \{P \vee (Q \wedge R)\}$$
$$\Leftrightarrow P \vee \{Q \wedge (Q \wedge R)\} \Leftrightarrow P \vee \{(Q \wedge Q) \wedge R\} \Leftrightarrow P \vee (Q \wedge R),$$
$$v \Leftrightarrow \neg(P \vee Q) \vee \neg(P \vee R) \Leftrightarrow \neg\{(P \vee Q) \wedge (P \vee R)\} \Leftrightarrow \neg\{P \vee (Q \wedge R)\} \equiv \neg u.$$

Therefore $w \equiv u \vee v \Leftrightarrow u \vee \neg u$, which is always true: the proposition is a tautology. (Also Problem 2 of

the solved university problems.)

4.10 Exercise Set 4B — the laws of logic

4B.1 Verify the identity, inverse and domination laws through truth tables.

4B.2 Prove De Morgan's law $\neg(p \wedge q) \Leftrightarrow \neg p \vee \neg q$.

4B.3 Prove the distributive law $p \wedge (q \vee r) \Leftrightarrow (p \wedge q) \vee (p \wedge r)$.

4B.4 Indicate the negations of the following compound propositions: (1) 3 is a prime and 4 is even. (2) $2 < 3$ or $4 < 3$. (3) If it rains, then I do not drive the car. (4) If $\sqrt{2}$ is rational, then $\sqrt{2}+1$ is rational. (5) If an odd integer is greater than 2 and less than 8, then it is a prime number. (6) $(p \vee q) \wedge r$ (7) $p \to (q \to r)$ (8) $(p \wedge r) \to (q \vee r)$ (9) $(p \vee q) \to r$ (10) $(p \wedge q) \to r$ (11) $p \to (\neg q \wedge r)$.

4B.5 Rewrite the following conditionals in the form of disjunctions [Hint: $(p \to q) \Leftrightarrow (\neg p \vee q)$]: (1) If I dream of home, then I will work hard and earn money. (2) If I am awake, then I will work on the computer or read a novel. (3) If I study mathematics and discrete structures, then I will not fail in the examination.

4B.6 Write down the negation of each of the conditionals in 4B.5.

4.11 Solutions to Exercise Set 4B

4B.1 – Identity, inverse and domination laws

p	$\neg p$	$p \vee F_0$	$p \wedge T_0$	$p \vee \neg p$	$p \wedge \neg p$	$p \vee T_0$	$p \wedge F_0$
T	F	T	T	T	F	T	F
F	T	F	F	T	F	T	F

The columns show $p \vee F_0 \equiv p$ and $p \wedge T_0 \equiv p$ (identity), $p \vee \neg p \equiv T_0$ and $p \wedge \neg p \equiv F_0$ (inverse), and $p \vee T_0 \equiv T_0$ and $p \wedge F_0 \equiv F_0$ (domination).

4B.2 – De Morgan's law

p	q	$p \wedge q$	$\neg(p \wedge q)$	$\neg p$	$\neg q$	$\neg p \vee \neg q$
T	T	T	F	F	F	F
T	F	F	T	F	T	T
F	T	F	T	T	F	T
F	F	F	T	T	T	T

The columns of $\neg(p \wedge q)$ and $\neg p \vee \neg q$ agree in every row.

4B.3 – Distributive law

p	q	r	$q \vee r$	$p \wedge (q \vee r)$	$p \wedge q$	$p \wedge r$	$(p \wedge q) \vee (p \wedge r)$
T	T	T	T	T	T	T	T
T	T	F	T	T	T	F	T
T	F	T	T	T	F	T	T
T	F	F	F	F	F	F	F
F	T	T	T	F	F	F	F
F	T	F	T	F	F	F	F
F	F	T	T	F	F	F	F
F	F	F	F	F	F	F	F

The columns of $p \wedge (q \vee r)$ and $(p \wedge q) \vee (p \wedge r)$ agree in all eight rows.

4B.4 – Negations

(1) 3 is not a prime or 4 is not even. (2) $2 \geq 3$ and $4 \geq 3$. (3) It rains and I drive the car. (4) $\sqrt{2}$ is rational and $\sqrt{2}+1$ is not rational. (5) An odd integer is greater than 2 and less than 8, and it is not a prime number.
(6) $\neg[(p \vee q) \wedge r] \equiv (\neg p \wedge \neg q) \vee \neg r$. (7) $\neg[p \to (q \to r)] \equiv p \wedge \neg(q \to r) \equiv p \wedge (q \wedge \neg r)$. (8) $(p \wedge r) \wedge \neg(q \vee r) \equiv (p \wedge r) \wedge (\neg q \wedge \neg r)$ (which is in fact a contradiction). (9) $(p \vee q) \wedge \neg r$. (10) $(p \wedge q) \wedge \neg r$. (11) $p \wedge \neg(\neg q \wedge r) \equiv p \wedge (q \vee \neg r)$.

4B.5 – Conditionals as disjunctions

(1) I do not dream of home, or I will work hard and earn money. (2) I am not awake, or I will work on the computer or read a novel. (3) I do not study mathematics or I do not study discrete structures, or I will not fail in the examination.

4B.6 – Negations of the conditionals

Using $\neg(p \to q) \equiv p \wedge \neg q$:
(1) I dream of home, and I will not work hard or will not earn money. (2) I am awake, and I will not work on the computer and I will not read a novel. (3) I study mathematics and discrete structures, and I will fail in the examination.

4.12 More examples on duality

Example 4.23 – Writing duals

Write down the duals of (i) $\neg(p \vee q) \wedge [p \vee \neg(q \wedge \neg s)]$ (ii) $(p \wedge q) \vee [(\neg p \vee q) \wedge (\neg r \vee s)] \vee (r \wedge s)$ (iii) $(p \wedge \neg q) \vee (r \wedge T_0)$ (iv) $[(p \vee T_0) \wedge (q \vee F_0)] \vee [(r \wedge s) \wedge T_0]$.

Solution. Interchanging $\wedge \leftrightarrow \vee$ and $T_0 \leftrightarrow F_0$:
(i) $\neg(p \wedge q) \vee [p \wedge \neg(q \vee \neg s)]$ (ii) $(p \vee q) \wedge [(\neg p \wedge q) \vee (\neg r \wedge s)] \wedge (r \vee s)$
(iii) $(p \vee \neg q) \wedge (r \vee F_0)$ (iv) $[(p \wedge F_0) \vee (q \wedge T_0)] \wedge [(r \vee s) \vee F_0]$.

Example 4.24 – Duals of conditionals

Write down the duals of (i) $p \to q$ (ii) $(p \to q) \to r$ (iii) $p \to (q \to r)$.

Solution. Recall $(u \to v) \Leftrightarrow (\neg u \vee v)$; eliminate $\to$ first.
(i) $(p \to q)^d \Leftrightarrow (\neg p \vee q)^d \equiv \neg p \wedge q$.

(ii) $[(p \to q) \to r]^d \Leftrightarrow [\neg(p \to q) \vee r]^d \Leftrightarrow [(p \wedge \neg q) \vee r]^d \equiv (p \vee \neg q) \wedge r$.
(iii) $[p \to (q \to r)]^d \Leftrightarrow [\neg p \vee (\neg q \vee r)]^d \equiv \neg p \wedge (\neg q \wedge r)$.

Example 4.25 – Prove, then deduce by duality

Prove that $[(\neg p \vee q) \wedge (p \wedge (p \wedge q))] \Leftrightarrow p \wedge q$. Hence deduce that $[(\neg p \wedge q) \vee (p \vee (p \vee q))] \Leftrightarrow p \vee q$.

Solution.

$$\begin{aligned}(\neg p \vee q) \wedge (p \wedge (p \wedge q)) &\Leftrightarrow (\neg p \vee q) \wedge ((p \wedge p) \wedge q) \Leftrightarrow (\neg p \vee q) \wedge (p \wedge q) \\ &\Leftrightarrow [\neg p \wedge (p \wedge q)] \vee [q \wedge (p \wedge q)] \Leftrightarrow (F_0 \wedge q) \vee (q \wedge p) \\ &\Leftrightarrow F_0 \vee (p \wedge q) \Leftrightarrow p \wedge q.\end{aligned}$$

This is the first result. Taking the dual on both sides (principle of duality) gives $(\neg p \wedge q) \vee (p \vee (p \vee q)) \Leftrightarrow p \vee q$, the second result.

Example 4.26 – Verifying the principle of duality

Verify the principle of duality for $[\neg(p \wedge q) \to \neg p \vee (\neg p \vee q)] \Leftrightarrow (\neg p \vee q)$.

Solution. The equivalence is $u \Leftrightarrow v$ with $u = \neg(p \wedge q) \to (\neg p \vee (\neg p \vee q))$ and $v = \neg p \vee q$. First $u \Leftrightarrow \neg\neg(p \wedge q) \vee (\neg p \vee (\neg p \vee q)) \Leftrightarrow (p \wedge q) \vee (\neg p \vee (\neg p \vee q))$. Therefore

$$\begin{aligned}u^d &\Leftrightarrow (p \vee q) \wedge (\neg p \wedge (\neg p \wedge q)) \Leftrightarrow (p \vee q) \wedge (\neg p \wedge q) \\ &\Leftrightarrow [p \wedge (\neg p \wedge q)] \vee [q \wedge (\neg p \wedge q)] \Leftrightarrow (F_0 \wedge q) \vee (q \wedge \neg p) \Leftrightarrow F_0 \vee (q \wedge \neg p) \Leftrightarrow q \wedge \neg p.\end{aligned}$$

Also $v^d \Leftrightarrow \neg p \wedge q \Leftrightarrow q \wedge \neg p$. We observe that $u^d \Leftrightarrow v^d$, which verifies the principle of duality for the given equivalence. (See also Part C1(b) of the exercises at the end.)

4.13 Exercise Set 4C — duality

4C.1 Write down the duals of (1) $q \to p$ (2) $(p \vee q) \wedge r$ (3) $(p \wedge q) \vee T_0$ (4) $p \to (q \wedge r)$ (5) $p \leftrightarrow q$ (6) $p \veebar q$.

4C.2 Using the results $\neg(p \vee q) \Leftrightarrow \neg p \wedge \neg q$ and $p \vee (q \wedge r) \Leftrightarrow (p \vee q) \wedge (p \vee r)$, prove: (1) $\neg(p \wedge q) \Leftrightarrow \neg p \vee \neg q$ (2) $p \wedge (q \vee r) \Leftrightarrow (p \wedge q) \vee (p \wedge r)$.

4C.3 Prove that $(\neg p \vee q) \wedge (p \wedge (p \wedge q)) \Leftrightarrow p \wedge q$. Hence deduce that $(\neg p \wedge q) \vee (p \vee (p \vee q)) \Leftrightarrow p \vee q$.

4C.4 Verify the principle of duality for the logical equivalence $(p \vee q) \wedge (\neg p \wedge (\neg p \wedge q)) \Leftrightarrow (\neg p \wedge q)$.

4.14 Solutions to Exercise Set 4C

4C.1 – Duals

Eliminate $\to$, $\leftrightarrow$ and $\veebar$ first, then interchange $\wedge \leftrightarrow \vee$ and $T_0 \leftrightarrow F_0$.
(1) $q \to p \equiv \neg q \vee p$; dual $\neg q \wedge p$. (2) $(p \wedge q) \vee r$. (3) $(p \vee q) \wedge F_0$. (4) $p \to (q \wedge r) \equiv \neg p \vee (q \wedge r)$; dual $\neg p \wedge (q \vee r)$.
(5) $p \leftrightarrow q \equiv (\neg p \vee q) \wedge (\neg q \vee p)$; dual $(\neg p \wedge q) \vee (\neg q \wedge p)$. (6) $p \veebar q \equiv (p \wedge \neg q) \vee (\neg p \wedge q)$; dual $(p \vee \neg q) \wedge (q \vee \neg p)$.

4C.2 – New laws from old by duality

(1) The dual of $\neg(p \vee q) \Leftrightarrow \neg p \wedge \neg q$ is $\neg(p \wedge q) \Leftrightarrow \neg p \vee \neg q$; by the principle of duality it holds.
(2) The dual of $p \vee (q \wedge r) \Leftrightarrow (p \vee q) \wedge (p \vee r)$ is $p \wedge (q \vee r) \Leftrightarrow (p \wedge q) \vee (p \wedge r)$; by the principle of duality it holds. ■

4C.3 – Prove, then deduce

This is exactly the example "Prove, then deduce by duality" above.

4C.4 – Verifying duality

The equivalence. $\neg p \wedge (\neg p \wedge q) \Leftrightarrow \neg p \wedge q$ (associative, idempotent), so the left side is $(p \vee q) \wedge (\neg p \wedge q) \Leftrightarrow [p \wedge (\neg p \wedge q)] \vee [q \wedge (\neg p \wedge q)] \Leftrightarrow F_0 \vee (\neg p \wedge q) \Leftrightarrow \neg p \wedge q$. So the equivalence holds.
The dual. The dual is $(p \wedge q) \vee (\neg p \vee (\neg p \vee q)) \Leftrightarrow \neg p \vee q$. Its left side is $(p \wedge q) \vee \neg p \vee q \Leftrightarrow \neg p \vee q$ (idempotent, and $p \wedge q$ is absorbed by q). So the dual also holds, verifying the principle of duality.

5 Logical Implication

Definition: Logical implication

A **logically implies** B, written $A \Rightarrow B$, when B is true in every row in which A is true; equivalently, $A \to B$ is a tautology.

- To show $A \Rightarrow B$: inspect only the rows where A is T and check that B is T there.
- To show $A \not\Rightarrow B$: exhibit one row with $A =$ T, $B =$ F.
- $A \equiv B$ exactly when $A \Rightarrow B$ and $B \Rightarrow A$.

Symbol	Kind	Meaning
$\to$	Connective	$p \to q$ may be T or F in a given row
$\Rightarrow$	Relation between formulas	$A \to B$ is T in **every** row
$\leftrightarrow$	Connective	$p \leftrightarrow q$ may be T or F in a given row
$\equiv$ / $\Leftrightarrow$	Relation between formulas	$A \leftrightarrow B$ is T in **every** row

Example 5.1 – Modus tollens as an implication

Show that $[(p \to q) \wedge \neg q] \Rightarrow \neg p$.

Solution. The premise is T only in the highlighted row, where $\neg p$ is also T.

p	q	$p \to q$	$\neg q$	$(p \to q) \wedge \neg q$	$\neg p$
T	T	T	F	F	**F**
T	F	F	T	F	**F**
F	T	T	F	F	**T**
F	F	T	T	T	**T**

Hence the implication holds.

Example 5.2 – Which are valid?

Decide: (a) $(P \wedge Q) \Rightarrow (P \leftrightarrow Q)$ (b) $(P \leftrightarrow Q) \Rightarrow (P \to Q)$ (c) $(P \leftrightarrow Q) \Rightarrow (P \to \neg Q)$ (d) $Q \Rightarrow (P \to Q)$.

Solution. (a) Valid: $P \wedge Q$ true forces $P = Q =$ T. (b) Valid: when P, Q are equal, $P \to Q$ is T. (c) Invalid: $P = Q =$ T gives T $\to$ F = F. (d) Valid: a true consequent makes the conditional true.

Example 5.3 – An implication that is not an equivalence

Show that $p \wedge q \Rightarrow p \vee q$ but $p \vee q \not\Rightarrow p \wedge q$.

Solution. If $p \wedge q$ is T then p is T, so $p \vee q$ is T. Conversely, $p =$ T, $q =$ F makes $p \vee q$ true and $p \wedge q$ false.

5.1 Converse, inverse and contrapositive: more examples

For a conditional $p \to q$: (1) $q \to p$ is its **converse**; (2) $\neg p \to \neg q$ is its **inverse** (or *opposite*); (3) $\neg q \to \neg p$ is its **contrapositive**. For example, let p: 2 is an integer, q: 9 is a multiple of 3. Then

- $p \to q$: If 2 is an integer, then 9 is a multiple of 3.
- Converse $q \to p$: If 9 is a multiple of 3, then 2 is an integer.
- Inverse $\neg p \to \neg q$: If 2 is not an integer, then 9 is not a multiple of 3.
- Contrapositive $\neg q \to \neg p$: If 9 is not a multiple of 3, then 2 is not an integer.

From the truth table in Section 2: (1) *a conditional and its contrapositive are logically equivalent*, $(p \to q) \Leftrightarrow (\neg q \to \neg p)$; (2) *the converse and the inverse of a conditional are logically equivalent*, $(q \to p) \Leftrightarrow (\neg p \to \neg q)$. Since $p \to q$ and $q \to p$ do not always have the same truth values, $(p \to q) \not\Leftrightarrow (q \to p)$: *if a conditional is true, its converse need not be true*, and vice versa.

5.2 Hypothetical statements; necessary and sufficient conditions

In the definition of $p \to q$, p and q can be any propositions. So we may form, for example, with p: 6 is a multiple of 2 and q: 3 is a prime number, the conditional "If 6 is a multiple of 2, then 3 is a prime number". It is *true* (both parts are true) but makes no sense, since there is no connection between p and q. Similarly, with p: 4 is an odd number and q: Bengaluru is not in Karnataka (both false), "If 4 is an odd number, then Bengaluru is not in Karnataka" makes no sense but is logically true.

Our real interest is in conditionals $p \to q$ where p and q are related, so that the truth value of q depends on that of p. Such conditionals are **hypothetical** (or **implicative**) statements. When q is true whenever p is true, we say that p (logically) **implies** q and write $p \Rightarrow q$; otherwise $p \not\Rightarrow q$ ("p does not imply q").

Definition: Necessary and sufficient conditions

Suppose $p \Rightarrow q$. Then, for q to be true it is **sufficient** that p is true; and if p is true it is **necessary** that q is true. So the following carry the same meaning:

(1) $p \Rightarrow q$; (2) p is sufficient for q; (3) q is necessary for p.

For example, "If a quadrilateral is a square, then it is a rectangle" means "(The fact that) a quadrilateral is a square is a sufficient condition for it to be a rectangle", or "(The fact that) a quadrilateral is a rectangle is a necessary condition for it to be a square". For two propositions p and q the following situations are possible:

Situation	Meaning
$p \Rightarrow q$, but $q \not\Rightarrow p$	p is sufficient but not necessary for q
$p \not\Rightarrow q$, but $q \Rightarrow p$	p is necessary but not sufficient for q
$p \Rightarrow q$ and $q \Rightarrow p$	p is necessary and sufficient for q (and vice versa): p if and only if q (p iff q)

A necessary and sufficient condition is also called a **criterion**; "iff" abbreviates "if and only if".

- Let A be a specified city, p: The city A is in Karnataka, q: The city A is in India. Then $p \Rightarrow q$ but $q \not\Rightarrow p$: p is a sufficient but not a necessary condition for q, and q is a necessary but not a sufficient condition for p.

- Let x be a specified integer, p: The integer x is even, q: The integer x is divisible by 2. Then $p \Rightarrow q$ and $q \Rightarrow p$: p is a necessary and sufficient condition for q; p iff q.

Example 5.4 – Contrapositive with fewer arrows

Write down the contrapositive of $p \to (q \to r)$ with (a) only one occurrence of the connective $\to$; (b) no occurrence of $\to$.

Solution. The contrapositive of $p \to (q \to r)$ is $\neg(q \to r) \to \neg p$. Now

$$\neg(q \to r) \to \neg p \Leftrightarrow \neg\{\neg(q \to r)\} \vee \neg p \Leftrightarrow (q \to r) \vee \neg p \quad \text{(a)} \qquad \Leftrightarrow (\neg q \vee r) \vee \neg p \quad \text{(b)}.$$

Example 5.5 – Three implications by truth table

Prove: (i) $[p \wedge (p \to q)] \Rightarrow q$ (ii) $[(p \to q) \wedge \neg q] \Rightarrow \neg p$ (iii) $[(p \vee q) \wedge \neg p] \Rightarrow q$.

p	q	$p \to q$	$p \wedge (p \to q)$	$(p \to q) \wedge \neg q$	$\neg p$	$(p \vee q) \wedge \neg p$
T	T	T	T	F	F	F
T	F	F	F	F	F	F
F	T	T	F	F	T	T
F	F	T	F	T	T	F

Solution. (i) In the only row where $p \wedge (p \to q)$ is true (first row), q is true. So $[p \wedge (p \to q)] \Rightarrow q$.
(ii) In the only row where $(p \to q) \wedge \neg q$ is true (last row), $\neg p$ is true. So $[(p \to q) \wedge \neg q] \Rightarrow \neg p$.
(iii) In the only row where $(p \vee q) \wedge \neg p$ is true (third row), q is true. So $[(p \vee q) \wedge \neg p] \Rightarrow q$.

Example 5.6 – Two three-variable implications

Prove: (i) $[p \wedge (p \to q) \wedge r] \Rightarrow [(p \vee q) \to r]$ (ii) $\{[p \vee (q \vee r)] \wedge \neg q\} \Rightarrow p \vee r$.

Solution. (i) The highlighted row is the only one in which p, $p \to q$ and r are all true; there $(p \vee q) \to r$ is true.

p	q	r	$p \wedge (p \to q) \wedge r$	$p \vee q$	$(p \vee q) \to r$
T	T	T	T	T	T
T	T	F	F	T	F
T	F	T	F	T	T
T	F	F	F	T	F
F	T	T	F	T	T
F	T	F	F	T	F
F	F	T	F	F	T
F	F	F	F	F	T

(ii) In every highlighted row, where $[p \vee (q \vee r)] \wedge \neg q$ is true, $p \vee r$ is true.

p	q	r	$p \vee (q \vee r)$	$[p \vee (q \vee r)] \wedge \neg q$	$p \vee r$
T	T	T	T	F	T
T	T	F	T	F	T
T	F	T	T	T	T
T	F	F	T	T	T
F	T	T	T	F	T
F	T	F	T	F	F
F	F	T	T	T	T
F	F	F	F	F	F

5.3 Exercise Set 5A — converse, inverse, contrapositive; implication

5A.1 State the converse, inverse and contrapositive of: (1) If a quadrilateral is a parallelogram, then its diagonals bisect each other. (2) If a real number x^2 is greater than zero, then x is not equal to zero. (3) If a triangle is not isosceles, then it is not equilateral.

5A.2 Write the conditionals of 5A.1 using (a) the "necessary condition" language; (b) the "sufficient condition" language.

5A.3 Write in the "necessary and sufficient condition" language: (1) An integer is odd if and only if it is not divisible by 2. (2) Two lines are parallel if and only if they are equidistant. (3) A square matrix is non-singular if and only if its determinant is not zero.

5A.4 Prove the following:
(1) $p \wedge q \Rightarrow p \vee q$ (2) $p \Rightarrow p \vee q$ (3) $\neg p \Rightarrow p \to q$ (4) $q \Rightarrow (p \to q)$
(5) $p \wedge (q \vee r) \Rightarrow (p \wedge q) \vee r$ (6) $(p \to q) \Rightarrow p \to (p \wedge q)$ (7) $[(p \to q) \to q] \Rightarrow p \vee q$
(8) $[(p \to q) \wedge (q \to r)] \Rightarrow [(p \wedge q) \to r]$ (9) $[(p \to r) \wedge (q \to r)] \Rightarrow [(p \vee q) \to r]$
(10) $(p \vee q) \wedge (p \to r) \wedge (q \to r) \Rightarrow r$ (11) $\{[(p \wedge q) \to r] \wedge \neg q \wedge (p \to \neg r)\} \Rightarrow \neg p \vee \neg q$
(12) $[(p \to q) \wedge (r \to s) \wedge (p \vee r)] \Rightarrow (q \vee s)$ (13) $[(p \to q) \wedge (r \to s) \wedge (\neg q \vee \neg s)] \Rightarrow (\neg p \vee \neg r)$.

5.4 Solutions to Exercise Set 5A

5A.1 – Converse, inverse, contrapositive

(1) Converse: If the diagonals of a quadrilateral bisect each other, then it is a parallelogram. Inverse: If a quadrilateral is not a parallelogram, then its diagonals do not bisect each other. Contrapositive: If the diagonals of a quadrilateral do not bisect each other, then it is not a parallelogram.
(2) Converse: If a real number x is not equal to zero, then x^2 is greater than zero. Inverse: If a real number x^2 is not greater than zero, then x is equal to zero. Contrapositive: If a real number x is equal to zero, then x^2 is not greater than zero.
(3) Converse: If a triangle is not equilateral, then it is not isosceles. Inverse: If a triangle is isosceles, then it is equilateral. Contrapositive: If a triangle is equilateral, then it is isosceles.

5A.2 – Necessary and sufficient language

In $p \to q$, q is necessary for p and p is sufficient for q.
(a)(1) A necessary condition for a quadrilateral to be a parallelogram is that its diagonals bisect each other. (2) A necessary condition for a real number x^2 to be greater than zero is that x is not equal to zero. (3) For a triangle to be non-isosceles it is necessary that it is not equilateral.
(b)(1) A sufficient condition for the diagonals of a quadrilateral to bisect each other is that the quadrilateral is a parallelogram. (2) For a real number x, the condition $x^2 > 0$ is sufficient for x to be not equal

to zero. (3) A sufficient condition for a triangle to be not equilateral is that it is not isosceles.

5A.3 – Criteria

(1) A necessary and sufficient condition that an integer is odd is that it is not divisible by 2. (2) A necessary and sufficient condition that two lines are parallel is that they are equidistant. (3) A necessary and sufficient condition that a square matrix is non-singular is that its determinant is not zero.

5A.4 – Thirteen implications

In each case assume the left side is true and show that the right side must be true (equivalently, the conditional is a tautology).
(1) $p \wedge q$ true makes p true, hence $p \vee q$ true. (2) p true makes $p \vee q$ true (disjunctive amplification).
(3) $\neg p$ true means p false, so $p \to q$ is true. (4) q true makes $p \to q$ true.
(5) $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$, and $p \wedge r \Rightarrow r$; so $p \wedge (q \vee r) \Rightarrow (p \wedge q) \vee r$.
(6) $p \to (p \wedge q) \equiv \neg p \vee (p \wedge q) \equiv (\neg p \vee p) \wedge (\neg p \vee q) \equiv \neg p \vee q \equiv p \to q$; so in fact the two are equivalent.
(7) $(p \to q) \to q \equiv \neg(\neg p \vee q) \vee q \equiv (p \wedge \neg q) \vee q \equiv (p \vee q) \wedge (\neg q \vee q) \equiv p \vee q$; so the implication holds (indeed it is an equivalence).
(8) If $p \wedge q$ is true, then q is true, and $q \to r$ gives r. So $(q \to r)$ alone implies $(p \wedge q) \to r$.
(9) $(p \to r) \wedge (q \to r) \equiv (p \vee q) \to r$ (proof by cases), so the implication holds.
(10) From (9), the premises give $(p \vee q) \to r$; with $p \vee q$, modus ponens gives r.
(11) $\neg q$ true makes $\neg p \vee \neg q$ true (disjunctive amplification).
(12) Constructive dilemma: $p \vee r \equiv \neg p \to r$; with $r \to s$, $\neg p \to s$; with $\neg q \to \neg p$ (contrapositive of $p \to q$), $\neg q \to s \equiv q \vee s$.
(13) Destructive dilemma: $\neg q \vee \neg s \equiv q \to \neg s$; with $p \to q$ and $\neg s \to \neg r$, syllogism gives $p \to \neg r \equiv \neg p \vee \neg r$. ■

6 Arguments and Rules of Inference

Definition: Argument

An **argument** consists of premises $P_1, \ldots, P_n$ and a conclusion Q. It is **valid** if Q cannot be false while all the premises are true.

$$\begin{array}{l} P_1 \\ P_2 \\ \vdots \\ P_n \\ \hline \therefore Q \end{array}$$

Five equivalent statements

1. The argument $P_1, \ldots, P_n \vdash Q$ is valid.
2. Q follows logically from $P_1, \ldots, P_n$.
3. $(P_1 \wedge \ldots \wedge P_n) \Rightarrow Q$.
4. $(P_1 \wedge \ldots \wedge P_n) \to Q$ is a tautology.
5. $P_1 \wedge \ldots \wedge P_n \wedge \neg Q$ is a contradiction.

Validity depends only on the **form** of the argument, not on what the statements say. A valid argument can have a false conclusion when one of its premises is false; but it can never have all premises true and the conclusion

false.

6.1 Rules of inference

No.	Rule	Premises	Conclusion
1	Modus Ponens (detachment)	$p,\ p \to q$	q
2	Modus Tollens	$p \to q,\ \neg q$	$\neg p$
3	Hypothetical Syllogism	$p \to q,\ q \to r$	$p \to r$
4	Disjunctive Syllogism	$p \lor q,\ \neg p$	q
5	Addition	p	$p \lor q$
6	Simplification	$p \land q$	p
7	Conjunction	$p,\ q$	$p \land q$
8	Conjunctive Syllogism	$\neg(p \land q),\ p$	$\neg q$
9	Constructive Dilemma	$p \to r,\ q \to s,\ p \lor q$	$r \lor s$
10	Simple Dilemma (cases)	$p \to r,\ q \to r,\ p \lor q$	r
11	Destructive Dilemma	$p \to r,\ q \to s,\ \neg r \lor \neg s$	$\neg p \lor \neg q$
12	Resolution	$p \lor q,\ \neg p \lor r$	$q \lor r$
13	Rule of contradiction	$\neg p \to F_0$	p

Memory aid: **Addition goes with $\lor$, Simplification goes with $\land$.** Note that "from p conclude $p \land q$" and "from $p \lor q$ conclude p" are *not* valid.

6.2 Fallacies

Fallacy	Form	Counterexample
Affirming the consequent	$p \to q,\ q \vdash p$	$p = \text{F},\ q = \text{T}$
Denying the antecedent	$p \to q,\ \neg p \vdash \neg q$	$p = \text{F},\ q = \text{T}$
Non sequitur	$p,\ q \vdash r$	$r = \text{F}$

6.3 Three methods of testing an argument

Method	How	Best when
Truth table	In every row where all premises are T, the conclusion must be T	≤ 3 variables
Direct derivation	Derive Q step by step, citing a rule or law each step	4 or more variables
Indirect (contradiction)	Add $\neg Q$ to the premises and derive F_0	conclusion is short, e.g. p or $\neg p$

To show an argument is **invalid**, give one assignment that makes every premise T and the conclusion F.

6.4 Conditional proof (CP) rule

When the conclusion is itself a conditional,

$$\{P_1, \dots, P_n\} \vdash (Q \to R) \iff \{P_1, \dots, P_n, Q\} \vdash R,$$

so move Q into the premises and derive R. This follows from exportation: $(A \land Q) \to R \equiv A \to (Q \to R)$.

6.5 Worked examples

Example 6.1 – Direct derivation

Show that u follows from $p \to q,\ q \to (r \wedge s),\ \neg r \vee (\neg t \vee u),\ p \wedge t$.

Solution.

Step	Statement	Reason
1	$p \wedge t$	Premise
2	p	Simplification (1)
3	$p \to q$	Premise
4	q	Modus ponens (2, 3)
5	$q \to (r \wedge s)$	Premise
6	$r \wedge s$	Modus ponens (4, 5)
7	r	Simplification (6)
8	$\neg r \vee (\neg t \vee u)$	Premise
9	$\neg t \vee u$	Disjunctive syllogism (7, 8)
10	t	Simplification (1)
11	u	Disjunctive syllogism (9, 10)

Example 6.2 – Chaining with contrapositives

Show that $\neg r \to s$ follows from $p \to r,\ \neg p \to q,\ q \to s$.

Solution.

Step	Statement	Reason
1	$p \to r$	Premise
2	$\neg r \to \neg p$	Contrapositive (1)
3	$\neg p \to q$	Premise
4	$\neg r \to q$	Hypothetical syllogism (2, 3)
5	$q \to s$	Premise
6	$\neg r \to s$	Hypothetical syllogism (4, 5)

Example 6.3 – Indirect method

Using the indirect method, show that $p \to q,\ q \to \neg r,\ r \vdash \neg p$.

Solution. Assume the negation of the conclusion, p.

Step	Statement	Reason
1	p	Assumed (negation of conclusion)
2	q	Modus ponens (1, $p \to q$)
3	$\neg r$	Modus ponens (2, $q \to \neg r$)
4	$r \wedge \neg r$	Conjunction (3, premise r) — contradiction

The assumption is false, so $\neg p$ follows; the argument is valid.

Example 6.4 – Indirect method detects an invalid argument

Does Q follow from $P \to Q$ and $\neg(P \wedge Q)$?

Solution. Add $\neg Q$ as an extra premise; modus tollens gives $\neg P$. But $P = Q = \mathrm{F}$ makes $P \to Q$, $\neg(P \wedge Q)$ and $\neg Q$ all true, so no contradiction can be reached. Hence Q does not follow: the argument is **invalid**. (On the other hand, $\neg P$ *does* follow: assuming P gives Q by modus ponens, while $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$ with P gives $\neg Q$ — a contradiction.)

Example 6.5 – Conditional proof

Show that $\{P \to (Q \to S),\ \neg R \vee P,\ Q\} \vdash R \to S$.

Solution.

Step	Statement	Reason
1	R	Additional premise (CP)
2	$\neg R \vee P \equiv R \to P$	Premise, conditional law
3	P	Modus ponens (1, 2)
4	$Q \to S$	Modus ponens (3, premise)
5	S	Modus ponens (4, premise Q)
6	$R \to S$	Conditional proof (1–5)

Example 6.6 – Recognising a fallacy

"If today is Vishal's birthday, then today is 1 April. Today is 1 April. Therefore today is Vishal's birthday."

Solution. Form $p \to q,\ q \vdash p$: affirming the consequent. Counterexample: $p = \mathrm{F}, q = \mathrm{T}$ (1 April, but not his birthday). **Invalid.** Replacing the second premise with "Today is not 1 April" and concluding "Today is not Vishal's birthday" gives a valid argument (modus tollens).

Examination tip

Write derivations in the three-column format (Step · Statement · Reason) and cite the line numbers used. Treat a compound part such as $C \vee D$ as a single unit; do not expand it unless needed.

6.6 More solved examples on rules of inference

Example 6.7 – Name the rule

Name the rule of inference used in each argument.

Argument	Rule
It is raining. ∴ It is raining or it is snowing.	Addition
The lab is cold and noisy. ∴ The lab is cold.	Simplification
If you have a password, you can log on. You have a password. ∴ You can log on.	Modus ponens
If it snows, the college closes. The college is not closed. ∴ It did not snow.	Modus tollens
If I eat spicy food, I dream strangely. If I dream strangely, I sleep badly. ∴ If I eat spicy food, I sleep badly.	Hypothetical syllogism
The server is down or the network is slow. The server is not down. ∴ The network is slow.	Disjunctive syllogism
Ravi codes or designs. Ravi does not code or he tests. ∴ Ravi designs or tests.	Resolution
If I study, I pass. If I play, I am happy. I study or I play. ∴ I pass or I am happy.	Constructive dilemma
Mohan cannot be both in class and in the canteen. He is in class. ∴ He is not in the canteen.	Conjunctive syllogism

Example 6.8 – An English argument with a conditional conclusion

"If you send me an email, I will finish the program. If you do not send me an email, I will sleep early. If I sleep early, I will wake up refreshed." Show that "If I do not finish the program, I will wake up refreshed" follows.

Solution. p: you send an email; q: I finish the program; r: I sleep early; s: I wake up refreshed. Premises $p \to q$, $\neg p \to r$, $r \to s$; conclusion $\neg q \to s$.

Step	Statement	Reason
1	$p \to q$	Premise
2	$\neg q \to \neg p$	Contrapositive (1)
3	$\neg p \to r$	Premise
4	$\neg q \to r$	Hypothetical syllogism (2, 3)
5	$r \to s$	Premise
6	$\neg q \to s$	Hypothetical syllogism (4, 5)

Example 6.9 – Resolution

Show that $(p \wedge q) \vee r$ and $r \to s$ imply $p \vee s$.

Solution.

Step	Statement	Reason
1	$(p \wedge q) \vee r$	Premise
2	$(p \vee r) \wedge (q \vee r)$	Distributive law (1)
3	$p \vee r$	Simplification (2)
4	$r \to s \equiv \neg r \vee s$	Premise, conditional law
5	$p \vee s$	Resolution (3, 4)

Example 6.10 – Using a combined conditional

Show that d follows from $(a \to b) \wedge (a \to c), \neg(b \wedge c), d \vee a$.

Solution.

Step	Statement	Reason
1	$(a \to b) \wedge (a \to c)$	Premise
2	$a \to (b \wedge c)$	Law 14 (1)
3	$\neg(b \wedge c)$	Premise
4	$\neg a$	Modus tollens (2, 3)
5	$d \vee a$	Premise
6	d	Disjunctive syllogism (4, 5)

Example 6.11 – A longer chain

Show that $\neg p$ follows from $(p \to q) \wedge (r \to s), (q \to t) \wedge (s \to u), \neg(t \wedge u), p \to r$.

Solution.

Step	Statement	Reason
1	$p \to q,\ r \to s$	Simplification of premise 1
2	$q \to t,\ s \to u$	Simplification of premise 2
3	$p \to t$	Hypothetical syllogism $(p \to q, q \to t)$
4	$p \to s$	Hypothetical syllogism $(p \to r, r \to s)$
5	$p \to u$	Hypothetical syllogism (4, $s \to u$)
6	$p \to (t \wedge u)$	Law 14 (3, 5)
7	$\neg(t \wedge u)$	Premise
8	$\neg p$	Modus tollens (6, 7)

Example 6.12 – Simple dilemma

Show that r follows from $p \to q, q \to r, \neg(p \wedge r), p \vee r$.

Solution.

Step	Statement	Reason
1	$p \to q$	Premise
2	$q \to r$	Premise
3	$p \to r$	Hypothetical syllogism (1, 2)
4	$r \to r$	Tautology $(\neg r \vee r)$
5	$p \vee r$	Premise
6	r	Simple dilemma (3, 4, 5)

The premise $\neg(p \wedge r)$ is not needed. A valid argument may contain premises that are never used.

Example 6.13 – Destructive dilemma in English

"If Kiran is in Patna, she is in Bihar. If Kiran is in Pune, she is in Maharashtra. Kiran is either not in Bihar or not in Maharashtra." What follows?

Solution. p: in Patna; q: in Bihar; r: in Pune; s: in Maharashtra. From $p \to q, r \to s, \neg q \vee \neg s$, the destructive dilemma gives $\neg p \vee \neg r$: "Kiran is not in Patna or she is not in Pune."

Example 6.14 – Conditional proof

Using the CP rule, show that $\neg p \vee q, \neg q \vee r, r \to s \vdash p \to s$.

Solution.

Step	Statement	Reason
1	p	Additional premise (CP)
2	$\neg p \vee q$	Premise
3	q	Disjunctive syllogism (1, 2)
4	$\neg q \vee r$	Premise
5	r	Disjunctive syllogism (3, 4)
6	s	Modus ponens (5, $r \to s$)
7	$p \to s$	Conditional proof (1–6)

Example 6.15 – Detecting an invalid argument

"If the professor is in the meeting, the lecture is cancelled. The lecture is cancelled or it is a holiday. It is a holiday. Therefore the professor is in the meeting."

Solution. p: professor in meeting; q: lecture cancelled; h: holiday. Form $p \to q,\ q \vee h,\ h \vdash p$.
Take $p = \text{F}$, $q = \text{T}$, $h = \text{T}$: $p \to q = \text{T}$, $q \vee h = \text{T}$, $h = \text{T}$, but $p = \text{F}$. All premises are true and the conclusion is false, so the argument is **invalid**.

Common pitfall

Find the error in this "derivation" of p from $p \to q$ and q: 1. $p \to q$ (premise) 2. q (premise) 3. p (modus ponens 1, 2).
Modus ponens needs the *antecedent* p, not the consequent q. Step 3 is the fallacy of affirming the consequent. Always check which part of the conditional you hold before citing MP or MT.

6.7 Rules of inference: other names

An argument $(p_1 \wedge p_2 \wedge \ldots \wedge p_n) \to Q$ is often written in tabular form: the premises above a line and the conclusion below it after "$\therefore$" ("therefore"). The argument is **valid** if whenever each premise is true the conclusion is likewise true, i.e. if $(p_1 \wedge \ldots \wedge p_n) \Rightarrow Q$. In an argument the premises are always taken to be true; the conclusion is true only in the case of a valid argument. Many books use the names below for the basic rules; the last column gives the name used in Section 6.1.

No.	Other name	Implication	Name in 6.1
1	Rule of Conjunctive Simplification	$(p \wedge q) \Rightarrow p$	Simplification
2	Rule of Disjunctive Amplification	$p \Rightarrow p \vee q$	Addition
3	Rule of Syllogism	$\{(p \to q) \wedge (q \to r)\} \Rightarrow (p \to r)$	Hypothetical syllogism
4	Modus Ponens (Rule of Detachment)	$\{p \wedge (p \to q)\} \Rightarrow q$	Modus ponens
5	Modus Tollens	$\{(p \to q) \wedge \neg q\} \Rightarrow \neg p$	Modus tollens
6	Rule of Disjunctive Syllogism	$\{(p \vee q) \wedge \neg p\} \Rightarrow q$	Disjunctive syllogism
7	Rule of Contradiction	$(\neg p \to F_0) \Rightarrow p$	Rule of contradiction

"Modus Ponens" is Latin for the "method of affirming" and "Modus Tollens" for the "method of denying". Rules 3–6 were proved by truth tables in Sections 3 and 5. The Rule of Contradiction (the basis of proof by contradiction) is proved by this table:

p	F_0	$\neg p$	$\neg p \to F_0$	$(\neg p \to F_0) \to p$
T	F	F	T	**T**
F	F	T	F	**T**

Remark. The validity (or otherwise) of a given argument may be established with the stated rules or their combinations, together with the laws of logic, logical equivalences and tautologies. When in doubt, use a truth table.

6.8 More worked examples on arguments

Example 6.16 – Affirming the antecedent

Test: "If Sachin hits a century, then he gets a free car. Sachin hits a century. $\therefore$ Sachin gets a free car."

Solution. Let p: Sachin hits a century, q: Sachin gets a free car. The argument reads

$$\begin{array}{l} p \to q \\ p \\ \hline \therefore q \end{array}$$

By the Modus Ponens rule it is **valid**.

Example 6.17 – Denying the consequent

Test: "If Sachin hits a century, he gets a free car. Sachin does not get a free car. $\therefore$ Sachin has not hit a century."

Solution. With p, q as before:

$$\begin{array}{l} p \to q \\ \neg q \\ \hline \therefore \neg p \end{array}$$

By the Modus Tollens rule it is **valid**.

Example 6.18 – Affirming the consequent

Test: "If Sachin hits a century, he gets a free car. Sachin gets a free car. $\therefore$ Sachin has hit a century."

Solution. The argument reads

$$\begin{array}{l} p \to q \\ q \\ \hline \therefore p \end{array}$$

No rule asserts that p must be true when $p \to q$ and q are true; indeed p can be false then:

p	q	$p \to q$	$(p \to q) \wedge q$	$[(p \to q) \wedge q] \to p$
T	T	T	T	**T**
T	F	F	F	**T**
F	T	T	T	**F**
F	F	T	F	**T**

So $[(p \to q) \wedge q] \to p$ is not a tautology, and the argument is **not valid**.

Example 6.19 – Driving to work

Test: "If I drive to work, then I will arrive tired. I am not tired (when I arrive at work). ∴ I do not drive to work."

Solution. p: I drive to work, q: I arrive tired. The argument is

$$\begin{array}{l} p \to q \\ \neg q \\ \hline \therefore \neg p \end{array}$$

Valid by Modus Tollens.

Example 6.20 – Famous or not a musician

Test: "I will become famous or I will not become a musician. I will become a musician. ∴ I will become famous."

Solution. p: I will become famous, q: I will become a musician. The argument is

$$\begin{array}{l} p \vee \neg q \\ q \\ \hline \therefore p \end{array}$$

Since $p \vee \neg q \Leftrightarrow \neg q \vee p \Leftrightarrow q \to p$, it is equivalent to

$$\begin{array}{l} q \to p \\ q \\ \hline \therefore p \end{array}$$

which is **valid** by Modus Ponens.

Example 6.21 – A chain of conditionals

Test: "If I study, then I do not fail in the examination. If I do not fail in the examination, my father gifts a two-wheeler to me. ∴ If I study then my father gifts a two-wheeler to me."

Solution. p: I study, q: I do not fail in the examination, r: My father gifts a two-wheeler to me. The argument is

$$\begin{array}{l} p \to q \\ q \to r \\ \hline \therefore p \to r \end{array}$$

Valid by the Rule of Syllogism.

Example 6.22 – Ravi and his friends

Test: "If Ravi goes out with friends, he will not study. If Ravi does not study, his father becomes angry. His father is not angry. ∴ Ravi has not gone out with friends."

Solution. p: Ravi goes out with friends, q: Ravi does not study, r: His father gets angry. The argument

$$\begin{array}{l} p \to q \\ q \to r \\ \neg r \\ \hline \therefore \neg p \end{array} \quad \text{is equivalent (syllogism) to} \quad \begin{array}{l} p \to r \\ \neg r \\ \hline \therefore \neg p \end{array}$$

Valid by Modus Tollens. (Problem 12 of the solved university problems.)

Example 6.23 – Watching TV

Test: "If I study, I will not fail in the examination. If I do not watch TV in the evenings, I will study. I failed in the examination. ∴ I must have watched TV in the evenings."

Solution. p: I study, q: I fail in the examination, r: I watch TV in the evenings. The argument is

$$\begin{array}{l} p \to \neg q \\ \neg r \to p \\ q \\ \hline \therefore r \end{array}$$

Using contrapositives, $p \to \neg q \Leftrightarrow q \to \neg p$ and $\neg r \to p \Leftrightarrow \neg p \to r$, so it is equivalent to

$$\begin{array}{l} q \to \neg p \\ \neg p \to r \\ q \\ \hline \therefore r \end{array} \quad \text{i.e. (syllogism)} \quad \begin{array}{l} q \to r \\ q \\ \hline \therefore r \end{array}$$

Valid by Modus Ponens. (Problem 3 of the solved university problems.)

Example 6.24 – Grade A and the army

"I will get grade A in this course or I will not graduate. If I do not graduate, I will join the army. I got grade A. ∴ I will not join the army." Is this a valid argument?

Solution. p: I get grade A in this course, q: I do not graduate, r: I join the army. The argument is

$$\begin{array}{l} p \vee q \\ q \to r \\ p \\ \hline \therefore \neg r \end{array}$$

Since $p \vee q \equiv \neg q \to p$ and $q \to r \equiv \neg r \to \neg q$, it is equivalent to

$$\begin{array}{l} \neg r \to \neg q \\ \neg q \to p \\ p \\ \hline \therefore \neg r \end{array} \quad \text{i.e.} \quad \begin{array}{l} \neg r \to p \\ p \\ \hline \therefore \neg r \end{array}$$

This is affirming the consequent. It is **not valid**: $p = \text{T}$, $q = \text{F}$, $r = \text{T}$ makes all premises true and the conclusion false. (Getting grade A does not stop me joining the army.)

Example 6.25 – Three short arguments

Test the validity of

(i) $\begin{array}{l} p \wedge q \\ p \to (q \to r) \\ \hline \therefore r \end{array}$ (ii) $\begin{array}{l} p \\ p \to \neg q \\ \neg q \to \neg r \\ \hline \therefore \neg r \end{array}$ (iii) $\begin{array}{l} p \to r \\ q \to r \\ \hline \therefore (p \vee q) \to r \end{array}$

Solution. (i) Since $p \wedge q$ is true, p and q are true. Since p and $p \to (q \to r)$ are true, $q \to r$ is true; with q true, r is true. **Valid.**
(ii) $p \to \neg q$ and $\neg q \to \neg r$ give $p \to \neg r$ (syllogism). Since p is true, $\neg r$ is true. **Valid.**
(iii) $(p \to r) \wedge (q \to r) \Leftrightarrow (\neg p \vee r) \wedge (\neg q \vee r) \Leftrightarrow r \vee (\neg p \wedge \neg q) \Leftrightarrow \neg(p \vee q) \vee r \Leftrightarrow (p \vee q) \to r$ (commutative, distributive, De Morgan). **Valid.**

Example 6.26 – Constructive and destructive dilemmas

Test

$$\text{(i)}\quad \begin{array}{l} p \to q \\ r \to s \\ p \vee r \\ \hline \therefore\ q \vee s \end{array} \qquad \text{(ii)}\quad \begin{array}{l} p \to q \\ r \to s \\ \neg q \vee \neg s \\ \hline \therefore\ \neg(p \wedge r) \end{array}$$

Solution. (i)

Step	Statement	Reason
1	$p \vee r \equiv \neg p \to r$	Premise, conditional law
2	$\neg p \to s$	Syllogism (1, $r \to s$)
3	$\neg q \to \neg p$	Contrapositive of $p \to q$
4	$\neg q \to s \equiv q \vee s$	Syllogism (3, 2), conditional law

(ii)

Step	Statement	Reason
1	$\neg q \vee \neg s \equiv q \to \neg s$	Premise, conditional law
2	$p \to \neg s$	Syllogism ($p \to q$, 1)
3	$\neg s \to \neg r$	Contrapositive of $r \to s$
4	$p \to \neg r \equiv \neg p \vee \neg r \equiv \neg(p \wedge r)$	Syllogism (2, 3), conditional law, De Morgan

Both arguments are **valid**.

Example 6.27 – Two more derivations

Prove the validity of

$$\text{(i)}\quad \begin{array}{l} p \to r \\ \neg p \to q \\ q \to s \\ \hline \therefore\ \neg r \to s \end{array} \qquad \text{(ii)}\quad \begin{array}{l} (\neg p \vee \neg q) \to (r \wedge s) \\ r \to t \\ \neg t \\ \hline \therefore\ p \end{array}$$

Solution. (i) $(\neg p \to q) \wedge (q \to s) \Rightarrow \neg p \to s$ (syllogism); $p \to r \Leftrightarrow \neg r \to \neg p$ (contrapositive); then $\neg r \to \neg p$ and $\neg p \to s$ give $\neg r \to s$ (syllogism). **Valid.** (Also the example "Chaining with contrapositives" above.)
(ii)

Step	Statement	Reason
1	$\neg r$	Modus tollens ($r \to t$, $\neg t$)
2	$\neg r \vee \neg s$	Disjunctive amplification (1)
3	$\neg(r \wedge s)$	De Morgan (2)
4	$\neg(\neg p \vee \neg q)$	Modus tollens (premise 1, 3)
5	$p \wedge q$	De Morgan (4)
6	p	Conjunctive simplification (5)

Example 6.28 – Using contrapositives and modus tollens

Prove that the following are valid:

$$\text{(i)}\quad \begin{array}{l} p \to (q \to r) \\ \neg q \to \neg p \\ p \\ \hline \therefore r \end{array} \qquad \text{(ii)}\quad \begin{array}{l} \neg p \leftrightarrow q \\ q \to r \\ \neg r \\ \hline \therefore p \end{array}$$

Solution. (i) p and $p \to (q \to r)$ give $q \to r$ (MP); $\neg q \to \neg p \Leftrightarrow p \to q$ (contrapositive); $p \to q$ and $q \to r$ give $p \to r$ (syllogism); since p is true, r is true. **Valid.**
(ii) $q \to r$ and $\neg r$ give $\neg q$ (MT). $\neg p \leftrightarrow q \Leftrightarrow (\neg p \to q) \wedge (q \to \neg p)$; by simplification $\neg p \to q$; with $\neg q$, MT gives $\neg\neg p \Leftrightarrow p$. **Valid.**

Example 6.29 – Conditional conclusions

Prove that the following are valid:

$$\text{(i)}\quad \begin{array}{l} p \to (q \to r) \\ p \vee \neg s \\ q \\ \hline \therefore s \to r \end{array} \qquad \text{(ii)}\quad \begin{array}{l} p \to (q \wedge r) \\ r \to s \\ \neg(q \wedge s) \\ \hline \therefore \neg p \end{array}$$

Solution. (i)

Step	Statement	Reason
1	$p \vee \neg s \equiv s \to p$	Premise, commutative and conditional laws
2	$s \to (q \to r)$	Syllogism (1, $p \to (q \to r)$)
3	$(s \wedge q) \to r$	Exportation (2)
4	$q \to (s \to r)$	Exportation and commutative law (3)
5	$s \to r$	Modus ponens (4, premise q)

(ii)

Step	Statement	Reason
1	$\neg(q \wedge s) \equiv \neg q \vee \neg s \equiv s \to \neg q$	Premise, De Morgan, conditional law
2	$r \to \neg q$	Syllogism ($r \to s$, 1)
3	$\neg r \vee \neg q \equiv \neg(q \wedge r)$	Conditional law, De Morgan
4	$\neg p$	Modus tollens ($p \to (q \wedge r)$, 3)

Example 6.30 – Longer arguments

Test the validity of

$$\text{(i)}\quad \begin{array}{l} (\neg p \vee q) \to r \\ r \to (s \vee t) \\ \neg s \wedge \neg u \\ \neg u \to \neg t \\ \hline \therefore p \end{array} \qquad \text{(ii)}\quad \begin{array}{l} p \to r \\ r \to s \\ t \vee \neg s \\ \neg t \vee u \\ \neg u \\ \hline \therefore \neg p \end{array}$$

Solution. (i)

Step	Statement	Reason
1	$(\neg p \vee q) \to (s \vee t)$	Syllogism (premises 1, 2)
2	$\neg s,\ \neg u$	Simplification (premise 3)
3	$\neg t$	Modus ponens ($\neg u$, $\neg u \to \neg t$)
4	$\neg s \wedge \neg t \equiv \neg(s \vee t)$	Conjunction (2, 3), De Morgan
5	$\neg(\neg p \vee q) \equiv p \wedge \neg q$	Modus tollens (1, 4), De Morgan
6	p	Simplification (5)

(ii) $t \vee \neg s \equiv s \to t$ and $\neg t \vee u \equiv t \to u$. Then $p \to r, r \to s, s \to t, t \to u$ give $p \to u$ (syllogism); with $\neg u$, MT gives $\neg p$. Both are **valid**.

Example 6.31 – Two harder arguments

Prove the validity of

(i) $p \to q$, $q \to (r \wedge s)$, $\neg r \vee (\neg t \vee u)$, $p \wedge t$ / $\therefore u$

(ii) $u \to r$, $(r \to s) \to (p \vee t)$, $q \to (u \wedge s)$, $\neg t$ / $\therefore q \to p$

Solution. (i) This is the example "Direct derivation" above: $p \wedge t$ gives p and t; $p \to q$, $q \to (r \wedge s)$ give $r \wedge s$, hence r; $\neg r \vee (\neg t \vee u) \equiv \neg(r \wedge t) \vee u$, and since r, t are true, $\neg(r \wedge t)$ is false, so u is true. **Valid.** (ii) Use conditional proof: assume q. Then $u \wedge s$ (MP), so u and s; $u \to r$ gives r; with s true, $r \to s$ is true; $(r \to s) \to (p \vee t)$ gives $p \vee t$; with $\neg t$, disjunctive syllogism gives p. Hence $q \to p$. **Valid.** (If q is false, $q \to p$ is true anyway.)

Example 6.32 – An exclusive-or premise

Test the validity of

$$\begin{array}{l} p \to q \\ q \to s \\ r \to \neg s \\ \neg p \veebar r \\ \hline \therefore \neg p \end{array}$$

Solution. $(p \to q) \wedge (q \to s) \Rightarrow p \to s$; $r \to \neg s \Leftrightarrow s \to \neg r$; so $p \to \neg r$, i.e. $r \to \neg p$ (contrapositive). Now $r \veebar \neg p$ is true only when (a) r is true and $\neg p$ false, or (b) r is false and $\neg p$ true. In case (a), $r \to \neg p$ would be false — impossible. So only case (b) remains, and $\neg p$ is true. The argument is **valid**.

Example 6.33 – Using the rules of inference

Establish the validity of $\{p \wedge (p \to q) \wedge (s \vee r) \wedge (r \to \neg q)\} \to (s \vee t)$.

Solution. The argument is

$$\begin{array}{l} p \\ p \to q \\ s \vee r \\ r \to \neg q \\ \hline \therefore s \vee t \end{array}$$

Step	Statement	Reason
1	$s \vee r \equiv \neg s \to r$	Premise, conditional law
2	$\neg s \to \neg q$	Syllogism (1, $r \to \neg q$)
3	$q \to s$	Contrapositive (2)
4	$p \to s$	Syllogism ($p \to q$, 3)
5	s	Rule of detachment (p, 4)
6	$s \vee t$	Disjunctive amplification (5)

Example 6.34 – An invalid argument

Show that the following argument is not valid:

$$\begin{array}{l} p \\ p \vee q \\ q \to (r \to s) \\ t \to r \\ \hline \therefore \neg s \to \neg t \end{array}$$

Solution. Look for an assignment making every premise true and the conclusion false. The conclusion $\neg s \to \neg t$ is false only when s is false and t is true. Then $t \to r$ forces r true, so $r \to s$ is false, and $q \to (r \to s)$ forces q false. With p true, $p \vee q$ is true. So $p = \text{T}$, $q = \text{F}$, $r = \text{T}$, $s = \text{F}$, $t = \text{T}$ makes all four premises true and the conclusion false: the argument is **not valid**.

Example 6.35 – Tautological implication

Show that $S \vee R$ is tautologically implied by $(P \vee Q) \wedge (P \to R) \wedge (Q \to S)$.

Solution. $(P \vee Q) \wedge (P \to R) \wedge (Q \to S) \Leftrightarrow (\neg P \to Q) \wedge (Q \to S) \wedge (P \to R) \Rightarrow (\neg P \to S) \wedge (P \to R) \Leftrightarrow (\neg S \to P) \wedge (P \to R) \Rightarrow \neg S \to R \Leftrightarrow S \vee R$.

Example 6.36 – Seven variables

Show that $R \vee S$ follows logically from $C \vee D$, $(C \vee D) \to \neg H$, $\neg H \to (A \wedge \neg B)$ and $(A \wedge \neg B) \to R \vee S$.

Solution. Using the Rule of Syllogism twice, $(C \vee D) \to (R \vee S)$; with $C \vee D$, Modus Ponens gives $R \vee S$. (Problem 4 of the solved university problems.)

Example 6.37 – A valid conclusion

Show that $R \wedge (P \vee Q)$ is a valid conclusion from $P \vee Q$, $Q \to R$, $P \to M$ and $\neg M$.

Solution. $P \to M$ and $\neg M$ give $\neg P$ (MT). $P \vee Q \Leftrightarrow \neg P \to Q$, and with $Q \to R$, syllogism gives $\neg P \to R$; MP gives R. Since $P \vee Q$ is a premise, $R \wedge (P \vee Q)$ follows (conjunction). (Part C4(a) of the exercises at the end.)

Example 6.38 – Conditional conclusion from three premises

Prove that $R \to S$ is a valid conclusion from the premises $P \to (Q \to S)$, $\neg R \vee P$ and Q.

Solution. $\neg R \vee P \Leftrightarrow R \to P$; with $P \to (Q \to S)$, syllogism gives $R \to (Q \to S) \Leftrightarrow \neg[R \wedge \neg(Q \to S)] \Leftrightarrow \neg[R \wedge Q \wedge \neg S] \Leftrightarrow \neg(R \wedge \neg S) \vee \neg Q$. Since Q is true (premise), $\neg(R \wedge \neg S)$ must be true, i.e. $\neg R \vee S \Leftrightarrow R \to S$. (The example "Conditional proof" above gives a shorter route.)

6.9 Exercise Set 6A — validity of arguments

6A.1 Test the validity of the following arguments:

(1) If a person is poor, he is unhappy. If a person is unhappy, he dies young. ∴ Poor persons die young.

(2) If there is a strike by students, the examination will be postponed. There was no strike by students. ∴ The examination was not postponed.

(3) If there is a strike by students, the examination will be postponed. The examination was not postponed. ∴ There was no strike by students.

(4) If I drive to work, then I will arrive tired. I do not drive to work. ∴ I will not arrive tired.

(5) If I have talent and work hard, then I will become successful in life. If I become successful in life then I will be happy. ∴ If I will not be happy, then I did not work hard or I do not have talent.

(6) If Ravi studies, then he will pass in Discrete Mathematics paper. If Ravi does not play cricket, then he will study. Ravi failed in Discrete Mathematics paper. ∴ Ravi played cricket.

6A.2 Prove that the following arguments are valid:

(1) $\neg p \to q$, $\neg q$ / $\therefore p$

(2) $(p \wedge r) \to s$, p / $\therefore r \to s$

(3) $p \vee q$, $\neg p \vee r$, $\neg r$ / $\therefore q$

(4) $p \to r$, $\neg q \to p$, $\neg r$ / $\therefore q$

(5) p, $p \to \neg q$, $r \to q$ / $\therefore \neg r$

(6) $p \to q$, $\neg q$, $\neg r$ / $\therefore \neg(p \vee r)$

(7) $p \to q$, $r \to \neg q$, r / $\therefore \neg p$

(8) $p \to q$, $\neg r \to \neg q$, p / $\therefore r$

(9) $p \to q$, $\neg r \vee s$, $p \vee r$ / $\therefore \neg q \to s$

(10) $p \to (q \to r)$, $\neg q \to \neg p$, p / $\therefore r$

(11) $(p \wedge q) \to r$, $\neg q$, $p \to \neg r$ / $\therefore \neg p \vee \neg q$

(12) $p \to q$, $r \to s$, $\neg q \vee \neg s$ / $\therefore \neg p \vee \neg r$

(13) $p \to (q \to r)$, $p \vee s$, $t \to q$, $\neg s$ / $\therefore \neg r \to \neg t$

(14) $p \wedge q$, $p \to (r \wedge q)$, $r \to (s \vee t)$, $\neg s$ / $\therefore t$

(15) p, $p \to q$, $s \vee r$, $r \to \neg q$ / $\therefore s$

(16) $(\neg p \vee q) \to r$, $r \to (s \vee t)$, $\neg s \wedge \neg u$, $\neg u \to \neg t$ / $\therefore p$

6A.3 Prove that the following arguments are *not* valid:

(1) $p \wedge \neg q$, $p \to (q \to r)$ / $\therefore \neg r$

(2) $(p \wedge q) \to r$, $\neg q \vee r$ / $\therefore p$

(3) p, $p \to r$, $p \to (q \vee r)$, $\neg q \vee \neg s$ / $\therefore s$

(4) $p \leftrightarrow q$, $q \to r$, $r \vee \neg s$, $\neg s \to q$ / $\therefore s$

6.10 Solutions to Exercise Set 6A

6A.1 – Arguments in words

(1) p: a person is poor, q: he is unhappy, r: he dies young. $p \to q, q \to r \therefore p \to r$: **valid** (syllogism).
(2) p: there is a strike, q: the examination is postponed. $p \to q, \neg p \therefore \neg q$: **not valid** (denying the antecedent; $p =$ F, $q =$ T — the examination may be postponed for another reason).
(3) $p \to q, \neg q \therefore \neg p$: **valid** (Modus Tollens).
(4) $p \to q, \neg p \therefore \neg q$: **not valid** (denying the antecedent).
(5) t: I have talent, w: I work hard, s: I become successful, h: I will be happy. $(t \wedge w) \to s, s \to h \therefore \neg h \to (\neg w \vee \neg t)$. By syllogism $(t \wedge w) \to h$, whose contrapositive is $\neg h \to \neg(t \wedge w) \equiv \neg h \to (\neg t \vee \neg w)$: **valid**.
(6) s: Ravi studies, p: he passes in DM, c: he plays cricket. $s \to p, \neg c \to s, \neg p \therefore c$. MT gives $\neg s$; MT on $\neg c \to s$ gives $\neg\neg c \equiv c$: **valid**.

Conclusion. (1), (3), (5), (6) are valid; (2), (4) are not valid.

6A.2 – Sixteen valid arguments (1)–(8)

(1) $\neg p \to q$ and $\neg q$ give $\neg\neg p \equiv p$ (MT).
(2) $(p \wedge r) \to s \equiv p \to (r \to s)$ (exportation); with p, MP gives $r \to s$.
(3) $\neg p \vee r$ and $\neg r$ give $\neg p$ (DS); $p \vee q$ and $\neg p$ give q (DS).
(4) $p \to r$ and $\neg r$ give $\neg p$ (MT); $\neg q \to p$ and $\neg p$ give q (MT).
(5) p and $p \to \neg q$ give $\neg q$ (MP); $r \to q$ and $\neg q$ give $\neg r$ (MT).
(6) $p \to q$ and $\neg q$ give $\neg p$ (MT); with $\neg r$, conjunction and De Morgan give $\neg p \wedge \neg r \equiv \neg(p \vee r)$.
(7) r and $r \to \neg q$ give $\neg q$ (MP); $p \to q$ and $\neg q$ give $\neg p$ (MT).
(8) p and $p \to q$ give q (MP); $\neg r \to \neg q \equiv q \to r$ gives r (MP).

6A.2 – (continued) (9)–(16)

(9) $p \vee r \equiv \neg p \to r$ and $\neg r \vee s \equiv r \to s$ give $\neg p \to s$; $\neg q \to \neg p$ (contrapositive of $p \to q$) then gives $\neg q \to s$ (syllogism).
(10) This is the example "Using contrapositives and modus tollens" (i).
(11) $\neg q$ gives $\neg p \vee \neg q$ at once (disjunctive amplification). (The other premises are not needed.)
(12) This is the destructive dilemma — example "Constructive and destructive dilemmas" (ii).
(13) $p \vee s$ and $\neg s$ give p (DS); $p \to (q \to r)$ gives $q \to r$ (MP); $t \to q$ and $q \to r$ give $t \to r$ (syllogism), whose contrapositive is $\neg r \to \neg t$.
(14) $p \wedge q$ gives p; $p \to (r \wedge q)$ gives $r \wedge q$, hence r; $r \to (s \vee t)$ gives $s \vee t$; with $\neg s$, DS gives t.
(15) p, $p \to q$ give q (MP); $r \to \neg q$ and q give $\neg r$ (MT); $s \vee r$ and $\neg r$ give s (DS).
(16) This is the example "Longer arguments" (i). ■

6A.3 – Four invalid arguments

For each, an assignment makes every premise true and the conclusion false.
(1) $p = \text{T}, q = \text{F}, r = \text{T}$: $p \wedge \neg q = \text{T}$; $p \to (q \to r) = \text{T} \to \text{T} = \text{T}$; but $\neg r = \text{F}$.
(2) $p = q = r = \text{F}$: $(p \wedge q) \to r = \text{T}$, $\neg q \vee r = \text{T}$, but $p = \text{F}$.
(3) $p = \text{T}, r = \text{T}, q = \text{F}, s = \text{F}$: p, $p \to r$, $p \to (q \vee r)$ and $\neg q \vee \neg s$ are all T, but $s = \text{F}$.
(4) $p = q = r = \text{T}, s = \text{F}$: $p \leftrightarrow q$, $q \to r$, $r \vee \neg s$ and $\neg s \to q$ are all T, but $s = \text{F}$.

Conclusion. None of the four arguments is valid.

7 The Use of Quantifiers

An open statement such as "$x + 3 = 6$" is neither true nor false until we know x. A **quantifier** ("for all" or "for some") turns it into a proposition by saying for how many elements of the universe it must hold.

7.1 Predicates and open statements

- An **open statement** $p(x)$ contains a free variable; it becomes a proposition when x is replaced by an element of the **universe of discourse** U. E.g. $p(x) : x + 3 = 6$ over $\mathbb{Z}$: $p(3)$ is T, $p(2)$ is F.
- In "Ram is a politician", *is a politician* is the **predicate**: with $P(x)$: x is a politician, $P(\text{Ram})$ is a proposition.
- Predicates may have several variables: $G(a, b) : a > b$ gives $G(7, 1) = \text{T}$, $G(2, 3) = \text{F}$.

7.2 Universal and existential quantifiers

Quantifier	Symbol	Read as	True when	False when
Universal	$\forall x$	for all, for every	$p(a)$ is T for every $a \in U$	one counterexample exists
Existential	$\exists x$	there exists, for some	$p(a)$ is T for at least one a	$p(a)$ is F for every a

On a finite universe $U = \{a_1, \dots, a_n\}$:

$$\forall x\, p(x) \equiv p(a_1) \wedge \dots \wedge p(a_n), \qquad \exists x\, p(x) \equiv p(a_1) \vee \dots \vee p(a_n).$$

- A variable governed by a quantifier is **bound**; otherwise it is **free**. In $\forall x\, S(x)$, $S(x)$ is the **scope**.
- For a non-empty universe, $\forall x\, p(x) \Rightarrow \exists x\, p(x)$.

7.3 Translating with quantifiers

English	Correct	Common error
All politicians are sportsmen	$\forall x\,[P(x) \to S(x)]$	$\forall x\,[P(x) \wedge S(x)]$
Some politicians are sportsmen	$\exists x\,[P(x) \wedge S(x)]$	$\exists x\,[P(x) \to S(x)]$
No politician is a sportsman	$\forall x\,[P(x) \to \neg S(x)]$	—
Some politicians are not sportsmen	$\exists x\,[P(x) \wedge \neg S(x)]$	—

Rule of thumb: $\forall$ **pairs with** $\to$, $\exists$ **pairs with** $\wedge$.

7.4 Negating quantified statements

De Morgan's laws for quantifiers

$$\neg[\forall x\, p(x)] \equiv \exists x\, \neg p(x), \qquad \neg[\exists x\, p(x)] \equiv \forall x\, \neg p(x).$$

Procedure: move $\neg$ inward past one quantifier at a time, changing $\forall$ to $\exists$ and $\exists$ to $\forall$; then negate the inner formula using the laws of logic (remember $\neg(p \to q) \equiv p \wedge \neg q$).

7.5 Quantifier laws

Law	Status
$\forall x\,[p(x) \wedge q(x)] \equiv \forall x\,p(x) \wedge \forall x\,q(x)$	Equivalence
$\exists x\,[p(x) \vee q(x)] \equiv \exists x\,p(x) \vee \exists x\,q(x)$	Equivalence
$\forall x\,p(x) \vee \forall x\,q(x) \Rightarrow \forall x\,[p(x) \vee q(x)]$	One direction only
$\exists x\,[p(x) \wedge q(x)] \Rightarrow \exists x\,p(x) \wedge \exists x\,q(x)$	One direction only

Counterexample for the converse of row 3: $U = \{1, 2\}$, $p(x) : x^2 = 4$, $q(x) : x^2 = 1$. Then $\forall x\,[p(x) \vee q(x)]$ is T but $\forall x\,p(x)$ and $\forall x\,q(x)$ are both F.

7.6 Nested quantifiers

Statement ($U = \mathbb{Z}$, $p(x, y) : x + y = 0$)	Meaning	Truth
$\forall x\,\exists y\,p(x, y)$	every integer has an additive inverse	T
$\exists y\,\forall x\,p(x, y)$	one fixed y cancels every x	F
$\forall x\,\forall y\,p(x, y)$	every pair sums to 0	F
$\exists x\,\exists y\,p(x, y)$	some pair sums to 0	T

Two quantifiers of the same kind ($\forall\forall$ or $\exists\exists$) can be swapped; a $\forall$ and an $\exists$ **cannot**: $\exists y\,\forall x\,p \Rightarrow \forall x\,\exists y\,p$, but not conversely.

7.7 Rules of inference for quantifiers

Rule	From	Infer
Universal specification (US)	$\forall x\,p(x)$	$p(a)$ for any particular a
Universal generalisation (UG)	$p(a)$ for an **arbitrary** a	$\forall x\,p(x)$
Existential specification (ES)	$\exists x\,p(x)$	$p(c)$ for some (new) c
Existential generalisation (EG)	$p(a)$ for a particular a	$\exists x\,p(x)$

7.8 Worked examples

Example 7.1 – Symbolising over the integers

$U = \mathbb{Z}$; $p(x)$: $x > 0$, $q(x)$: x is even, $r(x)$: x is a perfect square, $s(x)$: x is divisible by 4, $t(x)$: x is divisible by 5. Write in symbols: (a) At least one integer is even. (b) There is a positive even integer. (c) No even integer is divisible by 5. (d) If x is even and a perfect square, then x is divisible by 4. Negate (d).

Solution. (a) $\exists x\,q(x)$ (b) $\exists x\,[p(x) \wedge q(x)]$ (c) $\forall x\,[q(x) \to \neg t(x)]$ (d) $\forall x\,[(q(x) \wedge r(x)) \to s(x)]$. Negation of (d): $\exists x\,[q(x) \wedge r(x) \wedge \neg s(x)]$ — "some even perfect square is not divisible by 4".

Example 7.2 – Negating a conditional statement

Negate: "For every integer x, if x is odd then $x^2 - 1$ is even."

Solution. Given: $\forall x\,[p(x) \to q(x)]$. Negation: $\exists x\,\neg[p(x) \to q(x)] \equiv \exists x\,[p(x) \wedge \neg q(x)]$: "There is an integer x that is odd and $x^2 - 1$ is not even."

Example 7.3 – Negating nested quantifiers

Negate $\forall x\,\exists y\,[\,p(x,y)\wedge q(x,y)\to r(x,y)\,]$.

Solution.

$$\neg\forall x\,\exists y\,[\cdots]\equiv\exists x\,\forall y\,\neg[(p\wedge q)\to r]\equiv\exists x\,\forall y\,[\,p(x,y)\wedge q(x,y)\wedge\neg r(x,y)\,].$$

Example 7.4 – A quantified syllogism

"All men are mortal. Sachin is a man. Therefore Sachin is mortal."

Solution. $M(x)$: x is a man; $D(x)$: x is mortal; s = Sachin. Premises $\forall x\,[M(x)\to D(x)]$ and $M(s)$. By US, $M(s)\to D(s)$; by modus ponens, $D(s)$. **Valid.**

Common pitfall

One counterexample disproves a $\forall$ statement, but one example never proves it. When negating, students flip $\forall/\exists$ correctly yet forget to change $\wedge$ to $\vee$ (and vice versa) between the parts of a compound statement.

7.9 Open statements and quantifiers in detail

Declarative sentences such as (1) $x+3=6$ (2) $x^2<10$ (3) x divides 4 (4) $x=\sqrt{2}$ are not propositions unless x is specified. They are **open statements** (open sentences), and the unspecified symbols such as x are **free variables**. Sentence (1) becomes a proposition when x is replaced by any element of $\mathbb{R}$ — true for 3, false for 5. We say $\mathbb{R}$ is a **universe** (universe of discourse) for x; $\mathbb{R}$ is also a universe for (2)–(4), and so is $\mathbb{Z}$.

Open statements in x are written $p(x)$, $q(x)$, ... If U is the universe and $a\in U$, the proposition obtained by replacing x by a is written $p(a)$. For example, with $U=\mathbb{Z}$ and $p(x)$: $x+3=6$, $p(2)$ is the (false) proposition "$2+3=6$"; with $U=\mathbb{R}$ and $q(x)$: $x^2<10$, $q(\sqrt{2})$ is the (true) proposition "$2<10$". An open statement becomes a proposition only when x is replaced by a chosen element of the universe. Compound open statements $\neg p(x)$, $p(x)\wedge q(x)$, $p(x)\vee q(x)$, $p(x)\to q(x)$, $p(x)\leftrightarrow q(x)$ are formed and evaluated exactly as compound propositions.

Example 7.5 – Values of open statements

The universe is $\mathbb{Z}$; $p(x)$: $x\le 3$, $q(x)$: $x+1$ is odd, $r(x)$: $x>0$. Write down the truth values of (i) $p(2)$ (ii) $\neg q(4)$ (iii) $p(-1)\wedge q(1)$ (iv) $\neg p(3)\vee r(0)$ (v) $p(0)\to q(0)$ (vi) $p(1)\leftrightarrow\neg q(2)$ (vii) $p(4)\vee(q(1)\wedge r(2))$ (viii) $p(2)\wedge(q(0)\vee\neg r(2))$.

Solution. (i) $p(2)$: "$2\le 3$" — **T**. (ii) $q(4)$: "$4+1$ is odd" is true, so $\neg q(4)$ is **F**. (iii) $p(-1)$ is T but $q(1)$ ("2 is odd") is F: **F**. (iv) $p(3)$ is T so $\neg p(3)$ is F, and $r(0)$ is F: **F**. (v) $p(0)$ and $q(0)$ are both T: **T**. (vi) $p(1)$ is T and $q(2)$ is T, so $\neg q(2)$ is F: **F**. (vii) $p(4)$ is F; $q(1)$ is F so $q(1)\wedge r(2)$ is F: **F**. (viii) $p(2)$ is T; $q(0)$ is T, so $q(0)\vee\neg r(2)$ is T: **T**.

Quantifiers. In "All squares are rectangles", "For every integer x, x^2 is a non-negative integer", "Some determinants are equal to zero" and "There exists a real number whose square is equal to itself", the words *all, every, some, there exists* express a quantity; they are **quantifiers**. With S the set of all squares and $p(x)$: x is a rectangle, the first reads "For all $x\in S$, $p(x)$", written $\forall x\in S,\ p(x)$. The phrases "for all", "for every", "for each", "for any" are all written $\forall$ — the **universal quantifier**. With D the set of all determinants and $p(x)$: x is equal to zero, the third reads $\exists x\in D,\ p(x)$; "for some", "there exists", "for at least one" are all written $\exists$ — the **existential quantifier**. A proposition $\forall x\in S,\ p(x)$ or $\exists x\in S,\ p(x)$ is a **quantified statement**; when the universe is clear we write $\forall x,\ p(x)$ or $\exists x,\ p(x)$. The variable in a quantified statement is a **bound variable**. Some statements contain a quantifier *implicitly*: "If a triangle is equilateral then it is isosceles" means "For every triangle T, if T is equilateral then T is isosceles", and "9 is the square of an integer" means "There exists an integer whose square is 9".

Example 7.6 – Symbolic form

For the universe of all integers let $p(x)$: $x > 0$, $q(x)$: x is even, $r(x)$: x is a perfect square, $s(x)$: x is divisible by 3, $t(x)$: x is divisible by 7. Write in symbolic form: (i) At least one integer is even. (ii) There exists a positive integer that is even. (iii) Some even integers are divisible by 3. (iv) Every integer is either even or odd. (v) If x is even and a perfect square, then x is not divisible by 3. (vi) If x is odd or is not divisible by 7, then x is divisible by 3.

Solution. (i) $\exists x,\ q(x)$ (ii) $\exists x,\ [p(x) \wedge q(x)]$ (iii) $\exists x,\ [q(x) \wedge s(x)]$ (iv) $\forall x,\ [q(x) \veebar \neg q(x)]$ (v) $\forall x,\ [\{q(x) \wedge r(x)\} \to \neg s(x)]$ (vi) $\forall x,\ [\{\neg q(x) \vee \neg t(x)\} \to s(x)]$.

Truth value of a quantified statement

Rule 1: $\forall x \in S,\ p(x)$ is true only when $p(x)$ is true for each $x \in S$.
Rule 2: $\exists x \in S,\ p(x)$ is false only when $p(x)$ is false for every $x \in S$.

So to show that $\forall x \in S,\ p(x)$ is *false* it is enough to exhibit one $a \in S$ with $p(a)$ false; such an a is a **counterexample**. To show that $\exists x \in S,\ p(x)$ is *true* it is enough to exhibit one $a \in S$ with $p(a)$ true. The four quantified statements above are all true; "All rectangles are squares", "For every integer x, x^2 is a positive integer" and "The squares of some real numbers are negative" are false.

Two rules of inference

Rule 3 (Universal Specification): if $p(x)$ is true for all x in a universe S and $a \in S$, then $p(a)$ is true.
Rule 4 (Universal Generalization): if $p(x)$ is proved to be true for any (arbitrary) x chosen from S, then $\forall x \in S,\ p(x)$ is true.

Logical equivalence. Two quantified statements are logically equivalent when they have the same truth values in all possible situations. The following are easy to prove:

(i) $\forall x, [p(x) \wedge q(x)] \Leftrightarrow (\forall x, p(x)) \wedge (\forall x, q(x))$ (ii) $\exists x, [p(x) \vee q(x)] \Leftrightarrow (\exists x, p(x)) \vee (\exists x, q(x))$
(iii) $\exists x, [p(x) \to q(x)] \Leftrightarrow \exists x, [\neg p(x) \vee q(x)]$.

A statement "$\forall x,\ \neg p(x)$" is taken to be equivalent to "For no x, $p(x)$". For example, "For every integer x, x^2 is non-negative" $\Leftrightarrow$ "For no integer x, x^2 is negative"; "Every unit matrix is non-singular" $\Leftrightarrow$ "No unit matrix is singular".

Rule 5: negation of a quantified statement

To negate a quantified statement, change the quantifier from universal to existential and vice versa, and replace the open statement by its negation:

$$\neg\{\forall x,\ p(x)\} \equiv \exists x,\ \{\neg p(x)\}, \qquad \neg\{\exists x,\ p(x)\} \equiv \forall x,\ \{\neg p(x)\}.$$

- "All equilateral triangles are isosceles" is $\forall x \in T,\ p(x)$ (T: equilateral triangles, $p(x)$: x is isosceles). Its negation $\exists x \in T,\ \neg p(x)$ reads "Some equilateral triangles are not isosceles".
- "Some integers are even" is $\exists x \in \mathbb{Z},\ p(x)$. Its negation $\forall x \in \mathbb{Z},\ \neg p(x)$ reads "For every integer x, x is not even" — "No integer is even" or "All integers are noneven".
- "No even integer is divisible by 7" is $\forall x \in E,\ \neg p(x)$ (E: even integers, $p(x)$: x is divisible by 7). Its negation $\exists x \in E,\ p(x)$ reads "Some even integer is divisible by 7".

Note. For a statement "For some x, $p(x)$", the negation can be expressed as "For no x, $p(x)$", and vice versa.

Example 7.7 – Words and truth values

For the open statements of the previous symbolic-form example, express in words and give the truth value: (i) $\forall x, [r(x) \to p(x)]$ (ii) $\exists x, [s(x) \wedge \neg q(x)]$ (iii) $\forall x, [\neg r(x)]$ (iv) $\forall x, [r(x) \vee t(x)]$.

Solution. (i) For any integer x, if x is a perfect square, then $x > 0$. — **False** (take $x = 0$).
(ii) For some integer x, x is divisible by 3 and x is not even (there is an integer divisible by 3 that is not even). — **True** (take $x = 9$).
(iii) For any integer x, x is not a perfect square (no integer is a perfect square). — **False**.
(iv) For any integer x, x is a perfect square or x is divisible by 7. — **False** (take $x = 8$).

Example 7.8 – Converse, inverse and contrapositive of a quantified statement

With $\mathbb{R}$ as universe let $p(x)$: $|x| > 3$, $q(x)$: $x > 3$. Find the truth value of $\forall x, [p(x) \to q(x)]$. Also write down its converse, inverse and contrapositive, and find their truth values.

Solution. $p(-4) \equiv |-4| > 3$ is true and $q(-4) \equiv -4 > 3$ is false, so $p(x) \to q(x)$ is false for $x = -4$: the statement is **false** ($x = -4$ is a counterexample).
Converse $\forall x, [q(x) \to p(x)]$: "For every real number x, if $x > 3$ then $|x| > 3$", i.e. "Every real number greater than 3 has absolute value greater than 3" — **true**.
Inverse $\forall x, [\neg p(x) \to \neg q(x)]$: "For every real number x, if $|x| \leq 3$ then $x \leq 3$" — **true** (the converse and inverse are equivalent).
Contrapositive $\forall x, [\neg q(x) \to \neg p(x)]$: "Every real number which is less than or equal to 3 has magnitude less than or equal to 3" — **false** (it is equivalent to the original statement; again $x = -4$).

Example 7.9 – Six statements over $\mathbb{R}$

With $\mathbb{R}$ as universe let $p(x)$: $x \geq 0$, $q(x)$: $x^2 \geq 0$, $r(x)$: $x^2 - 3x - 4 = 0$, $s(x)$: $x^2 - 3 > 0$. Determine the truth values of (i) $\exists x,\ p(x) \wedge q(x)$ (ii) $\forall x,\ p(x) \to q(x)$ (iii) $\forall x,\ q(x) \to s(x)$ (iv) $\forall x,\ r(x) \vee s(x)$ (v) $\exists x,\ p(x) \wedge r(x)$ (vi) $\forall x,\ r(x) \to p(x)$.

Solution. This is Problem 5(a) of the solved university problems: (i) T (ii) T (iii) F (iv) F (v) T (vi) F. For example, (iv): $x^2 - 3x - 4 = (x-4)(x+1)$, so $r(x)$ is true only for $x = 4$ or $x = -1$; for $x = 1$ both $r(x)$ and $s(x)$ are false.

Example 7.10 – A two-element universe

Let $p(x)$: $x^2 - 7x + 10 = 0$, $q(x)$: $x^2 - 2x - 3 = 0$, $r(x)$: $x < 0$. Determine the truth of the following when the universe U contains only the integers 2 and 5; give a counterexample or explanation for a false statement: (i) $\forall x,\ p(x) \to \neg r(x)$ (ii) $\forall x,\ q(x) \to r(x)$ (iii) $\exists x,\ q(x) \to r(x)$ (iv) $\exists x,\ p(x) \to r(x)$.

Solution. $x^2 - 7x + 10 = (x-5)(x-2)$, so $p(x)$ is true for both elements of $U = \{2, 5\}$. $x^2 - 2x - 3 = (x-3)(x+1)$ is zero only for $x = 3, -1 \notin U$, so $q(x)$ is false for all $x \in U$. Obviously $r(x)$ is false for all $x \in U$.
(i) $p(x)$ and $\neg r(x)$ are true for all $x \in U$: **true**. (ii) $q(x)$ is false for all $x \in U$: **true**. (iii) $q(2)$ and $r(2)$ are false, so $q(2) \to r(2)$ is true: **true**. (iv) $p(x)$ is true but $r(x)$ false for every $x \in U$, so $p(x) \to r(x)$ is false for every x: **false**.

Example 7.11 – Negate and simplify

Negate and simplify: (i) $\exists x, [p(x) \vee q(x)]$ (ii) $\forall x, [p(x) \wedge \neg q(x)]$ (iii) $\forall x, [p(x) \to q(x)]$ (iv) $\exists x, [\{p(x) \vee q(x)\} \to r(x)]$.

Solution. (i) $\neg\{\exists x, [p(x) \vee q(x)]\} \equiv \forall x, [\neg\{p(x) \vee q(x)\}] \equiv \forall x, [\neg p(x) \wedge \neg q(x)]$.
(ii) $\neg\{\forall x, [p(x) \wedge \neg q(x)]\} \equiv \exists x, [\neg\{p(x) \wedge \neg q(x)\}] \equiv \exists x, [\neg p(x) \vee q(x)]$.
(iii) $\neg\{\forall x, [p(x) \to q(x)]\} \equiv \exists x, [\neg\{p(x) \to q(x)\}] \equiv \exists x, [p(x) \wedge \neg q(x)]$.
(iv) $\neg\{\exists x, [(p(x) \vee q(x)) \to r(x)]\} \equiv \forall x, [\neg\{(p(x) \vee q(x)) \to r(x)\}] \equiv \forall x, [\{p(x) \vee q(x)\} \wedge \neg r(x)]$.

Example 7.12 – Negating an existential statement

Let $\mathbb{Z}$ be the universe, $p(x)$: $2x+1=5$, $q(x)$: $x^2=9$. Obtain the negation of $\exists x \in \mathbb{Z}, [p(x) \wedge q(x)]$ and express it in words.

Solution. The negation is $\forall x \in \mathbb{Z}, [\neg p(x) \vee \neg q(x)]$: "For all integers x, $2x+1 \neq 5$ or $x^2 \neq 9$."

Example 7.13 – An implicit universal statement

Let the universe be all integers, $p(x)$: x is odd, $q(x)$: x^2-1 is even. Express "For any x, if x is odd, then x^2-1 is even" in symbolic form and negate it.

Solution. Symbolically $\forall x \in \mathbb{Z}, [p(x) \to q(x)]$; the negation is $\exists x \in \mathbb{Z}, [p(x) \wedge \neg q(x)]$: "For some integer x, x is odd and x^2-1 is not even."

Example 7.14 – Integers and rationals

Write in symbolic form and find the negation: "All integers are rational numbers and some rational numbers are not integers."

Solution. With $p(x)$: x is a rational number, $q(x)$: x is an integer, the proposition is $\{\forall x \in \mathbb{Z},\, p(x)\} \wedge \{\exists x \in \mathbb{Q},\, \neg q(x)\}$. Its negation is $\{\exists x \in \mathbb{Z},\, \neg p(x)\} \vee \{\forall x \in \mathbb{Q},\, q(x)\}$: "Some integers are not rational numbers or every rational number is an integer." The given proposition is true and its negation is false. (Problem 5(b) of the solved university problems.)

Example 7.15 – Right-angled triangles

Write in symbolic form and find the negation: "If all triangles are right-angled, then no triangle is equiangular."

Solution. Let T be the set of all triangles, $p(x)$: x is right-angled, $q(x)$: x is equiangular. The proposition is $\{\forall x \in T,\, p(x)\} \to \{\forall x \in T,\, \neg q(x)\}$. Its negation is $\{\forall x \in T,\, p(x)\} \wedge \{\exists x \in T,\, q(x)\}$: "All triangles are right-angled and some triangles are equiangular."

Example 7.16 – Four negations

Write down the negation of each statement: (i) For all integers n, if n is not divisible by 2, then n is odd. (ii) If k, m, n are any integers where $(k-m)$ and $(m-n)$ are odd, then $(k-n)$ is even. (iii) For all real numbers x, if $|x-3|<7$, then $-4<x<10$. (iv) If x is a real number where $x^2>16$, then $x<-4$ or $x>4$.

Solution. (i) $\forall n \in \mathbb{Z}, \neg p(n) \to q(n)$ with $p(n)$: n is divisible by 2, $q(n)$: n is odd. Negation $\exists n \in \mathbb{Z}, \neg p(n) \wedge \neg q(n)$: "For some integer n, n is not divisible by 2 and n is not odd."
(ii) $\forall k, m, n \in \mathbb{Z}, [p \wedge q] \to r$ with p: $k-m$ is odd, q: $m-n$ is odd, r: $k-n$ is even. Negation $\exists k, m, n \in \mathbb{Z}, [p \wedge q] \wedge \neg r$: "There exist integers k, m, n such that $k-m$, $m-n$ are odd and $k-n$ is not even."
(iii) $\forall x \in \mathbb{R}, p(x) \to q(x)$ with $p(x)$: $|x-3|<7$, $q(x)$: $-4<x<10$, i.e. $x \in (-4, 10)$. Negation $\exists x \in \mathbb{R}, p(x) \wedge \neg q(x)$: "For some real number x, $|x-3|<7$ and $x \notin (-4, 10)$."
(iv) $\forall x \in \mathbb{R}, p(x) \to q(x) \vee r(x)$ with $p(x)$: $x^2>16$, $q(x)$: $x<-4$, $r(x)$: $x>4$. Negation $\exists x \in \mathbb{R}, [p(x) \wedge \neg q(x) \wedge \neg r(x)]$: "For some real number x, $x^2>16$ and $x \geq -4$ and $x \leq 4$."

7.10 Exercise Set 7A — open statements and quantifiers

7A.1 The universe is the set of all integers; $p(x)$: $x>3$, $q(x)$: $x+1$ is even, $r(x)$: $x \leq 0$. Write down the truth values of (i) $p(2)$ (ii) $p(3) \vee \neg r(3)$ (iii) $\neg p(4) \wedge \neg q(5)$ (iv) $\neg\{p(2) \vee q(-3)\}$ (v) $\neg p(6) \wedge q(-6)$ (vi) $p(4) \to q(0) \wedge r(-1)$ (vii) $p(4) \leftrightarrow [q(1) \wedge r(0)]$ (viii) $[p(1) \wedge q(1)] \to \neg r(1)$.

7A.2 Write in symbolic form: (i) Some integers are divisible by 5. (ii) There exists a matrix whose transpose is itself. (iii) All real numbers are complex numbers. (iv) Every element of a group has an inverse. (v) No real number is greater than its square. (vi) At least one parallelogram is a rhombus. (vii) There is an integer which is not a perfect square. (viii) Every real number is rational or irrational but not both. (ix) An equilateral triangle has three angles of 60°, and conversely. (x) Every rational number is a real number and not every real number is a rational number.

7A.3 Let $\mathbb{Z}$ be the set of integers, $p(x)$: x is even, $q(x)$: x is a prime number. Write in words: (i) $\forall x,\ p(x)$ (ii) $\exists x,\ q(x)$ (iii) $\forall x,\ \neg q(x)$ (iv) $\exists x,\ p(x) \wedge q(x)$ (v) $[\forall x,\ p(x)] \to [\exists x,\ \neg q(x)]$.

7A.4 Let $p(x)$: $x^2 = 2x$ and $q(x)$: $x^3 = 4x$, with the integers as universe. Write down the truth values of (i) $\forall x,\ p(x)$ (ii) $\exists x,\ q(x)$ (iii) $\forall x,\ \neg p(x)$ (iv) $\exists x,\ \neg q(x)$ (v) $\forall x,\ p(x) \wedge q(x)$ (vi) $\exists x,\ p(x) \wedge q(x)$ (vii) $\forall x,\ p(x) \vee q(x)$ (viii) $\exists x,\ p(x) \vee q(x)$.

7A.5 Find the truth value of each quantified statement in the "Symbolic form" example above; for each false statement give a counterexample.

7A.6 Let $p(x)$: $x^2 - 8x + 15 = 0$, $q(x)$: x is odd, $r(x)$: $x > 0$, with the integers as universe. Determine the truth or falsity of the following; give a counterexample for a false one: (i) $\forall x, [p(x) \to q(x)]$ (ii) $\forall x, [q(x) \to p(x)]$ (iii) $\exists x, [p(x) \to q(x)]$ (iv) $\exists x, [q(x) \to p(x)]$ (v) $\exists x, [r(x) \to p(x)]$ (vi) $\forall x, [\neg q(x) \to \neg r(x)]$ (vii) $\exists x, [p(x) \to \{q(x) \wedge r(x)\}]$ (viii) $\forall x, [\{p(x) \vee q(x)\} \to r(x)]$.

7A.7 Write down the converse, inverse and contrapositive of each statement (universe: the real numbers), and indicate their truth values: (i) $\forall x, [(x > 3) \to (x^2 > 9)]$ (ii) $\forall x, [\{(x^2 + 4x - 21) > 0\} \to \{(x > 3) \vee (x < -7)\}]$.

7A.8 Let $p(x)$: $|x| > 3$, $q(x)$: $x > 3$, $r(x)$: $x < -3$, with the real numbers as universe. Prove that $\forall x, [p(x) \leftrightarrow \{r(x) \vee q(x)\}]$ is true.

7A.9 Write down the negations of (i) $\{\forall x, p(x)\} \vee \{\forall x, \neg q(x)\}$ (ii) $\{\exists x, \neg p(x)\} \wedge \{\forall x, q(x)\}$ (iii) $\forall x,\ p(x) \to q(x)$ (iv) $\exists x,\ \neg p(x) \to q(x)$ (v) $\{\exists x, p(x)\} \to \{\exists x, \neg q(x)\}$ (vi) $\{\forall x, p(x)\} \to \{\exists x, q(x)\}$.

7A.10 Write down the negations of (i) All even numbers are multiples of 4. (ii) At least one parallelogram is not a square. (iii) No real number is greater than its square. (iv) For all positive integers n, $(n^2 + 41n + 41)$ is a prime number. (v) There is some integer k for which $12 = 3k$. (vi) Some straight lines are parallel or all straight lines intersect. (vii) All rational numbers are real and some real numbers are not rational. (viii) For all real numbers x and y, if $x^2 > y^2$, then $x > y$. (ix) There exist real numbers x and y such that x and y are rational but $x + y$ is irrational. (x) There exist odd integers whose product is odd.

7A.11 Prove: (i) $\forall x, [p(x) \wedge q(x)] \Leftrightarrow \forall x, p(x) \wedge \forall x, q(x)$ (ii) $\exists x, [p(x) \vee q(x)] \Leftrightarrow \exists x, p(x) \vee \exists x, q(x)$ (iii) $\exists x, [p(x) \to q(x)] \Leftrightarrow \exists x, [\neg p(x) \vee q(x)]$.

7A.12 Prove: (i) $\neg[\forall x, \neg p(x)] \Leftrightarrow \exists x, p(x)$ (ii) $\neg[\exists x, \neg p(x)] \Leftrightarrow \forall x, p(x)$ (iii) $\exists x, [p(x) \to q(x)] \Leftrightarrow \forall x, p(x) \to \exists x, q(x)$ (iv) $[\exists x, p(x) \to \forall x, q(x)] \Rightarrow \forall x, [p(x) \to q(x)]$; is the converse true? (v) $\forall x, [p(x) \wedge \{q(x) \wedge r(x)\}] \Leftrightarrow \forall x, [\{p(x) \wedge q(x)\} \wedge r(x)]$.

7A.13 Prove that the following are tautologies: (i) $[\{\forall x, p(x)\} \vee \{\forall x, q(x)\}] \to \forall x, \{p(x) \vee q(x)\}$ (ii) $[\exists x, \{p(x) \wedge q(x)\}] \to [\{\exists x, p(x)\} \wedge \{\exists x, q(x)\}]$.

7.11 Solutions to Exercise Set 7A

7A.1 – Values of open statements

(i) $2 > 3$: **F**. (ii) $p(3)$ is F; $r(3)$ is F so $\neg r(3)$ is T: **T**. (iii) $p(4)$ is T so $\neg p(4)$ is F: **F**. (iv) $p(2)$ is F; $q(-3)$: "-2 is even" is T; so $\neg\{\text{F} \vee \text{T}\}$ is **F**. (v) $\neg p(6)$ is F: **F**. (vi) $p(4)$ is T; $q(0)$: "1 is even" is F, so $q(0) \wedge r(-1)$ is F: **F**. (vii) $p(4)$ is T; $q(1)$ is T and $r(0)$ is T: T $\leftrightarrow$ T, **T**. (viii) $p(1)$ is F, so the antecedent is F: **T**.

7A.2 – Symbolic form

(i) $\exists x \in \mathbb{Z},\ p(x)$; $p(x)$: x is divisible by 5. (ii) $\exists x \in M,\ p(x)$; M: all matrices, $p(x)$: x is the transpose of itself. (iii) $\forall x \in \mathbb{R},\ p(x)$; $p(x)$: x is a complex number. (iv) $\forall x \in G,\ p(x)$; G a group, $p(x)$: x has an inverse. (v) $\forall x \in \mathbb{R},\ \neg p(x)$; $p(x)$: x is greater than its square. (vi) $\exists x \in S,\ p(x)$; S: all parallelograms, $p(x)$: x is a rhombus. (vii) $\exists x \in \mathbb{Z},\ \neg p(x)$; $p(x)$: x is a perfect square. (viii) $\forall x \in \mathbb{R},\ p(x) \veebar q(x)$; $p(x)$: x is rational, $q(x)$: x is irrational. (ix) $\forall x \in T,\ [p(x) \leftrightarrow q(x)]$; T: all triangles, $p(x)$: x is equilateral, $q(x)$: x has three angles of 60°. (x) $\{\forall x, [p(x) \to q(x)]\} \wedge [\neg\{\forall x, [q(x) \to p(x)]\}]$; $p(x)$: x is a rational number, $q(x)$: x is a real number.

7A.3 – In words

(i) Every integer is even. (ii) Some integers are prime numbers. (iii) No integer is a prime number. (iv) Some integers are even and prime. (v) If every integer is even, then some integers are not prime.

7A.4 – Two polynomial conditions

$x^2 = 2x \iff x(x-2) = 0 \iff x \in \{0, 2\}$; $x^3 = 4x \iff x(x-2)(x+2) = 0 \iff x \in \{0, 2, -2\}$.
(i) **F** ($x = 1$). (ii) **T** ($x = 0$). (iii) **F** ($x = 0$). (iv) **T** ($x = 1$). (v) **F** ($x = 1$). (vi) **T** ($x = 2$). (vii) **F** ($x = 1$). (viii) **T** ($x = 2$).

7A.5 – Truth values in the "Symbolic form" example

(i) **T** ($x = 2$). (ii) **T** ($x = 2$). (iii) **T** ($x = 6$). (iv) **T** (every integer is either even or odd, not both). (v) **F**: $x = 36$ is even and a perfect square but divisible by 3. (vi) **F**: $x = 11$ is odd (and not divisible by 7) but not divisible by 3.

7A.6 – A quadratic condition

$x^2 - 8x + 15 = (x-3)(x-5)$, so $p(x)$ is true only for $x = 3, 5$.
(i) **T**: 3 and 5 are odd. (ii) **F**: $x = 7$ is odd but $p(7)$ is false. (iii) **T** ($x = 3$). (iv) **T** ($x = 3$). (v) **T** ($x = 3$, or any $x \leq 0$). (vi) **F**: $x = 2$ is not odd but is positive. (vii) **T** ($x = 3$: odd and positive). (viii) **F**: $x = -1$ is odd but not positive.

7A.7 – Converse, inverse, contrapositive over $\mathbb{R}$

(i) Original: **T**. Converse $\forall x, [(x^2 > 9) \to (x > 3)]$: **F** ($x = -4$). Inverse $\forall x, [(x \leq 3) \to (x^2 \leq 9)]$: **F** ($x = -4$). Contrapositive $\forall x, [(x^2 \leq 9) \to (x \leq 3)]$: **T**.
(ii) $x^2 + 4x - 21 = (x+7)(x-3) > 0 \iff x > 3$ or $x < -7$. Original **T**. Converse $\forall x, [\{(x > 3) \vee (x < -7)\} \to (x^2 + 4x - 21 > 0)]$: **T**. Inverse $\forall x, [(x^2 + 4x - 21 \leq 0) \to \{(x \leq 3) \wedge (x \geq -7)\}]$: **T**. Contrapositive $\forall x, [\{(x \leq 3) \wedge (x \geq -7)\} \to (x^2 + 4x - 21 \leq 0)]$: **T**.

7A.8 – $|x| > 3$

For any real x: if $x \geq 0$ then $|x| = x$, and $|x| > 3 \iff x > 3$; if $x < 0$ then $|x| = -x$, and $|x| > 3 \iff x < -3$. Hence $|x| > 3 \iff (x < -3) \vee (x > 3)$, i.e. $p(x) \leftrightarrow \{r(x) \vee q(x)\}$ is true for every x. By Universal Generalization, the statement is true. ■

7A.9 – Negations in symbols

(i) $\{\exists x, \neg p(x)\} \wedge \{\exists x, q(x)\}$ (ii) $\{\forall x, p(x)\} \vee \{\exists x, \neg q(x)\}$ (iii) $\exists x,\ p(x) \wedge \neg q(x)$ (iv) $\forall x,\ \neg p(x) \wedge \neg q(x)$ (v) $\{\exists x, p(x)\} \wedge \{\forall x, q(x)\}$ (vi) $\{\forall x, p(x)\} \wedge \{\forall x, \neg q(x)\}$.

7A.10 – Negations in words

(i) Some even numbers are not multiples of 4. (ii) All parallelograms are squares. (iii) At least one real number is greater than its square. (iv) There is at least one positive integer n for which $n^2 + 41n + 41$ is not prime. (v) For all integers k, $12 \neq 3k$ (there is no integer k for which $12 = 3k$). (vi) All straight lines are non-parallel and some straight lines do not intersect. (vii) Some rational numbers are not real or all real numbers are rational. (viii) There exist real numbers x and y such that $x^2 > y^2$ and $x \leq y$. (ix) For all real numbers x, y, if x and y are rational then $x + y$ is rational. (x) The product of any two odd integers is even.

7A.11 – Quantifier equivalences

(i) $\forall x, [p(x) \wedge q(x)]$ is true $\iff$ for every a, $p(a)$ and $q(a)$ are true $\iff$ $p(a)$ is true for every a and $q(a)$ is true for every a $\iff$ $\forall x, p(x) \wedge \forall x, q(x)$ is true.
(ii) $\exists x, [p(x) \vee q(x)]$ is true $\iff$ for some a, $p(a)$ or $q(a)$ is true $\iff$ $p(a)$ is true for some a, or $q(b)$ is true for some b $\iff$ $\exists x, p(x) \vee \exists x, q(x)$ is true.
(iii) $p(x) \to q(x) \equiv \neg p(x) \vee q(x)$ for each x, so the two existential statements have the same truth value. ■

7A.12 – More equivalences

(i) By Rule 5, $\neg[\forall x, \neg p(x)] \equiv \exists x, \neg\neg p(x) \equiv \exists x, p(x)$. (ii) Similarly $\neg[\exists x, \neg p(x)] \equiv \forall x, p(x)$.
(iii) $\exists x, [p(x) \to q(x)] \equiv \exists x, [\neg p(x) \vee q(x)] \equiv \exists x, \neg p(x) \vee \exists x, q(x)$ (7A.11(ii)) $\equiv \neg\forall x, p(x) \vee \exists x, q(x) \equiv \forall x, p(x) \to \exists x, q(x)$.
(iv) *Proof of* $\Rightarrow$: assume $\exists x, p(x) \to \forall x, q(x)$ and let a be arbitrary. If $p(a)$ is true, then $\exists x, p(x)$ is true, so $\forall x, q(x)$ is true and in particular $q(a)$ is true. Hence $p(a) \to q(a)$; by UG, $\forall x, [p(x) \to q(x)]$. *The converse is false.* Take the universe $\{1, 2\}$ and $p(x) = q(x)$: "$x = 1$". Then $\forall x, [p(x) \to q(x)]$ is true, but $\exists x, p(x)$ is true while $\forall x, q(x)$ is false, so $\exists x, p(x) \to \forall x, q(x)$ is false.
(v) For each x, $p(x) \wedge \{q(x) \wedge r(x)\} \equiv \{p(x) \wedge q(x)\} \wedge r(x)$ (associative law), so the universal statements are equivalent. ■

7A.13 – Two quantified tautologies

(i) Suppose $\{\forall x, p(x)\} \vee \{\forall x, q(x)\}$ is true. If $\forall x, p(x)$ is true, then for every a, $p(a)$ and hence $p(a) \vee q(a)$ is true; similarly if $\forall x, q(x)$ is true. Either way $\forall x, \{p(x) \vee q(x)\}$ is true, so the conditional is never false.
(ii) If $\exists x, \{p(x) \wedge q(x)\}$ is true, there is a with $p(a)$ and $q(a)$ true; then $\exists x, p(x)$ and $\exists x, q(x)$ are both true. So the conditional is never false. ■
(The converses are not tautologies — see the next subsection.)

7.12 Logical implication involving quantifiers

A quantified statement P **logically implies** a quantified statement Q, written $P \Rightarrow Q$, if Q is true whenever P is true. Given quantified statements $P_1, \ldots, P_n$ and Q, Q is a **valid conclusion** from the premises $P_1, \ldots, P_n$ (and $P_1 \wedge \ldots \wedge P_n \to Q$ is a **valid argument**) if Q is true whenever each P_i is true, i.e. if $P_1 \wedge \ldots \wedge P_n \Rightarrow Q$. Such arguments are analysed with the laws of logic and the rules of inference, together with Rules 3 and 4 (Universal Specification and Generalization).

Example 7.17 – Universal to existential

Prove: (i) $\forall x, p(x) \Rightarrow \exists x, p(x)$ (ii) $\forall x, [p(x) \vee q(x)] \Rightarrow \forall x, p(x) \vee \exists x, q(x)$.

Solution. Let S denote the (non-empty) universe.
(i) $\forall x, p(x)$ means $p(x)$ is true for every $x \in S$; so $p(a)$ is true for any particular $a \in S$, i.e. $p(x)$ is true

for some $x \in S$: $\exists x, p(x)$.
(ii) Assume $\forall x, [p(x) \vee q(x)]$. If $\forall x, p(x)$ is true we are done. Otherwise there is $a \in S$ with $p(a)$ false; since $p(a) \vee q(a)$ is true, $q(a)$ is true, so $\exists x, q(x)$ is true. Either way $\forall x, p(x) \vee \exists x, q(x)$ is true.

Example 7.18 – The converse fails

Prove that $\exists x, [p(x) \wedge q(x)] \Rightarrow \exists x, p(x) \wedge \exists x, q(x)$. Is the converse true?

Solution. $\exists x, [p(x) \wedge q(x)] \Rightarrow p(a) \wedge q(a)$ for some $a \in S \Rightarrow p(a)$ for some a and $q(a)$ for some a $\Rightarrow \exists x, p(x) \wedge \exists x, q(x)$.
Next, $\exists x, p(x) \Rightarrow p(a)$ for some $a \in S$ and $\exists x, q(x) \Rightarrow q(b)$ for some $b \in S$; but b need not be the same as a, so $p(a) \wedge q(b) \not\Rightarrow p(a) \wedge q(a)$. Thus $\exists x, p(x) \wedge \exists x, q(x) \not\Rightarrow \exists x, [p(x) \wedge q(x)]$: the converse is **not** necessarily true. (Over the integers: some integer is even and some integer is odd, but no integer is both.) Consequently $\exists x, [p(x) \wedge q(x)] \not\Leftrightarrow \exists x, p(x) \wedge \exists x, q(x)$.

Example 7.19 – A counterexample to a converse

Prove that $[\forall x, p(x) \vee \forall x, q(x)] \Rightarrow \forall x, [p(x) \vee q(x)]$, and through a counterexample show that the converse is not true.

Solution. The implication is 7A.13(i). For the converse take $p(x)$: $x^2 - 4 = 0$, $q(x)$: $x^2 - 1 = 0$ with $S = \{1, 2\}$. For $x = 1$, $p(x)$ is false but $q(x)$ is true; for $x = 2$, $p(x)$ is true. So $p(x) \vee q(x)$ is true for every $x \in S$: $\forall x, [p(x) \vee q(x)]$ is true. But $\forall x, p(x)$ is false ($x = 1$) and $\forall x, q(x)$ is false ($x = 2$), so $[\forall x, p(x) \vee \forall x, q(x)]$ is false. Thus $\forall x, [p(x) \vee q(x)] \not\Rightarrow [\forall x, p(x) \vee \forall x, q(x)]$, and $\forall x, p(x) \vee \forall x, q(x) \not\Leftrightarrow \forall x, [p(x) \vee q(x)]$.

Example 7.20 – An existential conclusion

Prove that $\exists x, q(x)$ follows logically from the premises $\forall x, [p(x) \to q(x)]$ and $\exists x, p(x)$.

Solution. $\exists x, p(x) \Rightarrow p(a)$ for some a in the universe. Therefore

$$[\exists x, p(x)] \wedge [\forall x, p(x) \to q(x)] \Rightarrow p(a) \wedge [p(a) \to q(a)] \Rightarrow q(a) \text{ (Modus Ponens)} \Rightarrow \exists x, q(x).$$

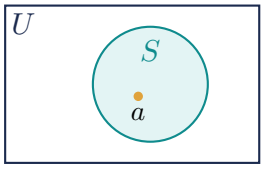
Example 7.21 – All men are mortal

Prove that the following argument is valid: "All men are mortal. Sachin is a man. $\therefore$ Sachin is mortal."

Solution. Let S be the set of all men, $p(x)$: x is mortal, and a: Sachin. The argument reads

$$\frac{\begin{array}{l}\forall x \in S,\ p(x) \\ a \in S\end{array}}{\therefore p(a)}$$

Since $\forall x \in S,\ p(x)$ is true and $a \in S$, the Rule of Universal Specification gives $p(a)$. **Valid.**
Aliter (Venn diagram): let U be the set of all mortal things. The premises say $S \subseteq U$ and $a \in S$; therefore $a \in U$, i.e. a is mortal.



Example 7.22 – An invalid quantified argument

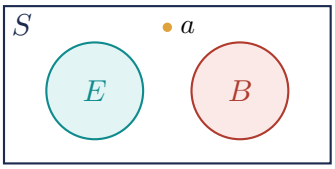
Find whether the following is a valid argument, the universe being the set of all students: "No Engineering student is bad in studies. Anil is not bad in studies. $\therefore$ Anil is an Engineering student."

Solution. Let $p(x)$: x is an engineering student, $q(x)$: x is bad in studies, a: Anil. The argument reads

$$\begin{array}{l} \forall x, [p(x) \to \neg q(x)] \\ \neg q(a) \\ \hline \therefore p(a) \end{array}$$

By Universal Specification $p(a) \to \neg q(a)$; but $\{p(a) \to \neg q(a)\} \wedge \neg q(a) \not\Rightarrow p(a)$, since $p(a)$ can be false when both $p(a) \to \neg q(a)$ and $\neg q(a)$ are true (affirming the consequent). The argument is **not valid** — a non-engineering student can also be not bad in studies!

Venn diagram: with S all students, E engineering students and B students bad in studies, E and B are disjoint, and Anil may lie outside both.



Example 7.23 – Anil and logic

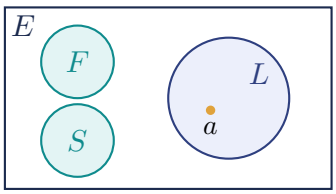
Find whether the following argument is valid: "No engineering student of First or Second Semester studies Logic. Anil is an engineering student who studies Logic. $\therefore$ Anil is not in Second Semester."

Solution. Take the universe to be the set of all engineering students; $p(x)$: x is in First Semester, $q(x)$: x is in Second Semester, $r(x)$: x studies logic, a: Anil. The argument reads

$$\begin{array}{l} \forall x, [\{p(x) \vee q(x)\} \to \neg r(x)] \\ r(a) \\ \hline \therefore \neg q(a) \end{array}$$

Step	Statement	Reason
1	$\{p(a) \vee q(a)\} \to \neg r(a)$	Universal specification
2	$r(a) \to \neg\{p(a) \vee q(a)\}$	Contrapositive (1)
3	$\neg\{p(a) \vee q(a)\}$	Modus ponens ($r(a)$, 2)
4	$\neg p(a) \wedge \neg q(a)$	De Morgan (3)
5	$\neg q(a)$	Conjunctive simplification (4)

The argument is **valid**. (Compare Problem 10 of the solved university problems.) *Illustration:* with E the engineering students, F the First-Semester and S the Second-Semester students and L those who study logic, L is disjoint from F and S, and $a \in L$.



Example 7.24 – A universal syllogism

Prove that

$$\begin{array}{l}\forall x, [p(x) \to q(x)] \\ \forall x, [q(x) \to r(x)] \\ \hline \therefore \forall x, [p(x) \to r(x)]\end{array}$$

is valid.

Solution. Take any a from the universe. Then $\{\forall x, [p(x) \to q(x)]\} \wedge \{\forall x, [q(x) \to r(x)]\} \Rightarrow \{p(a) \to q(a)\} \wedge \{q(a) \to r(a)\} \Rightarrow p(a) \to r(a)$ (Rule of Syllogism) $\Rightarrow \forall x, [p(x) \to r(x)]$ (Rule of Universal Generalization).

Example 7.25 – Detachment inside a universal

Prove that

$$\begin{array}{l}\forall x, [p(x) \to \{q(x) \wedge r(x)\}] \\ \forall x, [p(x) \wedge s(x)] \\ \hline \therefore \forall x, [r(x) \wedge s(x)]\end{array}$$

is valid.

Solution. Take any x from the universe. Then

$$\begin{aligned}[p(x) \to \{q(x) \wedge r(x)\}] \wedge [p(x) \wedge s(x)] &\Leftrightarrow p(x) \wedge [p(x) \to \{q(x) \wedge r(x)\}] \wedge s(x) \\ &\Rightarrow \{q(x) \wedge r(x)\} \wedge s(x) \Rightarrow r(x) \wedge s(x)\end{aligned}$$

(associative and commutative laws, Modus Ponens, conjunctive simplification). By Universal Generalization the argument is valid.

Example 7.26 – A case argument under $\forall$

Establish the validity of

$$\begin{array}{l}\forall x, [p(x) \vee q(x)] \\ \forall x, [\{\neg p(x) \wedge q(x)\} \to r(x)] \\ \hline \therefore \forall x, [\neg r(x) \to p(x)]\end{array}$$

Solution. Suppose $\neg r(x)$ is true for any x in the universe. Then $[\{\neg p(x) \wedge q(x)\} \to r(x)] \Leftrightarrow \neg r(x) \to \{p(x) \vee \neg q(x)\}$ (contrapositive and De Morgan) gives $p(x) \vee \neg q(x)$ by Modus Ponens. Therefore, for any x,

$$[p(x) \vee q(x)] \wedge [p(x) \vee \neg q(x)] \Leftrightarrow p(x) \vee \{q(x) \wedge \neg q(x)\} \Leftrightarrow p(x),$$

since $q(x) \wedge \neg q(x)$ is false. Thus when $\neg r(x)$ is true the premises imply $p(x)$, so $\neg r(x) \to p(x)$ is true; by Universal Generalization the argument is valid.

Example 7.27 – Triangle ABC

Find whether the following argument is valid: "If a triangle has two equal sides, then it is isosceles. If a triangle is isosceles, then it has two equal angles. A certain triangle ABC does not have two equal angles. $\therefore$ The triangle ABC does not have two equal sides."

Solution. Let the universe be the set of all triangles, $p(x)$: x has two equal sides, $q(x)$: x is isosceles, $r(x)$: x has two equal angles, and c: the triangle ABC. The argument reads

$$\begin{array}{l}\forall x, [p(x) \to q(x)] \\ \forall x, [q(x) \to r(x)] \\ \neg r(c) \\ \hline \therefore \neg p(c)\end{array}$$

By the universal syllogism example, the first two premises give $\forall x, [p(x) \to r(x)]$; by Universal Specification $p(c) \to r(c)$; with $\neg r(c)$, Modus Tollens gives $\neg p(c)$. **Valid.**

Example 7.28 – An existential premise

Prove that

$$\begin{array}{l} \forall x, [p(x) \vee q(x)] \\ \exists x, \neg p(x) \\ \forall x, [\neg q(x) \vee r(x)] \\ \forall x, [s(x) \to \neg r(x)] \\ \hline \therefore \exists x, \neg s(x) \end{array}$$

is valid.

Solution. $\exists x, \neg p(x)$ gives $\neg p(a)$ for some a in the universe; with $p(a) \vee q(a)$ (Universal Specification), disjunctive syllogism gives $q(a)$. With $\neg q(a) \vee r(a)$, disjunctive syllogism gives $r(a)$. With $s(a) \to \neg r(a)$, Modus Tollens gives $\neg s(a)$. Hence $\exists x, \neg s(x)$. **Valid.**

7.13 Exercise Set 7B — implication involving quantifiers

7B.1 Prove: (i) $\{\forall x, [p(x) \vee q(x)]\} \wedge \{\forall x, \neg p(x)\} \Rightarrow \exists x, q(x)$ (ii) $\neg\{\forall x, [p(x) \wedge q(x)]\} \wedge [\forall x, p(x)] \Rightarrow \neg\{\forall x, q(x)\}$ (iii) $\{\exists x, p(x) \to \forall x, q(x)\} \Rightarrow \forall x, [p(x) \to q(x)]$ (iv) $\forall x, [p(x) \to q(x)] \Rightarrow [\forall x, p(x) \to \forall x, q(x)]$.

7B.2 Prove that $\forall x, [p(x) \to \neg q(x)]$ is a valid conclusion from $\exists x, [p(x) \wedge q(x)] \to \forall y, [r(y) \to s(y)]$ and $\exists y, [r(y) \wedge \neg s(y)]$.

7B.3 Test the validity of the following arguments:

(i) Some intelligent boys are lazy. Ravi is an intelligent boy. $\therefore$ Ravi is lazy.

(ii) All squares have four sides. The quadrilateral $ABCD$ is not a square. $\therefore$ $ABCD$ does not have four sides.

(iii) Every square is a rectangle. Every rectangle is a parallelogram. $\therefore$ Every square is a parallelogram.

(iv) Some integers are powers of 3. All integers are rational numbers. $\therefore$ Some rational numbers are powers of 3.

(v) No real number has a negative square. All real numbers are complex numbers. Some complex numbers have negative squares. z is a number whose square is not negative. $\therefore$ z is a real number.

(vi) All Engineering students study Physics. All Engineering students of Computer Science study Logic. Ravi is an Engineering student who does not study Logic. Sachin studies Logic but does not study Physics. $\therefore$ Ravi is not a student of Computer Science and Sachin is not an Engineering student.

7.14 Solutions to Exercise Set 7B

7B.1 – Four implications

(i) Take any a (the universe is non-empty). By US, $p(a) \vee q(a)$ and $\neg p(a)$; disjunctive syllogism gives $q(a)$; hence $\exists x, q(x)$.
(ii) $\neg\{\forall x, [p(x) \wedge q(x)]\} \equiv \exists x, [\neg p(x) \vee \neg q(x)]$, so for some a, $\neg p(a) \vee \neg q(a)$. Since $\forall x, p(x)$ gives $p(a)$, disjunctive syllogism gives $\neg q(a)$, hence $\exists x, \neg q(x) \equiv \neg\{\forall x, q(x)\}$.
(iii) This is 7A.12(iv).
(iv) Assume $\forall x, [p(x) \to q(x)]$ and $\forall x, p(x)$. For any a, US gives $p(a) \to q(a)$ and $p(a)$, so $q(a)$ (MP); by UG, $\forall x, q(x)$. Hence $\forall x, p(x) \to \forall x, q(x)$. ■

7B.2 – A valid conclusion

$\exists y, [r(y) \wedge \neg s(y)] \equiv \neg\{\forall y, [r(y) \to s(y)]\}$ (Rule 5 and $\neg(r \to s) \equiv r \wedge \neg s$). With the first premise, Modus Tollens gives $\neg\{\exists x, [p(x) \wedge q(x)]\} \equiv \forall x, [\neg p(x) \vee \neg q(x)] \equiv \forall x, [p(x) \to \neg q(x)]$. ■

7B.3 – Six arguments in words

(i) $\exists x, [i(x) \wedge l(x)], i(r) \therefore l(r)$: **not valid**. The intelligent boy who is lazy need not be Ravi.
(ii) $\forall x, [s(x) \to f(x)], \neg s(a) \therefore \neg f(a)$: **not valid** (denying the antecedent; a rectangle that is not a square has four sides).
(iii) $\forall x, [s(x) \to r(x)], \forall x, [r(x) \to p(x)] \therefore \forall x, [s(x) \to p(x)]$: **valid** (universal syllogism).
(iv) $\exists x, [n(x) \wedge t(x)], \forall x, [n(x) \to q(x)] \therefore \exists x, [q(x) \wedge t(x)]$: **valid**. Take c with $n(c) \wedge t(c)$; US gives $n(c) \to q(c)$, so $q(c)$; hence $q(c) \wedge t(c)$ and $\exists x, [q(x) \wedge t(x)]$.
(v) **Not valid**. The premises say that every real number has a non-negative square; they do not say that every number with a non-negative square is real. Concluding "z is real" from "z^2 is not negative" affirms the consequent of $\forall x, [\text{real}(x) \to \neg\text{neg}(x)]$. Counterexample: $z = 1 + i$ has $z^2 = 2i$, which is not a negative number, yet z is not real. (vi) **Valid**. $\forall x, [(e(x) \wedge c(x)) \to l(x)]$ and $e(\text{Ravi}) \wedge \neg l(\text{Ravi})$ give, by US and MT, $\neg(e \wedge c)$ for Ravi, and since he is an engineering student, $\neg c(\text{Ravi})$. $\forall x, [e(x) \to ph(x)]$ and $\neg ph(\text{Sachin})$ give $\neg e(\text{Sachin})$ by US and MT.

Conclusion. (iii), (iv), (vi) are valid; (i), (ii), (v) are not valid.

7.15 Statements with more than one variable

Open statements such as (1) $x - 2y$ is a positive integer, (2) $x + y - z = 0$ contain more than one free variable. They become propositions when each variable is replaced by an element of a universe. With $\mathbb{Z}$ as universe, replacing x and y in (1) by 5 and -3 gives the proposition "$5 - 2(-3)$ is a positive integer" (true). With $\mathbb{Q}$ as universe, replacing x, y, z in (2) by $1/2, 1/4, 1/4$ gives "$1/2 + 1/4 - 1/4 = 0$" (false).

Open statements in two variables are written $p(x, y)$, $q(x, y)$, ..., and in three variables $p(x, y, z)$, ... The universe may be the same for all variables or different: for "$x - 2y$ is a positive integer", $\mathbb{Z}$ can be the universe for both, or $\mathbb{Z}$ for x and $\mathbb{Z}^+$ for y. If a and b are elements of the universes of x and y, $p(a, b)$ is the proposition obtained by replacing x by a and y by b. For example, with $p(x, y)$: $x - 2y$ is a positive integer and $\mathbb{Z}$ as universe, $p(6, 4)$ is "$6 - (2 \times 4)$ is a positive integer" and $p(-4, 2)$ is "$-4 - (2 \times 2)$ is a positive integer".

Example 7.29 – Evaluating two-variable statements

Let $p(x, y)$: $x^2 \geq y$, $q(x, y)$: $(x + 2) < y$, the universe for both x and y being $\mathbb{R}$. Find the truth values of (i) $p(2, 4)$ (ii) $q(1, \pi)$ (iii) $p(-3, 8) \wedge q(1, 3)$ (iv) $p(1/2, 1/3) \vee \neg q(-2, -3)$ (v) $p(2, 2) \leftrightarrow q(1, 1)$ (vi) $p(1, 2) \leftrightarrow \neg q(3, 8)$.

Solution. (i) $p(2, 4) \equiv 2^2 \geq 4$: **true**. (ii) $q(1, \pi) \equiv (1+2) < \pi$: **true**. (iii) $[(-3)^2 \geq 8] \wedge [(1+2) < 3]$: **false**. (iv) $[(1/2)^2 \geq 1/3] \vee [(-2 + 2) \geq -3]$: the second part is true, so **true**. (v) $(2^2 \geq 2) \leftrightarrow \{(1 + 2) < 1\}$: T $\leftrightarrow$ F, **false**. (vi) $(1^2 \geq 2) \leftrightarrow (3 + 2 \geq 8)$: F $\leftrightarrow$ F, **true**.

Quantified statements with more than one variable. If $p(x, y)$ has variables x, y we can form (1) $\forall x, \forall y,\ p(x, y)$ (2) $\exists x, \exists y,\ p(x, y)$ (3) $\forall x, \exists y,\ p(x, y)$ (4) $\exists x, \forall y,\ p(x, y)$. When x and y have the same universe, (1) and (2) are also written $\forall x, y,\ p(x, y)$ and $\exists x, y,\ p(x, y)$. From the meaning of the quantifiers,

$$\forall x, \forall y,\ p(x, y) \Leftrightarrow \forall y, \forall x,\ p(x, y), \qquad \exists x, \exists y,\ p(x, y) \Leftrightarrow \exists y, \exists x,\ p(x, y).$$

Now consider $p(x, y)$: $x + y = 1$ over $\mathbb{Z}$. The statement $\forall x, \exists y,\ p(x, y)$ reads "For every integer x, there exists an integer y such that $x + y = 1$" — "given any integer x, we can find a corresponding integer y such that $x + y = 1$". This is **true**: once we select any x, $y = 1 - x$ meets the requirement. On the other hand, $\exists y, \forall x,\ p(x, y)$ reads "For some integer y and for all integers x, $x + y = 1$" — **false**, because then every integer x would equal $1 - y$ for one fixed y. This illustrates the important result

$$\forall x, \exists y,\ p(x, y) \not\Leftrightarrow \exists y, \forall x,\ p(x, y); \qquad \text{similarly} \qquad \exists x, \forall y,\ p(x, y) \not\Leftrightarrow \forall y, \exists x,\ p(x, y).$$

Statements with more than two variables are analysed similarly, and all the rules for one variable extend in a natural way.

Example 7.30 – Reading nested quantifiers

Let x and y denote integers and $p(x,y)$: $x+y$ is even. Write in words: (i) $\forall x, \exists y,\ p(x,y)$ (ii) $\exists x, \forall y,\ p(x,y)$.

Solution. (i) For every integer x, there exists an integer y such that $x+y$ is even. (ii) There exists an integer x such that $x+y$ is even for every integer y.

Example 7.31 – Symbolising with nested quantifiers

Write in symbolic form: (1) Every real number has an additive inverse. (2) The set of real numbers has a multiplicative identity. (3) The integer 58 is equal to the sum of two perfect squares.

Solution. (1) "Given any real number x, there is a real number y such that $x+y=y+x=0$": $\forall x, \exists y,\ [x+y=y+x=0]$ (universe $\mathbb{R}$).
(2) "There exists a real number x such that $xy=yx=y$ for every y": $\exists x, \forall y,\ [xy=yx=y]$ (universe $\mathbb{R}$).
(3) "There exist integers m and n such that $58=m^2+n^2$": $\exists m, \exists n,\ 58=m^2+n^2$ (universe $\mathbb{Z}$).

Example 7.32 – Truth values over the non-zero integers

Determine the truth values, the universe being all non-zero integers: (i) $\exists x, \exists y, [xy=1]$ (ii) $\exists x, \forall y, [xy=1]$ (iii) $\forall x, \exists y, [xy=1]$ (iv) $\exists x, \exists y, [(2x+y=5)\wedge(x-3y=-8)]$ (v) $\exists x, \exists y, [(3x-y=17)\wedge(2x+4y=3)]$.

Solution. (i) **True** ($x=1, y=1$). (ii) **False**: for a fixed x, $xy=1$ cannot hold for every y. (iii) **False**: for $x=2$ there is no integer y with $xy=1$. (iv) **True** ($x=1, y=3$). (v) **False**: solving, $y=3x-17$ and $14x=71$, which has no integer solution.

Example 7.33 – Negating a nested statement

Find the negation of $\forall x, \exists y, [\{p(x,y)\wedge q(x,y)\}\to r(x,y)]$.

Solution. $\neg[\forall x, \exists y, \{p\wedge q\}\to r] \Leftrightarrow \exists x, \neg[\exists y, \{p\wedge q\}\to r] \Leftrightarrow \exists x, \forall y, \neg[\{p(x,y)\wedge q(x,y)\}\to r(x,y)] \Leftrightarrow \exists x, \forall y, [p(x,y)\wedge q(x,y)\wedge \neg r(x,y)]$. (Also the example "Negating nested quantifiers" above.)

Example 7.34 – Four negations with real numbers

Find the negation of (i) $\forall x, \forall y, [(x>y)\to((x-y)>0)]$ (ii) $\forall x, \forall y, [(x<y)\to\exists z, (x<z<y)]$ (iii) $\forall x, \forall y, [(|x|=|y|)\to(y=\pm x)]$ (iv) $[\forall x, \forall y, ((x<0)\wedge(y>0))]\to[\exists z, (xz>y)]$.

Solution. Using the rules of negation and the laws of logic:
(i) $\exists x, \exists y, [(x>y)\wedge((x-y)\le 0)]$ (ii) $\exists x, \exists y, [(x<y)\wedge\forall z, \{(x\ge z)\vee(z\ge y)\}]$
(iii) $\exists x, \exists y, [(|x|=|y|)\wedge(y\neq\pm x)]$ (iv) $[\forall x, \forall y, ((x<0)\wedge(y>0))]\wedge\forall z, (xz\le y)$.

Example 7.35 – A two-variable argument

Prove that the following argument is valid, where a and b are particular members of the universe:

$$\begin{array}{l}\forall x, \forall y, [p(x,y)\to q(x,y)]\\ \neg q(a,b)\\ \hline \therefore \exists x, \exists y, \{\neg p(x,y)\}\end{array}$$

Solution. $\{\forall x, \forall y, [p(x,y)\to q(x,y)]\}\wedge\neg q(a,b) \Rightarrow \{p(a,b)\to q(a,b)\}\wedge\neg q(a,b)$ (Universal

Specification) $\Rightarrow \neg p(a,b)$ (Modus Tollens) $\Rightarrow \exists x, \exists y, \{\neg p(x,y)\}$.

7.16 Exercise Set 7C — statements with more than one variable

7C.1 Identify the bound variables and free variables: (i) $\forall y, \exists z,\ \sin(x+y) = \cos(z-x)$ (ii) $\exists x, \exists y,\ [(x^2 + y^2) = z^2]$.

7C.2 Let $p(x,y)$: "x divides y", the universe being all integers. Find whether true or false: (i) $p(3,7)$ (ii) $p(3,27)$ (iii) $\forall y,\ p(1,y)$ (iv) $\forall x,\ p(x,0)$ (v) $\forall x,\ p(x,x)$ (vi) $\forall y, \exists x,\ p(x,y)$ (vii) $\exists y, \forall x,\ p(x,y)$ (viii) $\forall x, \forall y,\ [p(x,y) \wedge p(y,x) \rightarrow (x = y)]$.

7C.3 Let $p(x,y)$: "y is a multiple of x", the universe being all positive integers. Find whether true or false: (i) $\forall x, \exists y,\ p(x,y)$ (ii) $\forall y, \exists x,\ p(x,y)$ (iii) $\exists x, \forall y,\ p(x,y)$ (iv) $\exists y, \forall x,\ p(x,y)$.

7C.4 Find the negations of (i) $\forall x, \exists y, [(p(x,y) \wedge q(x,y)) \rightarrow r(x,y)]$ (ii) $\exists x, \forall y, [p(x,y) \rightarrow (q(x,y) \vee r(x,y))]$ (iii) $\exists x, \exists y, [\{p(x,y) \vee q(x,y)\} \wedge \neg r(x,y)]$.

7C.5 Write down the negations of (i) $\exists x, \forall y, [(x < y) \wedge \{(x-y) \leq 0\}]$ (ii) $\forall x, \forall y, [(x^2 + y^2 = 0) \vee (x = 0 \wedge y = 0)]$ (iii) $\exists x, \exists y, [(x + y < 0) \rightarrow \{(x > 0) \vee (y \leq 0)\}]$.

7C.6 Prove: (i) $\forall x, \forall y,\ p(x,y) \Rightarrow \exists y, \forall x,\ p(x,y)$ (ii) $\exists y, \forall x,\ p(x,y) \Rightarrow \forall x, \exists y,\ p(x,y)$ (iii) $\forall x, \exists y,\ p(x,y) \Rightarrow \exists y, \exists x,\ p(x,y)$ (iv) $\forall y, \exists x,\ p(x,y) \Rightarrow \exists x, \exists y,\ p(x,y)$.

7C.7 Prove that the following is a valid argument:

$$\frac{\begin{array}{l}\exists x, \exists y,\ [p(x,y) \rightarrow q(x,y)] \\ \forall x, y,\ \neg q(x,y)\end{array}}{\therefore \exists x, y,\ [\neg p(x,y)]}$$

7.17 Solutions to Exercise Set 7C

7C.1 – Bound and free variables

(i) y and z are bound (by $\forall y$ and $\exists z$); x is free. (ii) x and y are bound; z is free.

7C.2 – "x divides y" over $\mathbb{Z}$

(i) **F**. (ii) **T** ($27 = 3 \cdot 9$). (iii) **T** (1 divides every integer). (iv) **T** ($0 = x \cdot 0$). (v) **T** ($x = x \cdot 1$). (vi) **T** (take $x = 1$). (vii) **T** (take $y = 0$). (viii) **F** ($x = 2, y = -2$).

7C.3 – "y is a multiple of x" over $\mathbb{Z}^+$

(i) **T** (take $y = x$). (ii) **T** (take $x = 1$). (iii) **T** ($x = 1$ divides every y). (iv) **F**: no positive integer y is a multiple of every positive integer (e.g. $y + 1 \nmid y$).

7C.4 – Negations

(i) $\exists x, \forall y, [p(x,y) \wedge q(x,y) \wedge \neg r(x,y)]$ (ii) $\forall x, \exists y, [p(x,y) \wedge \{\neg q(x,y) \wedge \neg r(x,y)\}]$ (iii) $\forall x, \forall y, [\{\neg p(x,y) \wedge \neg q(x,y)\} \vee r(x,y)]$.

7C.5 – Negations with inequalities

(i) $\forall x, \exists y, [(x \geq y) \vee (x-y) > 0]$ (ii) $\exists x, \exists y, [(x^2 + y^2 \neq 0) \wedge (x \neq 0 \vee y \neq 0)]$ (iii) $\forall x, \forall y, [(x+y < 0) \wedge (x \leq 0) \wedge (y > 0)]$.

7C.6 – Nested implications

Let the universes be non-empty.
(i) If $p(x, y)$ holds for all x, y, pick any b; then $p(x, b)$ holds for all x, so $\exists y, \forall x,\ p(x, y)$.
(ii) If b satisfies $p(x, b)$ for all x, then for each x the same b works: $\forall x, \exists y,\ p(x, y)$.
(iii) Pick any a; some b satisfies $p(a, b)$; hence $\exists x, \exists y,\ p(x, y)$, i.e. $\exists y, \exists x,\ p(x, y)$.
(iv) Pick any b; some a satisfies $p(a, b)$; hence $\exists x, \exists y,\ p(x, y)$. ■

7C.7 – A two-variable argument

By the first premise there are a, b with $p(a, b) \to q(a, b)$. By the second premise (US), $\neg q(a, b)$. Modus Tollens gives $\neg p(a, b)$, hence $\exists x, y, [\neg p(x, y)]$. **Valid.**

8 Solved University Examination Problems

These twelve problems are typical of university examination questions. The marks shown are only a guide.

Problem 1: Tautology by truth table — 5 marks

Q. Prove that $[(p \vee q) \wedge (p \to r) \wedge (q \to r)] \to r$ is a tautology.

Solution. Three variables give 8 rows. Let $X = (p \vee q) \wedge (p \to r) \wedge (q \to r)$.

p	q	r	$p \vee q$	$p \to r$	$q \to r$	X	$X \to r$
T	T	T	T	T	T	T	**T**
T	T	F	T	F	F	F	**T**
T	F	T	T	T	T	T	**T**
T	F	F	T	F	T	F	**T**
F	T	T	T	T	T	T	**T**
F	T	F	T	T	F	F	**T**
F	F	T	F	T	T	F	**T**
F	F	F	F	T	T	F	**T**

The last column is all T, so the formula is a tautology. (This is the simple-dilemma rule.)

Common pitfall

Writing only 4 rows; evaluating $p \to r$ as F when p is F; omitting the concluding sentence.

Problem 2: Tautology by laws of logic — 10 marks

Q. Without truth tables, prove that $[(p \vee q) \wedge \neg\{\neg p \wedge (\neg q \vee \neg r)\}] \vee (\neg p \wedge \neg q) \vee (\neg p \wedge \neg r)$ is a tautology.

Solution. Write the formula as $u \vee v$, with $u = (p \vee q) \wedge \neg\{\neg p \wedge (\neg q \vee \neg r)\}$ and $v = (\neg p \wedge \neg q) \vee (\neg p \wedge \neg r)$.

$$
\begin{aligned}
u &\equiv (p \vee q) \wedge (p \vee (q \wedge r)) && \textit{De Morgan, Double negation}\\
&\equiv p \vee (q \wedge (q \wedge r)) && \textit{Distributive}\\
&\equiv p \vee (q \wedge r) && \textit{Associative, Idempotent}\\
v &\equiv \neg(p \vee q) \vee \neg(p \vee r) && \textit{De Morgan}\\
&\equiv \neg[(p \vee q) \wedge (p \vee r)] && \textit{De Morgan}\\
&\equiv \neg[p \vee (q \wedge r)] \equiv \neg u && \textit{Distributive}
\end{aligned}
$$

Hence $u \vee v \equiv u \vee \neg u \equiv T_0$ (inverse law).

Examination tip

The key insight is that $v = \neg u$. Simplify each half separately, then compare them.

Problem 3: Validity of an English argument — 7 marks

Q. Test the validity: "If I study, I will not fail in the examination. If I do not watch TV in the evenings, I will study. I failed in the examination. Therefore I must have watched TV in the evenings."

Solution. p: I study; q: I fail; r: I watch TV. Premises $p \to \neg q$, $\neg r \to p$, q; conclusion r.

Step	Statement	Reason
1	q	Premise
2	$p \to \neg q \equiv q \to \neg p$	Premise, contrapositive
3	$\neg p$	Modus ponens (1, 2)
4	$\neg r \to p \equiv \neg p \to r$	Premise, contrapositive
5	r	Modus ponens (3, 4)

The argument is **valid**.

Problem 4: Derivation with many variables — 5 marks

Q. Show that $R \vee S$ follows from $C \vee D$, $(C \vee D) \to \neg H$, $\neg H \to (A \wedge \neg B)$ and $(A \wedge \neg B) \to (R \vee S)$.

Solution. With seven variables a truth table would need 128 rows, so we use the rules of inference.

Step	Statement	Reason
1	$C \vee D$	Premise
2	$\neg H$	Modus ponens (1, premise)
3	$A \wedge \neg B$	Modus ponens (2, premise)
4	$R \vee S$	Modus ponens (3, premise)

Alternatively, hypothetical syllogism on the three conditionals gives $(C \vee D) \to (R \vee S)$; then modus ponens.

Problem 5: Quantified statements over $\mathbb{R}$ — 10 marks

Q. (a) Over $\mathbb{R}$, let $p(x) : x \geq 0$, $q(x) : x^2 \geq 0$, $r(x) : x^2 - 3x - 4 = 0$, $s(x) : x^2 - 3 > 0$. Find the truth values of (i) $\exists x\,[p(x) \wedge q(x)]$ (ii) $\forall x\,[p(x) \to q(x)]$ (iii) $\forall x\,[q(x) \to s(x)]$ (iv) $\forall x\,[r(x) \vee s(x)]$ (v) $\exists x\,[p(x) \wedge r(x)]$ (vi) $\forall x\,[r(x) \to p(x)]$.
(b) Write in symbols and negate: "All integers are rational numbers and some rational numbers are not integers."

Solution. (a) $x^2 - 3x - 4 = (x-4)(x+1)$, so $r(x)$ holds only for $x = 4$ and $x = -1$.

Part	Value	Justification
(i)	T	$x = 1$: $1 \geq 0$ and $1 \geq 0$
(ii)	T	$q(x)$ is true for every real x
(iii)	F	$x = 1$: $q(1)$ is T but $s(1) : -2 > 0$ is F
(iv)	F	$x = 1$: $r(1) = -6 \neq 0$ and $s(1)$ is F
(v)	T	$x = 4$: $4 \geq 0$ and $r(4) = 0$
(vi)	F	$x = -1$: $r(-1)$ is T but $p(-1)$ is F

(b) $p(x)$: x is rational; $q(x)$: x is an integer. Given: $[\forall x \in \mathbb{Z},\ p(x)] \wedge [\exists x \in \mathbb{Q},\ \neg q(x)]$.
Negation: $[\exists x \in \mathbb{Z},\ \neg p(x)] \vee [\forall x \in \mathbb{Q},\ q(x)]$ — "Some integer is not rational, or every rational number is an integer."

Problem 6: A canoe-trip argument — 7 marks

Q. "It is not sunny this afternoon and it is colder than yesterday. We will go swimming only if it is sunny. If we do not go swimming, then we will take a canoe trip. If we take a canoe trip, then we will be home by sunset." Show that these lead to "We will be home by sunset."

Solution. p: sunny; q: colder than yesterday; r: we swim; s: canoe trip; t: home by sunset. Premises $\neg p \wedge q$, $r \to p$, $\neg r \to s$, $s \to t$.

Step	Statement	Reason
1	$\neg p \wedge q$	Premise
2	$\neg p$	Simplification (1)
3	$r \to p$	Premise ("only if")
4	$\neg r$	Modus tollens (2, 3)
5	$\neg r \to s$	Premise
6	s	Modus ponens (4, 5)
7	$s \to t$	Premise
8	t	Modus ponens (6, 7)

Problem 7: Argument by truth table — 5 marks

Q. Using a truth table, test the validity of $p \to q,\ \neg q \vee r,\ \neg r \vdash \neg p$.

Solution. Highlighted rows are the *critical rows*, where all premises are true.

p	q	r	$p \to q$	$\neg q \vee r$	$\neg r$	$\neg p$
T	T	T	T	T	F	**F**
T	T	F	T	F	T	**F**
T	F	T	F	T	F	**F**
T	F	F	F	T	T	**F**
F	T	T	T	T	F	**T**
F	T	F	T	F	T	**T**
F	F	T	T	T	F	**T**
F	F	F	T	T	T	**T**

There is one critical row ($p = q = r =$ F) and there $\neg p$ is T. Hence the argument is **valid**.

Problem 8: Proving an argument invalid — 5 marks

Q. "If it rains, the ground is wet. Either it does not rain or the match is cancelled. The match is cancelled. Therefore the ground is wet." Is this valid?

Solution. p: it rains; q: the ground is wet; r: the match is cancelled. Form: $p \to q,\ \neg p \vee r,\ r \vdash q$.

p	q	r	$p \to q$	$\neg p \vee r$	r	q
T	T	T	T	T	T	T
T	T	F	T	F	F	T
T	F	T	F	T	T	F
T	F	F	F	F	F	F
F	T	T	T	T	T	T
F	T	F	T	T	F	T
F	F	T	T	T	T	F
F	F	F	T	T	F	F

In the highlighted critical rows, the row $p =$ F, $q =$ F, $r =$ T has all premises T and conclusion F. The argument is **invalid** (the match may be cancelled for another reason).

Problem 9: Simplification by laws — 5 marks

Q. Prove that $[\neg p \wedge (\neg q \wedge r)] \vee (q \wedge r) \vee (p \wedge r) \equiv r$.

Solution.

$$\begin{aligned}
\text{LHS} &\equiv [(\neg p \wedge \neg q) \wedge r] \vee [(q \vee p) \wedge r] && \textit{Associative, Distributive}\\
&\equiv [\neg(p \vee q) \wedge r] \vee [(p \vee q) \wedge r] && \textit{De Morgan, Commutative}\\
&\equiv [\neg(p \vee q) \vee (p \vee q)] \wedge r && \textit{Distributive}\\
&\equiv T_0 \wedge r \equiv r && \textit{Inverse, Identity}
\end{aligned}$$

Problem 10: Quantified argument — 7 marks

Q. "No engineering student of the I or II semester studies logic. Anil is an engineering student who studies logic. Therefore Anil is not in the II semester." Test validity.

Solution. $p(x)$: x is an engineering student; $q(x)$: x is in I semester; $r(x)$: x is in II semester; $s(x)$: x studies logic; a = Anil.
Premises $\forall x\,[(p(x) \wedge (q(x) \vee r(x))) \to \neg s(x)]$ and $p(a) \wedge s(a)$; conclusion $\neg r(a)$.

Step	Statement	Reason
1	$(p(a) \wedge (q(a) \vee r(a))) \to \neg s(a)$	Universal specification
2	$s(a)$	Simplification of premise
3	$\neg(p(a) \wedge (q(a) \vee r(a)))$	Modus tollens (1, 2)
4	$\neg p(a) \vee \neg(q(a) \vee r(a))$	De Morgan (3)
5	$p(a)$	Simplification of premise
6	$\neg q(a) \wedge \neg r(a)$	Disjunctive syllogism (4, 5), De Morgan
7	$\neg r(a)$	Simplification (6)

The argument is **valid**.

Problem 11: Resolution and dilemma — 5 marks

Q. Show that r follows from $p \vee q,\ \neg p \vee r,\ \neg q \vee r$.

Solution.

Step	Statement	Reason
1	$p \vee q$	Premise
2	$\neg p \vee r$	Premise
3	$q \vee r$	Resolution (1, 2)
4	$\neg q \vee r$	Premise
5	$r \vee r$	Resolution (3, 4)
6	r	Idempotent law (5)

(Equivalently: $p \to r$, $q \to r$, $p \vee q$ give r by the simple dilemma.)

Problem 12: Symbolise, then derive — 7 marks

Q. "If Ravi goes out with friends, he will not study. If Ravi does not study, his father becomes angry. His father is not angry. Therefore Ravi has not gone out with friends."

Solution. p: Ravi goes out; q: Ravi studies; r: father is angry. Premises $p \to \neg q$, $\neg q \to r$, $\neg r$; conclusion $\neg p$.

Step	Statement	Reason
1	$p \to \neg q$	Premise
2	$\neg q \to r$	Premise
3	$p \to r$	Hypothetical syllogism (1, 2)
4	$\neg r$	Premise
5	$\neg p$	Modus tollens (3, 4)

Valid.

9 Exercises

Try these yourself first; complete solutions are in Section 10.

9.1 Part A — Short answer (2 marks each)

A1. Define a proposition. Which are propositions: (a) 7 is prime (b) $x + 2 = 5$ (c) Shut the window?

A2. Distinguish between the syntax and the semantics of propositional logic.

A3. How many rows does the truth table of a formula with 5 distinct variables have?

A4. Define tautology, contradiction and contingency with one example each.

A5. True or false: "Every satisfiable formula is a contingency." Justify.

A6. Write the converse, inverse and contrapositive of "If it rains, the match is cancelled."

A7. Express "p only if q" and "p unless q" in symbols.

A8. State De Morgan's laws for propositions and for quantifiers.

A9. Write the dual of $(p \wedge T_0) \vee (q \wedge \neg r)$.

A10. Differentiate between $p \to q$ and $p \Rightarrow q$.

A11. State modus ponens and modus tollens.

A12. What is the fallacy of affirming the consequent? Give an example.

A13. Express $\neg p$ and $p \wedge q$ using NAND only.

A14. Negate: $\forall x \, \exists y \, (x + y = 0)$.

A15. Define bound and free variables with an example.

9.2 Part B — Medium answer (5 / 7 marks)

B1. Construct the truth table of $(p \vee q) \wedge \neg r \to (p \leftrightarrow r)$ and classify it.

B2. Prove that $p \to (q \to r) \equiv (p \wedge q) \to r$ without truth tables.

B3. Prove that $[(\neg p \vee \neg q) \to (p \wedge q \wedge r)] \equiv p \wedge q$ using the laws of logic.

B4. Simplify $\neg[\neg\{(p \vee q) \wedge r\} \vee \neg q]$, naming each law.

B5. Prove that $p \vee q \vee (\neg p \wedge \neg q \wedge r) \equiv p \vee q \vee r$.

B6. Prove that NAND is commutative but not associative.

B7. Using the conditional-proof rule, show that $\{p \to q, \; q \to (r \to s)\} \vdash p \to (r \to s)$.

B8. Using the indirect method, show that $\{p \vee q, \; p \to r, \; \neg r\} \vdash q$.

B9. Over $\mathbb{Z}$, write in symbols: (a) Some even integers are divisible by 3. (b) If x is even and a perfect square, then x is not divisible by 3.

B10. With $U = \{1, 2, 3, 4\}$, $p(x)$: x is even, $q(x)$: $x > 2$, find the truth values of $\forall x\,[p(x) \vee q(x)]$, $\exists x\,[p(x) \wedge q(x)]$, $\forall x\,[q(x) \to p(x)]$, $\exists x\,[\neg p(x) \wedge \neg q(x)]$.

B11. With $U = \{2, 5\}$, $p(x) : x^2 - 7x + 10 = 0$, $q(x) : x^2 - 2x - 3 = 0$, $r(x) : x < 0$, find the truth values of $\forall x\,[p(x) \to \neg r(x)]$, $\forall x\,[q(x) \to r(x)]$, $\exists x\,[p(x) \to r(x)]$.

B12. Show that $p \to q,\ r \to \neg q,\ r \vdash \neg p$ and express the argument in English with p: "Sunita clears GATE", q: "Sunita joins IIT", r: "Sunita takes a job".

9.3 Part C — Long answer (10 / 15 marks)

C1. (a) Prove $[(p \leftrightarrow q) \wedge (q \leftrightarrow r) \wedge (r \leftrightarrow p)] \equiv [(p \to q) \wedge (q \to r) \wedge (r \to p)]$ by truth table. (b) Verify the principle of duality for $[\neg(p \wedge q) \to \neg p \vee (\neg p \vee q)] \equiv (\neg p \vee q)$.

C2. (a) Prove $\neg(p \to q) \vee (p \wedge q) \equiv p$. (b) Derive the NOR-only forms of $\neg p$, $p \vee q$, $p \wedge q$ and $p \to q$.

C3. Test the validity of each; for an invalid one give a counterexample.
(a) $p \to r,\ q \to r \vdash (p \vee q) \to r$ (b) $p \to q,\ r \to s,\ \neg q \vee \neg s \vdash \neg(p \wedge r)$
(c) $p \wedge \neg q,\ p \to (q \to r) \vdash \neg r$ (d) $p,\ p \to q,\ s \vee r,\ r \to \neg q \vdash s \vee t$.

C4. (a) Show that $R \wedge (P \vee Q)$ follows from $P \vee Q,\ Q \to R,\ P \to M,\ \neg M$. (b) Show that $S \vee R$ is tautologically implied by $P \vee Q,\ P \to R,\ Q \to S$. (c) Is $p \vee r$ implied by $p \to q,\ r \to q,\ q$?

C5. (a) Write in symbols and negate: "If all triangles are right-angled, then no triangle is equiangular." (b) Test: "All integers are rational. Some integers are powers of 2. Therefore some rational numbers are powers of 2." (c) Prove $\forall x\,[p(x) \to q(x)],\ \forall x\,[q(x) \to r(x)] \vdash \forall x\,[p(x) \to r(x)]$.

C6. Which are true? Justify each. (a) R follows from $\{P \to (Q \to R),\ P \wedge Q\}$ (b) $\{P \to (R \to S),\ \neg R \to \neg P,\ P\} \Rightarrow S$ (c) $\{P \vee Q,\ \neg P\} \vdash P \wedge Q$ (d) $[(\neg T \to \neg R) \wedge \neg S \wedge (T \to W) \wedge (R \vee S)] \to W$ is a tautology.

9.4 Part D — Additional practice on rules of inference (5 marks each)

D1. Show that $r \wedge s$ follows from $p \wedge q, p \to r, q \to s$.

D2. Show that $\neg p$ follows from $\neg p \vee q, \neg q \vee r, \neg r$.

D3. Show that $\neg p$ follows from $p \to (q \vee r), \neg q, \neg r$.

D4. Show that $\neg p$ follows from $p \to q, q \to r, r \to \neg p$.

D5. "If the program compiles, it runs. If it runs, the output is correct. The output is not correct. Therefore the program did not compile." Test validity.

D6. Is $r \to p$ a valid conclusion from $p \to q$ and $\neg q \to \neg r$?

D7. Show that $\neg p \vee \neg r$ follows from $(p \to q) \wedge (r \to s)$ and $\neg q \vee \neg s$.

D8. Using the indirect method, show that $p \to q, q \to r, p \vdash r$.

D9. Show that p follows from $p \vee q, q \to r, r \to s, \neg s$.

D10. Show that $\neg(p \vee r)$ follows from $p \to q, r \to q, \neg q$.

10 Solutions to Exercises

10.1 Part A

A1. A declarative sentence that is either true or false but not both. (a) is a proposition (T); (b) is an open statement; (c) is a command.

A2. Syntax gives the rules for forming wffs (which strings are legal); semantics assigns truth values through interpretations (what a wff means).

A3. $2^5 = 32$.

A4. Tautology: always T, e.g. $p \vee \neg p$. Contradiction: always F, e.g. $p \wedge \neg p$. Contingency: both values occur, e.g. $p \to q$.

A5. False. A tautology is satisfiable but is not a contingency.

A6. Converse: if the match is cancelled, it rains. Inverse: if it does not rain, the match is not cancelled. Contrapositive: if the match is not cancelled, it does not rain.

A7. $p \to q$; $\neg q \to p \equiv p \vee q$.

A8. $\neg(p \vee q) \equiv \neg p \wedge \neg q$, $\neg(p \wedge q) \equiv \neg p \vee \neg q$; $\neg\forall x\, p(x) \equiv \exists x\, \neg p(x)$, $\neg\exists x\, p(x) \equiv \forall x\, \neg p(x)$.

A9. $(p \vee F_0) \wedge (q \vee \neg r)$.

A10. $p \to q$ is a compound proposition that may be T or F; $p \Rightarrow q$ asserts that $p \to q$ is a tautology.

A11. Modus ponens: $p,\ p \to q \vdash q$. Modus tollens: $p \to q,\ \neg q \vdash \neg p$.

A12. Concluding p from $p \to q$ and q. E.g. "If it rains the road is wet; the road is wet; so it rained" — the road may be wet for another reason.

A13. $\neg p \equiv p \uparrow p$; $p \wedge q \equiv (p \uparrow q) \uparrow (p \uparrow q)$.

A14. $\exists x\, \forall y\, (x + y \neq 0)$.

A15. In $\forall x\, P(x, y)$, x is bound (governed by $\forall$) and y is free.

10.2 Part B

B1.

p	q	r	$(p \vee q) \wedge \neg r$	$p \leftrightarrow r$	Given
T	T	T	F	T	**T**
T	T	F	T	F	**F**
T	F	T	F	T	**T**
T	F	F	T	F	**F**
F	T	T	F	F	**T**
F	T	F	T	T	**T**
F	F	T	F	F	**T**
F	F	F	F	T	**T**

The column has both T and F: a **contingency** (satisfiable, not valid).

B2.

$$
\begin{aligned}
p \to (q \to r) &\equiv \neg p \vee (\neg q \vee r) && \textit{Conditional law (twice)} \\
&\equiv (\neg p \vee \neg q) \vee r && \textit{Associative} \\
&\equiv \neg(p \wedge q) \vee r \equiv (p \wedge q) \to r && \textit{De Morgan, Conditional law}
\end{aligned}
$$

B3.

$$\begin{aligned}(\neg p \vee \neg q) \to (p \wedge q \wedge r) &\equiv \neg(\neg p \vee \neg q) \vee (p \wedge q \wedge r) && \textit{Conditional law}\\ &\equiv (p \wedge q) \vee ((p \wedge q) \wedge r) && \textit{De Morgan, Double negation, Associative}\\ &\equiv p \wedge q && \textit{Absorption}\end{aligned}$$

B4.

$$\begin{aligned}\neg[\neg\{(p \vee q) \wedge r\} \vee \neg q] &\equiv \neg\neg\{(p \vee q) \wedge r\} \wedge \neg\neg q && \textit{De Morgan}\\ &\equiv (p \vee q) \wedge r \wedge q && \textit{Double negation}\\ &\equiv [q \wedge (q \vee p)] \wedge r && \textit{Commutative, Associative}\\ &\equiv q \wedge r && \textit{Absorption}\end{aligned}$$

B5.

$$\begin{aligned}(p \vee q) \vee (\neg p \wedge \neg q \wedge r) &\equiv (p \vee q) \vee (\neg(p \vee q) \wedge r) && \textit{De Morgan}\\ &\equiv [(p \vee q) \vee \neg(p \vee q)] \wedge [(p \vee q) \vee r] && \textit{Distributive}\\ &\equiv T_0 \wedge (p \vee q \vee r) \equiv p \vee q \vee r && \textit{Inverse, Identity}\end{aligned}$$

B6. Commutative: $p \uparrow q = \neg(p \wedge q) = \neg(q \wedge p) = q \uparrow p$. Not associative: $p \uparrow (q \uparrow r) = \neg(p \wedge \neg(q \wedge r)) \equiv \neg p \vee (q \wedge r)$, whereas $(p \uparrow q) \uparrow r = \neg(\neg(p \wedge q) \wedge r) \equiv (p \wedge q) \vee \neg r$. With $p = q =$ F, $r =$ T: the first is T, the second F.

B7. By CP, take p as an extra premise and derive $r \to s$:

Step	Statement	Reason
1	p	Additional premise (CP)
2	q	Modus ponens (1, $p \to q$)
3	$r \to s$	Modus ponens (2, $q \to (r \to s)$)
4	$p \to (r \to s)$	Conditional proof (1–3)

B8.

Step	Statement	Reason
1	$\neg q$	Assumed (negation of conclusion)
2	p	Disjunctive syllogism ($p \vee q$, 1)
3	r	Modus ponens (2, $p \to r$)
4	$r \wedge \neg r$	Conjunction (3, premise $\neg r$) — contradiction

Hence q follows; the argument is valid.

B9. With $e(x)$: x is even, $d(x)$: x is divisible by 3, $r(x)$: x is a perfect square: (a) $\exists x\,[e(x) \wedge d(x)]$ (b) $\forall x\,[(e(x) \wedge r(x)) \to \neg d(x)]$.

B10. (i) $x = 1$: 1 is not even and $1 \not> 2$, so $\forall x\,[p \vee q]$ is **F**. (ii) $x = 4$ is even and > 2: **T**. (iii) $x = 3$: $3 > 2$ but 3 is odd: **F**. (iv) $x = 1$: odd and not > 2: **T**.

B11. $x^2 - 7x + 10 = (x-2)(x-5)$, so $p(2), p(5)$ are both T. $x^2 - 2x - 3 = (x-3)(x+1)$, so $q(2), q(5)$ are both F. $r(2), r(5)$ are both F.
$\forall x\,[p \to \neg r]$: T $\to$ T for both, **T**. $\forall x\,[q \to r]$: false antecedent for both, **T**. $\exists x\,[p \to r]$: T $\to$ F for both, **F**.

B12.

Step	Statement	Reason
1	r	Premise
2	$r \to \neg q$	Premise
3	$\neg q$	Modus ponens (1, 2)
4	$p \to q$	Premise
5	$\neg p$	Modus tollens (3, 4)

English: "If Sunita clears GATE, she joins IIT. If she takes a job, she does not join IIT. She takes a job. Therefore she has not cleared GATE." Valid.

10.3 Part C

C1. (a) Let A and B denote the left and right sides.

p	q	r	A	B	$A \leftrightarrow B$
T	T	T	T	T	**T**
T	T	F	F	F	**T**
T	F	T	F	F	**T**
T	F	F	F	F	**T**
F	T	T	F	F	**T**
F	T	F	F	F	**T**
F	F	T	F	F	**T**
F	F	F	T	T	**T**

$A \leftrightarrow B$ is a tautology, so $A \equiv B$. (Both are true exactly when p, q, r all have the same value.)

(b) Eliminating $\to$: LHS $\equiv (p \wedge q) \vee \neg p \vee (\neg p \vee q) \equiv (p \wedge q) \vee q \vee \neg p \equiv q \vee \neg p$ (idempotent, absorption) = RHS.

Dual statement: $(p \vee q) \wedge [\neg p \wedge (\neg p \wedge q)] \equiv \neg p \wedge q$. Now $\neg p \wedge (\neg p \wedge q) \equiv \neg p \wedge q$ (associative, idempotent), and $(p \vee q) \wedge (\neg p \wedge q) \equiv \neg p \wedge q$ by absorption, since $q \wedge (q \vee p) \equiv q$. The dual also holds, verifying the principle.

C2. (a)

$$\begin{aligned} \neg(p \to q) \vee (p \wedge q) &\equiv (p \wedge \neg q) \vee (p \wedge q) && \textit{Negation of conditional} \\ &\equiv p \wedge (\neg q \vee q) \equiv p \wedge T_0 \equiv p && \textit{Distributive, Inverse, Identity} \end{aligned}$$

(b) $p \downarrow p = \neg(p \vee p) = \neg p$. $(p \downarrow q) \downarrow (p \downarrow q) = \neg(p \downarrow q) = p \vee q$. $(p \downarrow p) \downarrow (q \downarrow q) = \neg(\neg p \vee \neg q) = p \wedge q$. $p \to q \equiv \neg p \vee q$, which by the $\vee$ form with $\neg p = p \downarrow p$ is $((p \downarrow p) \downarrow q) \downarrow ((p \downarrow p) \downarrow q)$.

C3.

(a) **Valid**: $(p \to r) \wedge (q \to r) \equiv (p \vee q) \to r$ (law 15); or by CP, assume $p \vee q$ and apply the simple dilemma.

(b) **Valid**: destructive dilemma gives $\neg p \vee \neg r \equiv \neg(p \wedge r)$ (De Morgan).

(c) **Invalid**: $p = \text{T}, q = \text{F}, r = \text{T}$ makes $p \wedge \neg q = \text{T}$ and $p \to (q \to r) = \text{T} \to \text{T} = \text{T}$, but $\neg r = \text{F}$.

(d) **Valid**:

Step	Statement	Reason
1	q	Modus ponens (p, $p \to q$)
2	$\neg r$	Modus tollens (1, $r \to \neg q$)
3	s	Disjunctive syllogism ($s \vee r$, 2)
4	$s \vee t$	Addition (3)

C4. (a)

Step	Statement	Reason
1	$\neg P$	Modus tollens ($P \to M, \neg M$)
2	Q	Disjunctive syllogism ($P \vee Q$, 1)
3	R	Modus ponens (2, $Q \to R$)
4	$R \wedge (P \vee Q)$	Conjunction (3, premise)

(b) Constructive dilemma on $P \to R$, $Q \to S$, $P \vee Q$ gives $R \vee S$; by commutativity, $S \vee R$.
(c) **No**: $p = r =$ F, $q =$ T makes all premises T but $p \vee r$ F (this is the fallacy of affirming the consequent).

C5. (a) $r(x)$: x is right-angled; $e(x)$: x is equiangular (universe: triangles). Statement: $\forall x\, r(x) \to \forall x\, \neg e(x)$. Negation, using $\neg(A \to B) \equiv A \wedge \neg B$: $\forall x\, r(x) \wedge \exists x\, e(x)$ — "All triangles are right-angled and some triangle is equiangular."

(b) $I(x)$: integer; $Q(x)$: rational; $P(x)$: power of 2. Premises $\forall x\,[I(x) \to Q(x)]$, $\exists x\,[I(x) \wedge P(x)]$.

Step	Statement	Reason
1	$I(c) \wedge P(c)$	Existential specification
2	$I(c) \to Q(c)$	Universal specification
3	$Q(c)$	Modus ponens (simplification of 1, 2)
4	$Q(c) \wedge P(c)$	Conjunction (3, simplification of 1)
5	$\exists x\,[Q(x) \wedge P(x)]$	Existential generalisation

Valid. (Apply existential specification first, so that c is a new constant not used earlier.)

(c) Let a be arbitrary. By US, $p(a) \to q(a)$ and $q(a) \to r(a)$; by hypothetical syllogism $p(a) \to r(a)$; since a was arbitrary, UG gives $\forall x\,[p(x) \to r(x)]$.

C6.
(a) **True**: $P \wedge Q$ gives P and Q; modus ponens twice gives R.
(b) **True**: P with $P \to (R \to S)$ gives $R \to S$; $\neg R \to \neg P \equiv P \to R$ gives R; modus ponens gives S.
(c) **False**: $\neg P$ makes $P \wedge Q$ false whenever the premises hold ($P =$ F, $Q =$ T).
(d) **True**: $R \vee S$ and $\neg S$ give R; $\neg T \to \neg R \equiv R \to T$ gives T; $T \to W$ gives W. So the conjunction of the premises implies W, and the conditional is a tautology.

10.4 Part D

D1. Simplification of $p \wedge q$ gives p and q. Modus ponens gives r (from $p \to r$) and s (from $q \to s$). Conjunction gives $r \wedge s$. Valid.

D2. $\neg r$ with $\neg q \vee r$ gives $\neg q$ by disjunctive syllogism. $\neg q$ with $\neg p \vee q$ gives $\neg p$ by disjunctive syllogism. Valid.

D3. Conjunction gives $\neg q \wedge \neg r \equiv \neg(q \vee r)$ (De Morgan). Modus tollens with $p \to (q \vee r)$ gives $\neg p$. Valid.

D4. Hypothetical syllogism twice gives $p \to \neg p \equiv \neg p \vee \neg p \equiv \neg p$ (conditional law, idempotent law). Valid. (Note that no premise asserts p or $\neg p$ directly.)

D5. c: compiles; r: runs; o: output correct. $c \to r$ and $r \to o$ give $c \to o$ (HS); with $\neg o$, modus tollens gives $\neg c$. **Valid.**

D6. Invalid. $\neg q \to \neg r \equiv r \to q$, so the premises are $p \to q$ and $r \to q$, which say nothing about p when q is true. Counterexample: $p =$ F, $q =$ T, $r =$ T makes both premises T and $r \to p =$ T $\to$ F $=$ F.

D7. Simplification gives $p \to q$ and $r \to s$; with $\neg q \vee \neg s$ the destructive dilemma gives $\neg p \vee \neg r$. Valid.

D8. Assume $\neg r$. Modus tollens with $q \to r$ gives $\neg q$; modus tollens with $p \to q$ gives $\neg p$. This contradicts the premise p, so r follows. Valid.

D9.

Step	Statement	Reason
1	$\neg s$	Premise
2	$\neg r$	Modus tollens (1, $r \to s$)
3	$\neg q$	Modus tollens (2, $q \to r$)
4	p	Disjunctive syllogism ($p \vee q$, 3)

D10. $(p \to q) \wedge (r \to q) \equiv (p \vee r) \to q$ (law 15). With $\neg q$, modus tollens gives $\neg(p \vee r)$. Valid. (Equivalently: MT twice gives $\neg p$ and $\neg r$; conjunction and De Morgan give $\neg(p \vee r)$.)

11 One-Page Revision Sheet

Connectives
$p \to q$ is F only for T→F.
$p \leftrightarrow q$ is T when values agree.
$p \uparrow q = \neg(p \wedge q)$, $p \downarrow q = \neg(p \vee q)$.
$p \veebar q$ is T when exactly one is T.
Rows $= 2^n$; truth functions $= 2^{2^n}$.

English to symbols
p only if q: $p \to q$. q is necessary for p: $p \to q$.
p unless q: $\neg q \to p \equiv p \vee q$. "but" $= \wedge$.
p is sufficient for q: $p \to q$.

Conditional family
$p \to q \equiv \neg q \to \neg p$ (contrapositive).
Converse $q \to p \equiv$ inverse $\neg p \to \neg q$.
$\neg(p \to q) \equiv p \wedge \neg q$.

Classification
Tautology = valid; contradiction = unsatisfiable;
A valid $\iff \neg A$ unsatisfiable.

Key laws
$p \to q \equiv \neg p \vee q$
$\neg(p \vee q) \equiv \neg p \wedge \neg q$; $\neg(p \wedge q) \equiv \neg p \vee \neg q$
$p \vee (p \wedge q) \equiv p$; $p \wedge (p \vee q) \equiv p$
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
$p \to (q \to r) \equiv (p \wedge q) \to r$
$(p \to r) \wedge (q \to r) \equiv (p \vee q) \to r$
$(p \to r) \vee (q \to r) \equiv (p \wedge q) \to r$

Duality
Swap $\wedge \leftrightarrow \vee$, $T_0 \leftrightarrow F_0$ after removing $\to, \leftrightarrow$.

Rules of inference
MP: $p,\ p \to q \vdash q$
MT: $p \to q,\ \neg q \vdash \neg p$
HS: $p \to q,\ q \to r \vdash p \to r$
DS: $p \vee q,\ \neg p \vdash q$
Addition: $p \vdash p \vee q$; Simplification: $p \wedge q \vdash p$
Resolution: $p \vee q,\ \neg p \vee r \vdash q \vee r$
Constructive dilemma: $p \to r,\ q \to s,\ p \vee q \vdash r \vee s$
Destructive dilemma: $p \to r,\ q \to s,\ \neg r \vee \neg s \vdash \neg p \vee \neg q$

Fallacies
$p \to q,\ q \vdash p$ (affirming the consequent)
$p \to q,\ \neg p \vdash \neg q$ (denying the antecedent)

Testing arguments
Valid $\iff (P_1 \wedge \ldots \wedge P_n) \to Q$ is a tautology
$\iff P_1 \wedge \ldots \wedge P_n \wedge \neg Q$ is a contradiction.
Invalid: one row with premises T, conclusion F.
CP: to get $Q \to R$, assume Q, derive R.

Quantifiers
$\forall$ with $\to$; $\exists$ with $\wedge$.
$\neg\forall x\, p \equiv \exists x\, \neg p$; $\neg\exists x\, p \equiv \forall x\, \neg p$.
$\forall$ distributes over $\wedge$; $\exists$ over $\vee$.
$\exists y\, \forall x \Rightarrow \forall x\, \exists y$ (not conversely).
US, UG (arbitrary a), ES (new c), EG.
One counterexample disproves $\forall$; one example proves $\exists$.

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